Swirl-String-Theory Canon-v0.8.3

Canonical Reference and Research Framework

Omar Iskandarani*

Independent Researcher
Groningen, The Netherlands

Version v0.8.3
July 2, 2026

Abstract

This document presents Swirl-String Theory (SST) as a structured canonical synthesis rather than as a claim that every phenomenological sector has already been derived to theorem level. The text consolidates the primitive constant chain linking the effective fluid density, characteristic swirl speed, Compton anchoring, and core scale; records delay-induced mode selection as the preferred non-operator route to spectral discreteness; and organizes the atomic bridge, spectroscopic filter, relational-time sector, and topology-based mass program within one explicitly stratified framework. Throughout, orthodox inputs, SST-internal derivations, bridge-level constructions, and conjectural extensions are kept distinct so that the canon remains internally coherent, externally comparable, and open to falsification.

Keywords: Swirl String Theory, Canon, Theoretical Physics, Formal Systems

---

Canon reading guide.

[**ORTHODOX**] established physics or mathematics.
[**DERIVED**] consequences obtained from definitions and assumptions of this SST layer.
[**SPECULATIVE**] conjectural extensions or identifications not yet closed as theorems.

Optional *role* labels (*e.g.*, *Definition*, *Benchmark*, *Constraint*) qualify how to read a statement but do not replace the primary tag. Orthodox constraints—including quantum predictions, atomic spectroscopy, and relativistic consistency—are not overridden: bridge constructions, benchmark sections, and research-track material remain subordinate to the consistency stratification introduced later in the canon.

---

* Email: info@omariskandarani.com
ORCID: 0009-0006-1686-3961
DOI: 10.5281/zenodo.21137961

Contents

1 Philosophical Prelude

1.1 Why a philosophical prelude is necessary

The SST canon is not only a collection of equations but also a statement about what counts as fundamental. Standard quantum theory and general relativity are highly successful formalisms, yet they begin from different ontological primitives: operator-valued states in one case and spacetime geometry in the other. SST instead adopts a single material substrate as the underlying level of description and treats particles, radiation, and clock structure as organized states of that substrate.

1.2 The ontological shift

The canon therefore makes the following conceptual replacement:

$$\text{point particles} \longrightarrow \text{stable swirl-string configurations,} \tag{1}$$

$$\text{vacuum as background} \longrightarrow \text{vacuum as structured medium,} \tag{2}$$

$$\text{quantization as postulate} \longrightarrow \text{discreteness from dynamics and topology.} \tag{3}$$

This is not a denial of effective particle language. Rather, it is a claim that particle language is secondary and emerges from a more primitive hydrodynamic and topological layer.

1.3 Relation to orthodox physics

SST is not introduced as a replacement for the empirical content of orthodox physics. Its canonical role is instead threefold:

1. to reproduce known low-energy structures in the appropriate limits,
2. to identify which apparently fundamental constants or rules may be structurally redundant,
3. to supply a dynamical mechanism for phenomena that are otherwise postulated axiomatically.

In this sense the canon follows a *Rosetta principle*: whenever a standard description already works, SST must map to it before claiming any deeper reinterpretation.

1.4 Why delay and topology matter

Two themes recur across the development of the canon. First, finite circulation delay on closed carriers provides a natural source of temporal nonlocality. Second, topological invariants constrain admissible configurations and stabilize structures that would otherwise collapse or disperse. Together they motivate the canonical SST slogan

delay selects modes, topology protects them.

This is the philosophical reason why the canon treats quantization neither as a primitive operator rule nor as a purely statistical statement.

1.5 Scope of the present canon

The present version of the canon should be read as a structured reference document rather than a claim that every sector has already been derived to theorem level. Some components are mature and reusable, such as the primitive constant chain, circulation quantization, and the Swirl-Clock. Other components are deliberately retained as research-track layers, such as full gauge emergence, topology-to-charge assignments, and complete many-body atomic closure. The point of the canon is therefore not maximal rhetorical closure, but explicit logical organization.

1.6 Interpretive summary

The philosophical stance of SST can be summarized in one sentence:

Physics is interpreted as the dynamics of a continuous medium whose stable, delayed, and topologically organized motions appear effectively as particles, clocks, fields, and spectra.

2 Formal Foundations

2.1 Primitive Structure of the Theory

We define the Swirl-String Theory (SST) as a continuum-based framework built on a minimal set of primitive quantities.

Primitive set:

$$\mathcal{P} = \{\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c\} \tag{4}$$

where:

- ρ_f is the effective background fluid density,
- $\mathbf{v}_{\circlearrowleft}$ is the characteristic swirl velocity,
- ω_c is the Compton angular frequency.

[**ORTHODOX**]: $\omega_c = \frac{m_e c^2}{\hbar}$ is a standard physical constant.

[**SPECULATIVE**]: The identification of ω_c as a primitive dynamical frequency of the vortex core is an SST postulate.

2.2 Derived Horn/Circulation Radius

The quantity previously denoted as the characteristic "core radius" is fixed more precisely as a Compton-locked horn/circulation radius:

$$r_c \equiv R_{\text{horn}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{\omega_c} = \frac{\hbar\|\mathbf{v}_{\circlearrowleft}\|}{m_e c^2} = \frac{\alpha\hbar}{2m_e c} = \frac{r_e}{2}. \tag{5}$$

Here

$$\omega_c \equiv \frac{m_e c^2}{\hbar} \tag{6}$$

is the electron Compton angular frequency. Thus r_c is the radius at which the canonical tangential swirl speed $\|\mathbf{v}_{\circlearrowleft}\|$ is obtained at the Compton angular frequency.

[**CANON / NOTATION**]: ω_c denotes the Compton angular frequency. It must not be confused with the local vorticity scale. If $\omega_{\text{vort},c}$ denotes the local vorticity of a rigid local rotation, then

$$\omega_{\text{vort},c} = 2\omega_c, \qquad R_{\text{horn}} = \frac{2\|\mathbf{v}_{\circlearrowleft}\|}{\omega_{\text{vort},c}}. \tag{7}$$

[**CANON / SEMANTIC RULE**]: r_c is not the resolved physical vortex-tube radius. The local tube/core thickness is denoted by a_{core} and remains a separate geometric or variational quantity:

$$a_{\text{core}} \neq r_c. \tag{8}$$

[**DERIVED**] **Consequence**: Eq. (5) removes r_c as an independent parameter while avoiding the earlier ambiguity between horn-envelope scale, tube radius, and vorticity.

2.3 Circulation Quantization

The circulation is defined as:

$$\Gamma = \oint \mathbf{v} \cdot d\ell \tag{9}$$

We introduce the fundamental circulation quantum:

$$\Gamma_0 = 2\pi r_c \, \|\mathbf{v}_{\circlearrowleft}\| \tag{10}$$

and impose:

$$\Gamma = n\Gamma_0, \quad n \in \mathbb{Z} \tag{11}$$

[ORTHODOX]: Circulation quantization appears in superfluid systems.
[DERIVED]: In SST, Γ_0 is no longer an input but follows from Eq. 5.

2.4 Energy and Mass Relation

Mass is interpreted as stored rotational energy:

$$E = mc^2 \tag{12}$$

[ORTHODOX]: Standard relativistic energy relation.
[SPECULATIVE]: SST interprets this energy as localized vortex kinetic energy.

2.5 Swirl Velocity Constraint

We define the dimensionless ratio:

$$\eta_0 = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{c} \tag{13}$$

Empirically:

$$\eta_0 \approx \frac{\alpha}{2} \tag{14}$$

[DERIVED] Empirical closure: Within the present canonical calibration, this connects the characteristic swirl velocity to the fine-structure constant.

[CRITICAL NOTE] This relation should be read as a calibrated SST closure unless and until it is re-derived from a deeper dynamical sector.

2.6 Swirl-Clock Definition

We introduce the Swirl-Clock as a local scalar field encoding relativistic time dilation:

$$S_{(t)}^{\circlearrowleft} = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}\|^2}{c^2}} \tag{15}$$

[ORTHODOX] Functional form: This matches the standard Lorentz time-dilation factor.

[SPECULATIVE] Identification: In SST, the relevant velocity is interpreted as a local medium-kinematic state rather than only a standard worldline speed.

[DERIVED] Consequence: Once this identification is adopted, the Swirl-Clock provides a local scalar measure of kinematic time modulation within the medium description.

2.7 Density Hierarchy

We distinguish the following densities:

$$\rho_f \quad \text{(effective background fluid density)}, \tag{16}$$
$$\rho_E = \tfrac{1}{2}\rho_f \,\|\mathbf{v}_{\circlearrowleft}\|^2 \quad \text{(swirl energy density)}, \tag{17}$$
$$\rho_m = \rho_E/c^2 \quad \text{(mass-equivalent density)}, \tag{18}$$
$$\rho_{\text{core}} \quad \text{(canonical saturated envelope-equivalent density)}. \tag{19}$$

[ORTHODOX] Definition: $\rho_m = \rho_E/c^2$ is the mass-equivalent density associated with the swirl kinetic accounting above; in orthodox limits, total energy E in a region still yields mass density E/c^2.

[SPECULATIVE] Interpretation: The distinction between background, energetic, and saturated-core sectors is elevated in SST to a structural ontology of the medium.

2.8 Core Density Closure

The core density follows from a force constraint:

$$\rho_{\text{core}} \sim \frac{m_e c^2}{2\pi \|\mathbf{v}_{\circlearrowleft}\|^2 {r_c}^3} \tag{20}$$

[SPECULATIVE] Closure ansatz: Equation (20) is retained as the canonical core-density closure used by the present SST calibration program.

[DERIVED] Consequence: Within that closure, ρ_{core} is no longer an independent parameter. **[CRITICAL NOTE AFTER RADIUS DISAMBIGUATION].** Equation (20) defines a canonical density associated with the neutral circulation envelope at scale r_c. It should not be read as the material density of the microscopic rotation-dominated tube. If a local tube mass density is required, it must be introduced through the separate tube radius a_{core} and a corresponding local tube-volume model. Thus ρ_{core} is retained as a calibrated envelope-equivalent density, while a_{core} controls local filament cutoffs and reconnection-scale modelling.

2.9 Chronos–Kelvin Invariant and Clock–Radius Transport

For a thin material loop of radius $R(t)$ carried by an incompressible, inviscid flow with no reconnection events, Kelvin's theorem implies conservation of circulation. In the near-solid-body core approximation this gives

$$\frac{D}{Dt}(R^2\omega) = 0, \tag{21}$$

where ω is the characteristic local vorticity on the loop. Using the core relation $v_t = \omega r_c$ together with Eq. (15), one may rewrite

$$\omega = \frac{c}{r_c}\sqrt{1 - S_{(t)}^2}, \tag{22}$$

so that Eq. (21) takes the equivalent form

$$\frac{D}{Dt}\left(\frac{c}{r_c} R^2 \sqrt{1 - S_{(t)}^2}\right) = 0. \tag{23}$$

[ORTHODOX]: Eq. (21) is the thin-loop form of Kelvin circulation conservation in inviscid barotropic flow [6, 23].

[DERIVED]: Eq. (23) is the SST rewriting of the same invariant in terms of the Swirl-Clock.

Differentiating Eq. (23) gives a useful transport identity:

$$\frac{dS_{(t)}}{dt} = \frac{2(1 - S_{(t)}^2)}{S_{(t)}} \frac{1}{R} \frac{dR}{dt}. \tag{24}$$

[DERIVED]: contraction of a material loop drives stronger clock suppression, while expansion relaxes it.

2.10 Reparametrized Zero-Parameter Corollary

Older SST canon layers often used the triplet (Γ_0, ρ_f, r_c) as the primitive parameterization. In the present v0.8.1 architecture, Eq. (5) and Eq. (10) show that this is a reparameterization of the present primitive set $(\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c)$ rather than a distinct axiom.

Explicitly,

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\| = 2\pi \frac{\|\mathbf{v}_{\circlearrowleft}\|^2}{\omega_c}. \tag{25}$$

Hence any dimensional SST quantity written in terms of (Γ_0, ρ_f, r_c) can be rewritten in terms of $(\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c)$ with no new input.

A coarse-grained density law retained from earlier canon layers is

$$\rho_f = K\,\Omega, \qquad K = \frac{\rho_{\rm core}\, r_c}{\|\mathbf{v}_{\circlearrowleft}\|}, \tag{26}$$

where Ω is a coarse-grained local rotation rate and K has dimensions $\mathrm{kg\,m^{-3}\,s}$. This relation records the canonical bridge between a sparse effective bulk density and a high-density localized core.

[DERIVED COROLLARY]: the older "zero-parameter" language is retained only as a statement of reparameterization discipline: once one primitive triplet is fixed, no additional dimensional constants should be introduced without explicit justification.

2.11 Canonical Master Equation

The preceding subsections define the four ingredients that later SST papers often treat separately: a local mass-equivalent density, a local Swirl-Clock, a geometric gate, and a topological kernel. We therefore introduce a single canonical equation that organizes these sectors in one place.

First define the local swirl energy density and its mass-equivalent form:

$$\rho_E(x) = \frac{1}{2}\rho_f(x)\,\|\mathbf{v}_{\circlearrowleft}(x)\|^2 \tag{27}$$

$$\rho_m(x) = \frac{\rho_E(x)}{c^2} \tag{28}$$

Using Eq. 15, the local Swirl-Clock factor is

$$S^{\circlearrowleft}_{(t)}(x) = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}(x)\|^2}{c^2}} \tag{29}$$

We then define a binary geometric exposure gate

$$G(K) \in \{0, 1\} \tag{30}$$

together with the canonical geometric factor

$$\Pi_K = \left(\frac{\lambda_c}{\pi r_c}\right)^{G(K)} = \left(\frac{4}{\alpha}\right)^{G(K)}. \tag{31}$$

The canonical topological kernel is taken to be

$$\Xi_K = [\alpha_C\, C(K) + \beta_L\, L(K) + \gamma_H\, \mathcal{H}(K)]\, \varphi^{-2k(K)}. \tag{32}$$

The canonical SST master equation is therefore

$$M_K(x) = \rho_m(x)\, {r_c}^3\, \Pi_K\, \Xi_K\, S^{\circlearrowleft}_{(t)}(x)^{-2} \tag{33}$$

$$= \left(\frac{\rho_f(x)\, \|\mathbf{v}_{\circlearrowleft}(x)\|^2}{2c^2}\right) {r_c}^3 \left(\frac{\lambda_c}{\pi r_c}\right)^{G(K)} [\alpha_C\, C(K) + \beta_L\, L(K) + \gamma_H\, \mathcal{H}(K)]\, \varphi^{-2k(K)} S^{\circlearrowleft}_{(t)}(x)^{-2} \tag{34}$$

[SPECULATIVE] Organizing principle: This equation factorizes SST into four sectors:

$$\text{medium scale} \times \text{geometric gate} \times \text{topological kernel} \times \text{clock impedance}. \tag{35}$$

[DERIVED] Dimensional check:

$$\left[\frac{\rho_f\, \|\mathbf{v}_{\circlearrowleft}\|^2}{c^2} {r_c}^3\right] = \text{kg} \tag{36}$$

and all remaining factors are dimensionless, so Eq. 34 has the correct mass dimension.

[SPECULATIVE] Interpretation: The same local kinematic state that slows the Swirl-Clock also enters the effective mass scale; mass and local proper-time modulation are therefore treated as two aspects of one medium configuration.

[CRITICAL NOTE] Equation (34) is presently a canonical synthesis formula, not yet a theorem derived from first-principles filament dynamics alone.

In the stationary weak-swirl limit $S^{\circlearrowleft}_{(t)} \to 1$, it reduces to the static mass-functional form; in the local-kinematic limit at fixed K, it reduces to a pure clock-impedance correction. The hierarchy is therefore carried predominantly by the geometric and topological factors, while the Swirl-Clock supplies a coherent local correction rather than the primary source of mass separation.

2.12 Geometric Representation

A swirl-string is represented as a closed curve:

$$X(t, \sigma) = (x(t, \sigma), y(t, \sigma), z(t, \sigma)) \tag{37}$$

with $\sigma \in [0, 2\pi)$.

2.13 Summary of Dependency Structure

The full dependency chain is:

$$(\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c) \to r_c \to \Gamma_0 \to (\rho_E, \rho_m, \rho_{\mathrm{core}}) \to S^{\circlearrowleft}_{(t)} \to M_K. \tag{38}$$

[DERIVED] Consistency statement: No quantity is defined circularly. The canonical closure is summarized by Eq. 34.

2.14 Consistency Statement

- All derived quantities depend only on $\mathcal{P}$
- No external parameters are introduced
- The theory is closed under its primitive set

2.15 Interpretation

The Formal Foundations establish SST as:

- a continuum theory with minimal primitives
- a geometrically grounded framework
- a non-circular parameter system

2.16 Maximal Swirl Tension

2.16.1 Definition and canonical form

Within the primitive set $\mathcal{P} = \{\rho_f, \mathbf{v}_{\circlearrowleft}, \omega_c\}$, a characteristic force scale emerges naturally from the Compton energy stored over the canonical circulation cross-section. We define the *maximal swirl tension* as

$$F^{\max}_{\mathrm{swirl}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|\,\hbar}{2{r_c}^2} = \frac{\hbar\omega_c}{2r_c}. \tag{39}$$

[DERIVED] Canonical form: substituting $\|\mathbf{v}_{\circlearrowleft}\| = \omega_c r_c$ into Eq. (39) gives $F^{\max}_{\mathrm{swirl}} = \hbar\omega_c/2r_c$, the Compton energy scale divided by twice the canonical circulation radius. Numerically, $F^{\max}_{\mathrm{swirl}} = 29.053507$ N.

[CRITICAL NOTE] $F^{\max}_{\mathrm{swirl}}$ is a mechanical reformulation of the Compton energy scale over the canonical circulation cross-section. It is *not* independent of $\hbar$ and m_e; its role in the canon is interpretive and organisational rather than that of an independent primitive.

2.16.2 Internal consistency identities

The following expressions are algebraically equivalent within the canonical constant chain and serve as coherence verifications rather than independent derivations:

$$\begin{aligned} F^{\max}_{\mathrm{swirl}} &= \frac{e^2}{16\pi\varepsilon_0 {r_c}^2}, && (40)\\ &= \frac{h\alpha c}{8\pi {r_c}^2}, && (41)\\ &= \pi {r_c}^2\,\rho_{\mathrm{calc}}\,\|\mathbf{v}_{\circlearrowleft}\|^2, && (42) \end{aligned}$$

where $\rho_{\text{calc}} = 4F^{\max}_{\text{swirl}}/(\pi\alpha^2 c^2 r_c{}^2)$ is the derived bulk density consistent with the canonical neutral-circulation closure.

[DERIVED] Consistency check: Eqs. (40)–(42) are equivalent given $\|\mathbf{v}_{\circlearrowleft}\| = \alpha c/2$ and the standard identity $e^2/(4\pi\varepsilon_0) = \alpha\hbar c$. Their mutual agreement constitutes a non-trivial numerical self-consistency test of the primitive set.

2.16.3 Strongest non-circular identity: the Rydberg constant

[DERIVED] The Rydberg constant is recoverable directly from the three SST primitives without e, ε_0, or m_e as explicit input:

$$R_\infty = \frac{\|\mathbf{v}_{\circlearrowleft}\|^3}{\pi\, r_c\, c^3}. \tag{43}$$

This is the canonical benchmark for the internal non-redundancy of the primitive set. Combining Eq. (43) with Eq. (39) yields the compact relation

$$\boxed{F^{\max}_{\text{swirl}} = \frac{32\pi^2\hbar R_\infty^2 c}{\alpha^5},} \tag{44}$$

which expresses $F^{\max}_{\text{swirl}}$ in terms of spectroscopic observables alone, with no reference to m_e, r_c, or e as separate inputs. A further identity connecting atomic and force scales is

$$R_\infty = F^{\max}_{\text{swirl}} \cdot \frac{\alpha^3}{4\pi m_e c^2}, \tag{45}$$

which states that the Rydberg constant is the maximal swirl tension scaled by the Compton energy and a purely geometric factor $\alpha^3/4\pi$.

2.16.4 Threefold physical interpretation

The quantity $F^{\max}_{\text{swirl}}$ admits three distinct physical interpretations, each arising from a different sector of physics. Their numerical coincidence within the canonical constant chain is a non-trivial coherence result.

(i) Gravitational analogue: the Planck force ceiling. In general relativity, the combination $c^4/4G$ sets the maximum force that any physical process can transmit without forming a causal horizon [29]. We denote this scale the maximal gravitational tension:

$$F^{\max}_{\text{gr}} = \frac{c^4}{4G}. \tag{46}$$

[ORTHODOX] $F^{\max}_{\text{gr}}$ is the Planck force, an invariant upper bound in classical general relativity [29].

[DERIVED] The ratio of the two tension scales reduces exactly to

$$\boxed{\frac{F^{\max}_{\text{swirl}}}{F^{\max}_{\text{gr}}} = \frac{4\alpha_g}{\alpha},} \tag{47}$$

where $\alpha_g = Gm_e^2/\hbar c$ is the gravitational fine-structure constant of the electron and α is the electromagnetic fine-structure constant. Numerically, $4\alpha_g/\alpha \approx 9.6 \times 10^{-43}$, in agreement with the directly computed ratio 29.053507 N / 3.026×10^{43} N.

[SPECULATIVE] Interpretation: the hierarchy between the electromagnetic and gravitational force scales in SST is not an unexplained numerical coincidence but the ratio of the two fundamental coupling constants of the electron. The structured medium is $\sim 10^{42}$ times stiffer than the space-time geometry at the canonical circulation scale, and this stiffness ratio equals $\alpha/(4\alpha_g)$ exactly.

(ii) Electrostatic interpretation: the Coulomb ceiling at core scale. When two protons are compressed to a centre-to-centre separation equal to the canonical circulation radius r_c, the Coulomb repulsion is

$$F_{pp}(r_c) \;=\; \frac{e^2}{4\pi\varepsilon_0 {r_c}^2} \;=\; 4\,F_{\text{swirl}}^{\max}. \tag{48}$$

[DERIVED] The maximal Coulomb repulsion at the canonical circulation scale is exactly four times $F_{\text{swirl}}^{\max}$. The factor of four matches the denominator structure of Eq. (46) and is not numerically coincidental within the canonical chain.

[SPECULATIVE] Interpretation: $F_{\text{swirl}}^{\max}$ is the quarter-Coulomb force at the canonical circulation scale. The medium sustains one quarter of the maximum proton–proton Coulomb repulsion before the neutral circulation envelope reaches its structural limit. This provides a physical ceiling on the compression of charged vortex configurations.

(iii) Mechanical interpretation: the spring constant of photon–electron interaction. The electron's internal vortex is modelled as a linear spring with stiffness

$$K_e \;=\; \frac{F_{\text{swirl}}^{\max}}{n\,r_c}, \qquad n = 2, \tag{49}$$

where n counts the number of half-wavelength segments on the core and is fixed to $n = 2$ by the photon–electron swirl matching condition. The natural oscillation frequency of the electron core under this spring is

$$\omega_{\text{spring}} \;=\; \sqrt{\frac{K_e}{m_e}} \;=\; \sqrt{\frac{F_{\text{swirl}}^{\max}}{2\,r_c\,m_e}} \;=\; \frac{\omega_c}{\alpha} \;=\; \frac{m_e c^2}{\alpha\hbar}, \tag{50}$$

a frequency intermediate between the atomic Rydberg scale ($\sim \alpha^2\omega_c$) and the bare Compton scale (ω_c).

[DERIVED] The energy stored in this spring at maximum compression $\delta = r_c$ is

$$E_{\text{spring}} \;=\; \tfrac{1}{2}\,K_e\,{r_c}^2 \;=\; \frac{F_{\text{swirl}}^{\max}\,r_c}{4} \;=\; \frac{\hbar\omega_c}{8} \;=\; \frac{m_e c^2}{8}. \tag{51}$$

One eighth of the electron rest energy is stored as internal vortex-spring potential at maximum compression.

[SPECULATIVE] Physical picture: during photon absorption, the electron vortex core is compressed for a duration of order one Planck time t_p into a transient mass-like configuration. The restoring force scale of this compression is $F_{\text{swirl}}^{\max}$, and the energy scale of the transient is $m_e c^2/8$. The spring releases at the Compton period, generating the recoil that initiates the Coulomb interaction with the proton.

[CRITICAL NOTE] Equation (50) assumes $n = 2$ from empirical matching. A first-principles derivation of the photon wrap number from the delay dynamics of Section 7 is an open derivational step.

2.16.5 Unified summary

The three interpretations are collected in Table 1. Together they identify $F_{\rm swirl}^{\max}$ as the *structural tension threshold of the medium*: the scale at which a physical system — gravitational, electrostatic, or mechanical — exhausts its capacity to store energy in the current configuration and must either fail topologically, form a horizon, or release energy as radiation.

Table 1: Threefold physical interpretation of $F_{\rm swirl}^{\max}$ within the SST canonical framework.

Context	Expression	Physical meaning	Status
Gravitational ceiling	$F_{\rm gr}^{\max}/F_{\rm swirl}^{\max} = \alpha/(4\alpha_g)$	Medium stiffer than spacetime by $\alpha/4\alpha_g$	[Derived]
Coulomb ceiling	$F_{pp}(r_c) = 4\,F_{\rm swirl}^{\max}$	Quarter of max. Coulomb force at core scale	[Derived]
Vortex spring	$K_e = F_{\rm swirl}^{\max}/2r_c$	Spring constant of photon–electron interaction	[Speculative]

[SPECULATIVE] Canonical interpretation: $F_{\rm swirl}^{\max}$ is the tension at which the structured æther medium reaches its topological integrity limit. It plays the same organisational role for the electromagnetic and mechanical sectors that $F_{\rm gr}^{\max} = c^4/4G$ plays for the gravitational sector. Their ratio is precisely the ratio of the two fundamental coupling constants of the electron.

3 Axiomatic Framework

3.1 Preliminaries

We formalize Swirl-String Theory (SST) as an axiomatic continuum theory defined over a three-dimensional spatial domain $\mathcal{M} \subset \mathbb{R}^3$ equipped with a preferred time parameter $t \in \mathbb{R}$.

The theory is constructed from a minimal primitive set $\mathcal{P}$ (Section 2) and a set of axioms $\mathcal{A}$ governing admissible configurations and dynamics.

3.2 Axiom 1: Continuum Substrate

Statement. There exists a continuous, incompressible, inviscid medium filling $\mathcal{M}$, characterized by a constant background density ρ_f.

$$\nabla \cdot \mathbf{v} = 0 \tag{52}$$

[ORTHODOX]: This corresponds to the standard incompressible Euler framework.

Interpretation. All physical structures arise as configurations within this medium; no discrete particles are assumed at the fundamental level.

3.3 Axiom 2: Existence of Stable Vortex Filaments

Statement. The medium admits stable, localized, topologically nontrivial vortex filament solutions.

[ORTHODOX] Physics input: vortex filaments are known idealized structures in fluid dynamics.

[SPECULATIVE] Extension: such filaments are assumed here to be dynamically stable and persist as fundamental carriers.

Mathematical form. A filament is represented as a closed embedding:

$$X : S^1 \to \mathcal{M}, \quad \sigma \mapsto X(t, \sigma) \tag{53}$$

3.4 Axiom 3: Circulation Quantization

Statement. Circulation around any stable filament is quantized:

$$\Gamma = \oint \mathbf{v} \cdot d\ell = n\Gamma_0, \quad n \in \mathbb{Z} \tag{54}$$

[**ORTHODOX**]: Quantized circulation appears in superfluid systems.
[**SPECULATIVE**]: Elevated here to a universal invariant.

3.5 Axiom 4: Compton Anchoring

Statement. The characteristic angular frequency of the vortex core satisfies:

$$\frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = \omega_c \tag{55}$$

[**ORTHODOX**]: ω_c is the Compton frequency.
[**SPECULATIVE**]: Identified as the intrinsic rotational frequency of the vortex core.
Consequence. The core radius is not independent but derived (Eq. 5).

3.6 Axiom 5: Delay as a Constitutive Principle

Statement. Finite propagation time along closed vortex structures is a fundamental dynamical ingredient.

$$\tau = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|} \tag{56}$$

[**DERIVED**] Kinematic consequence: the delay follows from finite signal speed along a closed carrier.

Axiomatic role. The delay is not treated as negligible, but as structurally relevant to the dynamics.

Interpretation. Temporal nonlocality is intrinsic to the dynamics.

3.7 Axiom 6: Energy Localization

Statement. Energy is localized in vortex structures and determines mass via:

$$E = mc^2 \tag{57}$$

[**ORTHODOX**]: Relativistic energy relation.
[**SPECULATIVE**]: Interpreted as rotational energy of the vortex configuration.

3.8 Axiom 7: Swirl-Clock Relational Time

Statement. Local time is encoded in a scalar field derived from fluid kinematics:

$$S_{(t)}^{\circlearrowleft} = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}\|^2}{c^2}} \tag{58}$$

[**ORTHODOX**] Functional form: this matches Lorentz time dilation.

[**SPECULATIVE**] Identification: the relevant velocity is taken to be the local medium kinematic state.

[**DERIVED**] Consequence: within SST this yields a relational clock field tied to local motion.

Interpretation. Time is not fundamental but relational to the state of motion in the medium.

3.9 Axiom 8: Density Saturation and Core Structure

Statement. The vortex core exhibits a saturated density ρ_{core} determined by mechanical constraints of the medium.

[**SPECULATIVE**] Closure hypothesis: the saturated density is fixed by the closure relation of Eq. (20).

[**DERIVED**] Consequence: once that closure is adopted, matter corresponds to regions where energy density reaches a structural limit.

3.10 Logical Structure of the Axioms

The axioms form a dependency hierarchy:

$$\text{Continuum} \to \text{Vortex Structures} \to \text{Quantization} \to \text{Delay Dynamics} \to \text{Mass/Energy Emergence} \tag{59}$$

3.11 Minimality Principle

Statement. The axioms are minimal in the sense that removing any one of them destroys at least one of the following:

- existence of stable structures
- emergence of discrete spectra
- compatibility with known physics

3.12 Philosophical Interpretation

The axiomatic structure implies:

- Particles are not fundamental objects but stable dynamical configurations
- Discreteness is emergent, not postulated
- Time is relational, not absolute

3.13 Consistency Condition

A valid SST realization must satisfy:

- Compatibility with orthodox physics in appropriate limits
- Internal logical closure
- Explicit traceability from axioms to predictions

3.14 Canonical Statement

Swirl-String Theory is defined as the minimal continuum theory in which stable vortex structures, governed by quantized circulation and delay dynamics, give rise to discrete physical phenomena consistent with known physics.

4 Consistency and Classification Framework

4.1 Classification Scheme

All statements are labeled as:

- [**ORTHODOX**] — established physics
- [**DERIVED**] — logically deduced within SST
- [**SPECULATIVE**] — novel or unverified

4.2 Extended Labeling Convention

We distinguish between *epistemic status* and *editorial role.*

Epistemic status. The only primary status labels used in the present Canon are:

- [**ORTHODOX**] — established in standard physics or mathematics
- [**DERIVED**] — obtained internally from the present SST assumptions, definitions, or closures
- [**SPECULATIVE**] — conjectural SST extension, interpretation, or unverified identification

Editorial role. Additional descriptors such as *Definition*, *Constraint*, *Interpretation*, *Benchmark*, *Roadmap*, and *Dimensional check* are not independent epistemic classes. They indicate only the role played by the statement.

Usage rule. When both are needed, the Canon uses the format

$$\texttt{[STATUS] Role: statement.} \tag{60}$$

For example, one may write [**ORTHODOX**] **Constraint**, [**DERIVED**] **Consequence**, or [**SPECULATIVE**] **Interpretation**.

4.3 Orthodox Constraints

The following must remain intact:

- Quantum mechanics predictions
- Atomic spectroscopy
- Relativistic consistency

4.4 Internal Consistency Rules

- No circular definitions
- All primitives explicitly defined
- Derived quantities must follow from axioms

4.5 Conflict Resolution Strategy

Priority order:

1. Logical consistency
2. Experimental agreement
3. Canonical simplicity

4.6 Epistemic Status Tracking

Each major result must explicitly state:

- Assumptions
- Derivation status
- Empirical testability

In particular, a definitional choice must not be presented as a theorem, and an interpretive bridge must not be presented as an orthodox result unless the underlying mathematical structure is itself standard.

4.7 Canon Integrity Principle

A valid SST canon must be internally consistent, externally compatible, and explicitly stratified by epistemic status.

5 Geometric and Topological Sector

5.1 Role of geometry and topology in the canon

This sector collects those parts of SST in which the identity of a state depends not only on local field amplitudes but on the global organization of the carrier. Geometry enters through lengths, radii, and closure constraints; topology enters through linking, crossing, chirality, and other invariant features that survive smooth deformations. At canon level, the main role of this sector is selective rather than exhaustive: it provides the organizing variables that feed the topological kernel Ξ_K and the geometric gate Π_K in Eq. (34).

The canon therefore treats geometry and topology as follows:

1. geometry sets admissible scales and shielding ratios,
2. topology labels persistent identity classes,
3. only those quantities that survive smooth deformation are allowed to enter the long-lived state classification.

This is also why layer or dressing indices are not treated as primary identity labels: they may modify a state, but they do not replace the topology-first classification of the carrier itself.

5.2 Spectral Structure of Locked Closed Carriers (Secondary Result)

Scope and status. This subsection records a *secondary consequence* of the kinematics of rationally locked, closed carriers. It does not introduce new primitives. All results are conditional on the existence of a well-defined phase field and a regular mode spectrum. Status: **derived under spectral assumptions; experimentally unverified**.

Kinematic reduction. Consider a closed carrier with a single central driving phase $\phi(t)$ and N_s sector variables $\theta_i(t)$ $(i = 1, \dots, N_s)$ subject to a rational locking relation

$$\theta_i = \sigma_i \kappa_i \phi + \delta_i, \qquad \sigma_i \in \{+1, -1\}, \quad \kappa_i \in \mathbb{Q}. \tag{61}$$

In the rigid-lock limit, the dynamics reduce to a single collective coordinate $\phi(t)$ with effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{I_{\text{eff}}}{2}\dot{\phi}^2 - V(\phi), \tag{62}$$

where I_{eff} is an effective inertia and $V(\phi)$ is a periodic locking potential.

Distributed phase field. On a closed carrier of length L, allow small phase fluctuations

$$\phi(t) \longrightarrow \phi_0 + \eta(s,t), \qquad s \in [0, L], \tag{63}$$

with periodic boundary condition

$$\eta(s + L, t) = \eta(s, t). \tag{64}$$

A minimal quadratic field model consistent with Eq. (62) is

$$\mathcal{L}_{\text{field}} = \int_0^L \left[\frac{\mu_{\text{eff}}}{2}(\partial_t \eta)^2 - \frac{T_{\text{eff}}}{2}(\partial_s \eta)^2 - \frac{\mu_{\text{eff}}\omega_0^2}{2}\eta^2 \right] ds, \tag{65}$$

where μ_{eff} is an effective phase inertia density, T_{eff} an effective stiffness, and ω_0 the small-oscillation frequency set by $V(\phi)$.

Dimensional consistency:

$$[\mu_{\text{eff}}] = \text{kg}\,\text{m}, \qquad [T_{\text{eff}}] = \text{N}, \qquad [\omega_0] = \text{s}^{-1}.$$

Mode spectrum. Expanding in normal modes

$$\eta(s,t) = \sum_{n \in \mathbb{Z}} a_n(t)\, e^{i k_n s}, \qquad k_n = \frac{2\pi n}{L}, \tag{66}$$

yields the dispersion relation

$$\boxed{\omega_n^2 = \omega_0^2 + c_\phi^2 k_n^2, \qquad c_\phi^2 = \frac{T_{\text{eff}}}{\mu_{\text{eff}}}.} \tag{67}$$

For large n,

$$\omega_n \sim \frac{2\pi c_\phi}{L}\, n. \tag{68}$$

Cubic spectral sums and $\zeta(3)$. Consider a cubic-weighted spectral contribution of the form

$$\Delta\mathcal{E}^{(3)} = A_3 \sum_{n=1}^{\infty} \frac{1}{\omega_n^3}, \qquad [A_3] = \mathrm{J\,s^{-3}}. \tag{69}$$

Using the asymptotic scaling (68),

$$\Delta\mathcal{E}^{(3)} \approx A_3 \left(\frac{L}{2\pi c_\phi}\right)^3 \sum_{n=1}^{\infty} \frac{1}{n^3}. \tag{70}$$

Hence

$$\boxed{\Delta\mathcal{E}^{(3)} \approx A_3 \left(\frac{L}{2\pi c_\phi}\right)^3 \zeta(3).} \tag{71}$$

Interpretation. Equation (71) shows that $\zeta(3)$ arises from the *high-mode spectral tail* of a closed carrier, not from the geometric threefold structure alone. The role of the locking relation (61) is indirect: it fixes I_{eff}, ω_0, and admissible boundary conditions, thereby determining the spectral scale (L, c_ϕ).

Universality. Different geometric realizations (e.g., distinct multi-sector carriers) that reduce to the same class of effective phase dynamics and share the asymptotic linear spectrum (68) will produce the same $\zeta(3)$ factor in Eq. (71), up to system-dependent prefactors.

$$\boxed{\zeta(3)\text{ is a spectral invariant of the mode tower, not a direct invariant of the carrier geometry.}} \tag{72}$$

Limitations and falsifiers. The result fails if:

1. the mode spectrum is not asymptotically linear in n,
2. strong damping or nonlocal coupling invalidates the expansion (65),
3. the system does not admit a well-defined periodic phase field on a closed domain.

Minimal experimental test. Measure mode frequencies ω_n for $n = 1, 2, 3, \ldots$ and verify

$$\frac{\omega_n}{n} \to \text{constant} \quad \text{as } n \to \infty. \tag{73}$$

Deviation from this scaling excludes Eq. (71).

Status. **Secondary derived result.** Valid under spectral assumptions; not a defining postulate.

5.3 Symmetry-Sensitive Dark-Knot Classification

A compact classification retained from earlier SST topology layers distinguishes visible, quasi-dark, and dark sectors by chirality, helicity, and low-order instability content.

Let $\chi(K)$ denote a chirality indicator, let

$$\mathcal{H}[K] = \int \mathbf{v} \cdot (\nabla \times \mathbf{v})\, dV \tag{74}$$

be the helicity functional, and let $N_u^{(1)}(K)$ count low-order unstable modes in a chosen perturbative stability analysis. Then the retained taxonomy is

$$\text{dark:} \quad \chi(K) = 0, \quad \mathcal{H}[K] \approx 0, \quad N_u^{(1)}(K) = 0, \tag{75}$$

$$\text{quasi-dark:} \quad |\chi(K)| \ll 1, \quad \mathcal{H}[K] \approx 0, \quad 0 < N_u^{(1)}(K) \leq 2, \tag{76}$$

$$\text{visible:} \quad \chi(K) \neq 0 \quad \text{or} \quad \mathcal{H}[K] \neq 0. \tag{77}$$

[**ORTHODOX**]: helicity is a standard invariant of ideal vortex dynamics [26, 6].

[**SPECULATIVE**]: the interpretation of Eq. (77) as a matter-sector taxonomy is an SST model choice.

Heuristic balance diagnostic from the SST helicity archive. As a supporting diagnostic (not a stand-alone classifier), define

$$\Delta_H(K) = \left| \widehat{\mathcal{H}}_b(K) + \frac{1}{2} \right|, \tag{78}$$

where $\widehat{\mathcal{H}}_b(K)$ is the archived normalized helicity-like base indicator for class K. Near-balanced values with $\Delta_H(K) \ll 1$ are treated only as candidate evidence for dark or quasi-dark sectors.

[**CRITICAL NOTE**]: in canon usage, the diagnostic above is subordinate to Eq. (77); no single scalar is sufficient for dark-sector membership without the symmetry and instability filters.

[**CANON WORKFLOW NOTE**]: if SST-labelled and parallel legacy-labelled helicity exports are numerically identical, only the SST-labelled archive is required for canon workflows.

Symmetry-first refinement (v0.8.x). In the taxonomy program, candidate sectors are labeled symmetry-first, then filtered by helicity balance and low-order instability content. Amphichiral knot types furnish the preferred *candidate* pool for the dark bin in symmetry-first tables; assigning "dark" still requires the conjunction of near-zero chirality indicators, helicity balance in the sense of Eq. (77), and vanishing low-order instability count. The implication "amphichiral $\Rightarrow$ dark" is a classifier proposal tested against observables and archive diagnostics, not a closed theorem.

Ambiguity tie-break. If rule conjunctions overlap, precedence is

$$\text{symmetry gate} \;\rightarrow\; \text{helicity balance gate} \;\rightarrow\; \text{stability gate}.$$

6 Swirl–Electromagnetic Bridge

6.1 Minimal transverse field sector

A conservative bridge from the fluid sector to a radiation sector is obtained by introducing a divergence-free transverse field $\mathbf{A}_{\text{eff}}$ with

$$\nabla \cdot \mathbf{A}_{\text{eff}} = 0, \tag{79}$$

and effective Lagrangian density

$$\mathcal{L}_{\text{rad}} = \frac{\rho_f}{2} \, \|\partial_t \mathbf{A}_{\text{eff}}\|^2 - \frac{K_\gamma}{2} \, \|\nabla \times \mathbf{A}_{\text{eff}}\|^2. \tag{80}$$

Varying Eq. (80) and using Eq. (79) gives

$$\frac{1}{c_\gamma^2}\partial_t^2 \mathbf{A}_{\text{eff}} - \nabla^2 \mathbf{A}_{\text{eff}} = 0, \qquad c_\gamma^2 = \frac{K_\gamma}{\rho_f}. \tag{81}$$

Define

$$\mathbf{E}_{\text{eff}} = -\partial_t \mathbf{A}_{\text{eff}}, \qquad \mathbf{B}_{\text{eff}} = \nabla \times \mathbf{A}_{\text{eff}}. \tag{82}$$

Then one immediately recovers the source-free identities

$$\nabla \cdot \mathbf{B}_{\text{eff}} = 0, \tag{83}$$
$$\nabla \times \mathbf{E}_{\text{eff}} = -\partial_t \mathbf{B}_{\text{eff}}. \tag{84}$$

[ORTHODOX FORM]: Eq. (81) and Eq. (84) are the standard transverse-wave and source-free Faraday identities in vector-potential form [16].

[SPECULATIVE IDENTIFICATION]: SST identifies $\mathbf{A}_{\text{eff}}$ with a coarse-grained torsional or director displacement field of the medium.

6.2 Photon / unknot sector

The minimal retained radiation picture is that an R-phase excitation is an unknotted, finite-duration torsional packet propagating in the field $\mathbf{A}_{\text{eff}}$. At the Rosetta level this corresponds to a source-free transverse wave packet with dispersion inherited from Eq. (81).

[SPECULATIVE]: handedness of the local torsional packet is taken to encode polarization, while additional azimuthal phase winding encodes orbital angular momentum.

6.3 Swirl pressure law

For steady axisymmetric swirl, Euler radial balance gives

$$\frac{1}{\rho_f}\frac{dp_{\text{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}. \tag{85}$$

For a locally rigid rotation profile $v_\theta(r) = \Omega r$, Eq. (85) integrates to

$$p_{\text{swirl}}(r) = p_0 + \frac{1}{2}\rho_f \Omega^2 r^2, \tag{86}$$

while for an exterior irrotational profile $v_\theta(r) = \Gamma/(2\pi r)$ one finds

$$p_{\text{swirl}}(r) = p_\infty - \frac{\rho_f \Gamma^2}{8\pi^2 r^2}. \tag{87}$$

In a flat rotation curve with asymptotically constant azimuthal speed $v_\theta(r) \to v_0$ as r grows in a window where the radial balance (85) remains valid, integration yields the logarithmic tail

$$p_{\text{swirl}}(r) = p_0 + \rho_f v_0^2 \ln\left(\frac{r}{r_0}\right), \tag{88}$$

for suitable reference radius r_0 and offset p_0 within that window.

[ORTHODOX] Assumption block: Eq. (85) is taken in the usual incompressible, inviscid, stationary, axisymmetric regime with azimuthal-dominant flow, so that radial drift, viscosity, and stratification corrections are subleading in the domain of validity.

[ORTHODOX] Dimensional check: $[\rho_f{}^{-1}\, dp/dr] = \mathrm{m\,s^{-2}} = [v_\theta^2/r]$.

[ORTHODOX]: Eq. (85) is the ordinary radial Euler balance for steady swirl [6, 23].

[DERIVED SST ROLE]: this pressure law is the canonical bridge between localized circulation and large-scale force analogies. Interpreting $p_{\rm swirl}$ as a dark-sector classifier or support layer is an SST model step, logically separate from the orthodox fluid derivation.

[DERIVED] Observables / falsifiers: (i) consistency of a pressure profile reconstructed from archived $v_\theta(r)$ with the radial gradient implied by (85); (ii) null test: systematic sign or scale mismatch of the reconstructed $dp_{\rm swirl}/dr$ relative to v_θ^2/r across validated radial windows excludes the applicability of the balance in those windows.

6.4 Minimal effective Lagrangian bridge

A compact field-theoretic summary retaining the clock, radiation, and matter sectors is

$$\mathcal{L}_{\rm eff} = \mathcal{L}_T + \mathcal{L}_{\rm rad} + \sum_K \bar{\Psi}_K\big(i\gamma^\mu D_\mu - m_K^{(\rm sol)}\big)\Psi_K + \cdots\,, \tag{89}$$

where $\mathcal{L}_T$ is the clock/foliation sector introduced in Section 10, $\mathcal{L}_{\rm rad}$ is given by Eq. (80), and the ellipsis denotes higher-order self-interaction, topological, or medium-response operators.

[SPECULATIVE BRIDGE]: Eq. (89) is a bookkeeping layer, not yet a completed first-principles derivation.

7 Delay and Mode Selection Theory

7.1 Circulation Delay as a Physical Primitive

We introduce the circulation delay as a constitutive element of the theory. For a closed vortex filament of effective length L and characteristic propagation speed $\|\mathbf{v}_\circlearrowleft\|$, the circulation time is defined as:

$$\tau = \frac{L}{\|\mathbf{v}_\circlearrowleft\|} \tag{90}$$

In the special case of a minimal closed loop associated with the neutral circulation scale, this reduces to:

$$\tau_c = \frac{2\pi r_c}{\|\mathbf{v}_\circlearrowleft\|} = \frac{2\pi}{\omega_c} \tag{91}$$

This establishes a direct identification between the circulation delay and the inverse Compton frequency.

[DERIVED]: The intrinsic delay of the neutral circulation loop is identical to the Compton period.

7.2 Delayed Self-Interaction Dynamics

We model the phase $\phi(t)$ of a circulating excitation as a delayed self-interacting oscillator:

$$\dot{\phi}(t) = \omega_0 + \kappa \sin\left(\phi(t-\tau) - \phi(t)\right) \tag{92}$$

This equation represents a minimal effective description of a circulating field with finite propagation time.

[ORTHODOX]: Delay-differential equations of this type are well-established in feedback systems [1].

7.3 Emergence of Discrete Stable Modes

We seek steady-state solutions of the form:

$$\phi(t) = \Omega t \tag{93}$$

Substituting into Eq. 92 yields:

$$\Omega = \omega_0 + \kappa \sin(\Omega\tau) \tag{94}$$

This transcendental equation admits a discrete set of stable solutions Ω_n determined by the delay parameter τ.

[DERIVED]: Discrete mode selection arises from temporal nonlocality alone.

7.4 Delay-Induced Quantization Mechanism

Unlike conventional quantum mechanics, where discreteness is postulated, here it emerges dynamically:

- No boundary condition required
- No operator quantization assumed
- Discreteness arises from stability selection

[SPECULATIVE]: Particle spectra may be interpreted as delay-selected stable circulation modes.

7.5 Extension to Nonlinear Schrödinger Dynamics

To connect with field-theoretic descriptions, we extend the delay mechanism to a nonlinear Schrödinger framework:

$$i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + F_{\text{delay}}[\psi(t-\tau)] \tag{95}$$

where the delayed functional introduces temporal nonlocality into the field evolution.

[SPECULATIVE]: This provides a bridge between classical delay systems and quantum wave dynamics.

7.6 Physical Interpretation

The delay τ acts as:

- a memory kernel
- a nonlocal temporal coupling
- a selection mechanism for stable modes

Thus:

$$\textit{Delay} \Rightarrow \textit{Stability} \Rightarrow \textit{Discreteness}$$

This mechanism forms one of the central pillars of the SST canon.

8 Atomic Bridge Model

8.1 Introduction and epistemic framing

This section records the minimal SST atomic bridge rather than a full derivation of atomic structure. Its canon role is deliberately narrow: recover the orthodox Coulomb limit when the SST-specific sector is switched off, define a branch-resolved circulation coupling fixed by canonical medium scales, and make explicit which many-electron and shell-structure claims remain open. It is therefore a benchmark layer, not yet a complete atomic theory.

8.2 Baseline Hamiltonian

We adopt the standard Coulomb Hamiltonian as the orthodox baseline:

$$H_0 = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \tag{96}$$

[ORTHODOX]: This reproduces the hydrogenic spectrum.

8.3 Legacy Soft-Core Regularization

Early canon layers introduced a finite-core regularization of the hydrogenic potential. In the present document this is retained only as a controlled bridge ansatz, not as a replacement for Eq. (96).

Define the effective coupling scale

$$\Lambda_{\mathrm{SST}} = 4\pi\, \rho_{\mathrm{core}}\, \|\mathbf{v}_{\circlearrowleft}\|^2\, {r_c}^4 \tag{97}$$

with dimensions

$$[\Lambda_{\mathrm{SST}}] = \mathrm{J\,m} = \mathrm{N\,m^2}. \tag{98}$$

A finite-core bridge potential is then

$$V_{\mathrm{soft}}(r) = -\frac{\Lambda_{\mathrm{SST}}}{\sqrt{r^2 + r_c^2}} \tag{99}$$

satisfying the large-radius limit

$$V_{\mathrm{soft}}(r) \xrightarrow{r \gg r_c} -\frac{\Lambda_{\mathrm{SST}}}{r}. \tag{100}$$

[ORTHODOX] Effective-model input: finite-core regularizations of singular central potentials are standard tools.
[DERIVED] Canonical consequence: the SST scale in Eq. (97) is fixed by canonical constants rather than introduced phenomenologically.
[ORTHODOX] Constraint: this sector is admissible only if the large-radius limit reproduces the standard Coulomb problem to spectroscopic accuracy.

In particular, Eq. (99) is retained as a bridge ansatz only. It is not promoted here to the primary atomic potential.

8.4 SST Circulation Correction

We introduce a circulation-dependent perturbation:

$$\delta H^{(\sigma)} = -2\sigma E_R \eta_0 \gamma + \frac{E_R \eta_0 \sigma}{2} \left\{ \frac{\hat{L}_z}{\hbar}, \gamma \right\} \tag{101}$$

[DERIVED]: Interaction depends on circulation amplitude $\gamma(r, t)$.

8.5 Circulation source chain

The atomic perturbation is not introduced as an arbitrary extra scalar, but is intended to descend from a circulation-bearing source chain,

$$\xi \to \omega_z \to u_\theta \to \Gamma_{\mathrm{loc}} \to \gamma, \tag{102}$$

where an axial displacement field ξ generates local vorticity, the vorticity sources azimuthal swirl, and the loop circulation is normalized by the canonical quantum Γ_0. This makes the circulation amplitude γ the natural canon-level coupling variable rather than a free phenomenological insertion.

8.6 Branch Structure

Two chiral sectors exist:

$$\sigma = \pm 1 \tag{103}$$

These correspond to:

- ↺ (left-handed)
- ↻ (right-handed)

8.7 Consistency Requirement

To remain compatible with atomic physics:

$$|\delta E_n| \ll |E_n| \tag{104}$$

[ORTHODOX] Constraint: SST corrections must not violate known spectroscopy.

8.8 Interpretation

[STATUS]: The atomic bridge is a controlled benchmark sector. The orthodox Coulomb problem is imported as the baseline; the circulation-dependent term is the SST addition; shell counting, exclusion-like behavior, and heavy-atom extensions remain programmatic rather than derived here. The purpose of the section is therefore structural compatibility, not yet a complete replacement of atomic quantum mechanics.

9 Spectroscopic Constraints

9.1 Motivation

Any extension of atomic structure must remain consistent with high-precision spectroscopic measurements.

[**ORTHODOX**]: Atomic energy levels are experimentally verified to extremely high precision.

Requirement:

$$|\delta E_n| \ll |E_n| \tag{105}$$

where δE_n represents any SST-induced correction.

9.2 Internal Mode Hypothesis

We consider the possibility that atomic bound states possess an additional internal excitation sector associated with vortex filament dynamics.

[**SPECULATIVE**]: Each bound state carries an internal mode spectrum $\{E_{m,n}\}$.

The effective energy becomes:

$$E_n^{\text{eff}} = E_n^{(0)} + \delta E_n \tag{106}$$

9.3 Generic Coupling Structure

We parameterize the coupling between internal modes and observable energy levels:

$$|\delta E_n| \leq \lambda_n \sum_m E_{m,n}\, \nu_{m,n} \tag{107}$$

where:

- λ_n is the coupling efficiency
- $\nu_{m,n}$ is the occupation factor

[**DERIVED**]: This form captures a general upper bound independent of microscopic details.

9.4 Ungapped Spectrum: Instability

If the internal spectrum is ungapped:

- arbitrarily low-energy excitations exist
- no universal suppression mechanism applies

Consequence:

$$\delta E_n \not\to 0 \quad \text{generically} \tag{108}$$

[**CRITICAL**]: An ungapped internal sector is generically incompatible with spectroscopy unless:

- $\lambda_n \ll 1$ (extremely weak coupling), or
- the density of states is highly suppressed

9.5 Gapped Spectrum: Decoupling Mechanism

We introduce an excitation gap Δ_K:

$$E_{m,n} \geq \Delta_K \tag{109}$$

Then occupation factors follow Boltzmann suppression:

$$\nu_{m,n} \sim \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \tag{110}$$

Substituting into Eq. 107:

$$\delta E_n \propto \exp\left(-\frac{\Delta_K}{k_B T_{\text{eff}}}\right) \tag{111}$$

[DERIVED]: Exponential suppression ensures compatibility with spectroscopy.

9.6 Physical Origin of the Gap

[SPECULATIVE]: The gap Δ_K may arise from:

- geometric constraints on filament curvature
- torsional rigidity
- topological restrictions
- core structure of the vortex

9.7 Order-of-Magnitude Estimate

Using filament parameters:

$$\Delta_K \sim \frac{\Gamma^2}{4\pi\xi L} \tag{112}$$

where:

- ξ is the core scale
- L is the loop length

For typical SST scales:

$$\Delta_K = \mathcal{O}(10^2 - 10^3\,\text{eV}) \tag{113}$$

[DERIVED]: This places the internal sector well above atomic energy scales ($\sim$ eV).

9.8 Consistency Condition

To preserve atomic physics:

$$\frac{\Delta_K}{k_B T_{\text{eff}}} \gg 1 \tag{114}$$

Interpretation: The internal sector is effectively frozen at atomic energies.

9.9 Observable Consequences

[**SPECULATIVE**]: Residual effects may include:

- tiny level shifts
- suppressed line broadening
- weak temperature dependence

9.10 Falsifiability

The theory is falsifiable via:

- detection of unexplained spectral shifts
- absence of expected suppression
- deviation from Boltzmann scaling

9.11 Integration with SST Framework

The spectroscopic constraint enforces:

- compatibility with the Atomic Bridge (Section 8)
- constraints on delay-induced excitations (Section 7)
- bounds on allowable vortex configurations

9.12 Canonical Statement

Any viable SST extension must include a mechanism—such as an excitation gap—that ensures exponential suppression of internal modes at atomic energy scales.

9.13 Precision electroweak bridge: lessons from the CMS W-mass measurement

Status. [**ORTHODOX**] External input: the CMS measurement is a precision constraint on any SST effective weak-sector realization. [**DERIVED**] Methodological consequence: the result supports a forward-modelling strategy for SST phenomenology. [**CRITICAL NOTE**] The paper does *not* provide direct evidence for a structured-substrate ontology of the W boson.

Empirical anchor. Let the experimentally inferred resonance mass be denoted by

$$m_W^{\mathrm{exp}} = 80.3602 \pm 0.0099 \text{ GeV}. \tag{115}$$

In the CMS analysis this quantity is not read off from a single observable, but extracted from a profiled likelihood over finely binned charged-lepton kinematics. This motivates the following SST principle:

> *Collider observables constrain SST through forward maps from latent swirl-string states to detector-level distributions, not through direct identification of a single internal parameter with a measured mass.*

Latent-to-observable map. Introduce a latent SST parameter set for an effective spin-1 weak excitation,

$$\Theta_W^{\rm SST} = \Big(K_W,\, \chi_W,\, \mathcal{L}_W,\, \mathcal{H}_W,\, S_{(t)W}^{\circlearrowleft},\, {\rho_f}^{\rm eff},\, \mathbf{v}_{\circlearrowleft}^{\rm eff}\Big), \tag{116}$$

where:

- K_W is the topology class or excitation family,
- χ_W is chirality / handedness label,
- $\mathcal{L}_W$ is an effective geometric length scale,
- $\mathcal{H}_W$ is a helicity/linkage content,
- $S_{(t)W}^{\circlearrowleft}$ is the local clock factor for the mode,
- ${\rho_f}^{\rm eff}$ and $\mathbf{v}_{\circlearrowleft}^{\rm eff}$ are the coarse-grained medium parameters seen by the excitation.

The experimentally accessible distribution is then not a bare function of m_W alone, but a projected density

$$\mathcal{P}_{\rm meas}(x) = \int dz\; R(x|z)\, \mathcal{P}_{\rm prod}(z \mid \Theta_W^{\rm SST}), \tag{117}$$

where z denotes truth-level production/decay variables and $R(x|z)$ is the detector response kernel.

Mass as an effective resonance parameter. In SST one should distinguish:

$$m_W^{\rm internal} \neq m_W^{\rm eff} \tag{118}$$

in general, where:

- $m_W^{\rm internal}$ is the internal excitation-energy scale of the structured mode,
- $m_W^{\rm eff}$ is the resonance parameter inferred from scattering distributions.

The experimentally constrained quantity is $m_W^{\rm eff}$.

A minimal canon-safe matching condition is therefore

$$m_W^{\rm eff}(\Theta_W^{\rm SST}) = m_W^{\rm exp} + \delta_W, \qquad |\delta_W| \lesssim 10\ \text{MeV}, \tag{119}$$

with the current precision scale set by experiment.

Distribution-level fit target. The correct SST comparison object is a likelihood built from binned observables:

$$\mathcal{L}(\Theta_W^{\rm SST}, \nu) = \prod_{b=1}^{N_{\rm bins}} \text{Poisson}\Big(n_b \,\Big|\, \mu_b(\Theta_W^{\rm SST}, \nu)\Big)\; \Pi(\nu), \tag{120}$$

where:

- n_b is the observed count in bin b,
- μ_b is the SST prediction after production, decay, and detector folding,
- ν denotes nuisance parameters,
- $\Pi(\nu)$ is the nuisance prior / constraint factor.

The profile likelihood for the physically interesting SST parameters is then

$$\mathcal{L}_{\rm prof}(\Theta_W^{\rm SST}) = \max_{\nu}\; \mathcal{L}(\Theta_W^{\rm SST}, \nu). \tag{121}$$

Helicity decomposition as a falsifier. A particularly sharp SST test is not the central value of m_W alone, but whether the effective spin-1 mode reproduces the standard angular basis. Write

$$\frac{d\sigma}{d\Omega} = \sum_{i=0}^{8} A_i\, f_i(\theta, \phi) + \epsilon_{\mathrm{SST}}\, \Delta(\theta, \phi), \tag{122}$$

where the $A_i f_i$ are the standard helicity structures and $\epsilon_{\mathrm{SST}}\Delta$ is the first SST correction. Then:

- if $\epsilon_{\mathrm{SST}} = 0$ within experimental precision, SST survives this channel,
- if $\epsilon_{\mathrm{SST}} \neq 0$, the deviation is directly falsifiable through angular distributions.

Canon interpretation. This leads to the following conservative reading:

1. The weak sector must be *SM-like at the effective distribution level.*
2. Large $\mathcal{O}(10^{-4})$–$\mathcal{O}(10^{-3})$ relative distortions in the effective W mass are disfavoured.
3. The best SST discovery channels are therefore likely to be

   $$\text{small angular deviations}, \quad \text{polarization asymmetries}, \quad \text{off-shell tails}, \quad \text{rare differential distortions}, \tag{123}$$

   rather than a large shift in the central value of m_W.

Minimal bridge claim. The CMS result does *not* force the W boson to be fundamental in the ontological sense. It does force any SST realization of the weak sector to satisfy the stronger requirement:

$$\mathcal{P}^{\mathrm{SST}}_{\mathrm{prod}}(z \mid \Theta^{\mathrm{SST}}_W) \approx \mathcal{P}^{\mathrm{SM}}_{\mathrm{prod}}(z \mid m_W, \text{SM nuisances}) \tag{124}$$

to current experimental precision after detector folding and nuisance profiling.

Practical SST program. A precision-compatible SST weak-sector program should therefore proceed in three layers:

$$\text{(i) effective resonance matching:} \quad m_W^{\mathrm{eff}} \to m_W^{\mathrm{exp}}, \tag{125}$$

$$\text{(ii) helicity matching:} \quad A_i^{\mathrm{SST}} \to A_i^{\mathrm{SM}}, \tag{126}$$

$$\text{(iii) residual search:} \quad \Delta \neq 0 \text{ only in subleading differential structure.} \tag{127}$$

Conclusion. The CMS W-mass result is best understood in SST as a *precision filter*: it does not supply ontology, but it strongly constrains phenomenology. Any viable SST weak-sector realization must be distributionally SM-like to present precision, while leaving room only for small, structured residuals in higher-order observables.

10 Relational Time Framework

10.1 Swirl-Clock and local proper time

The canonical time variable of SST is not introduced as an independent primitive. It is defined relationally from the local kinematic state of the medium through the Swirl-Clock already introduced in Eq. (15). In its coarse-grained form,

$$dt_{\mathrm{loc}} = S_{(t)}(x)\, dt_{\infty}, \qquad S_{(t)}(x) = \sqrt{1 - \frac{v(x)^2}{c^2}}. \tag{128}$$

Here t_∞ denotes the asymptotic laboratory time and t_{loc} the effective proper time associated with a localized process in the medium. In SST, this factor is interpreted as a fluid-kinematic clock field rather than as a purely geometric spacetime postulate.

10.2 Carrier-local clocking and the turnover scale

For localized carriers the natural microscopic rate is the turnover frequency

$$\Omega_0 = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = \omega_c, \tag{129}$$

which coincides with the Compton anchoring scale. A complete SST clock description therefore involves two linked but distinct levels:

- the *microscopic turnover scale* Ω_0, which counts circulation cycles of the carrier,
- the *coarse-grained Swirl-Clock* $S_{(t)}$, which relates local rates to asymptotic time.

This resolves a recurring ambiguity in earlier canon versions: local cycle counting and coarse-grained time dilation are related, but they are not identical observables.

10.3 Relational observables from conserved currents

A purely operator-based notion of time is avoided in SST by defining time observables through conserved currents. Let J^μ denote a matter current and let j^μ_{ev} denote a coarse-grained event current associated with structured activity of the medium. The relational clock field $T(x)$ is then recovered only after coarse graining,

$$\partial_\mu T(x) \approx \frac{1}{\nu_0} \langle j^{\text{ev}}_\mu \rangle_\ell, \tag{130}$$

where ℓ is the coarse-graining scale and ν_0 a reference event frequency. The canon interpretation is that time is read from correlations between currents, not from an abstract external operator.

10.4 Time-of-arrival as a relational field observable

Let $\mathcal{W}$ be a detector world-tube with boundary $\Sigma = \partial\mathcal{W}$. Then the arrival-time distribution associated with a detector label Θ is written as

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_\Sigma d\sigma_\mu \, J^\mu(x) \, \delta(T(x) - \Theta). \tag{131}$$

This formulation is canonically important because it bypasses the need for a self-adjoint time operator conjugate to a bounded Hamiltonian. The measured time is instead the value of a physical clock field at the point where matter flux crosses the detector boundary.

10.5 Clock broadening and continuum limits

If the event current has a finite variance, then the observed time-of-arrival distribution is broadened relative to the ideal classical prediction. Writing P_{cl} for the classical arrival distribution and σ_τ^2 for the effective clock variance, one expects a convolution structure of the form

$$P_{\text{obs}}(\Theta) = \int dt \, P_{\text{cl}}(t) \frac{1}{\sqrt{2\pi\sigma_\tau^2}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma_\tau^2}\right]. \tag{132}$$

In the limit $\ell \gg r_c$, the event structure is smoothed and one recovers a continuum clock field. In the opposite limit $\ell \to r_c$, the underlying discreteness of carrier events becomes operationally relevant.

10.6 Connection to gravity and geometry

In the canon, the relational-time sector and the geometric-response sector are not independent. Variations in the Swirl-Clock track variations in local swirl energy density and therefore provide the time component of the effective geometric response of the medium. At this level, the SST reading is:

gravity and time are two coarse-grained aspects of the same clock-bearing medium.

This statement should be read as a canon-level organizing principle, not yet as a completed field-theoretic proof.

10.7 Clock Field and Foliation Variable

[DERIVED] EFT bridge: the scalar Swirl-Clock field is promoted from the kinematic relation of Eq. (15) to a long-wavelength foliation variable

$$\chi(x), \tag{133}$$

whose normalized gradient defines a preferred local clock frame:

$$u_\mu = \frac{\nabla_\mu \chi}{\sqrt{g^{\alpha\beta} \nabla_\alpha \chi \, \nabla_\beta \chi}}. \tag{134}$$

[ORTHODOX] Equation (134) matches the standard hypersurface-orthogonal foliation construction used in khronometric / Einstein–Æther effective field theory [7, 8].

[DERIVED] Within SST, the interpretation is different: u_μ is not taken as a fundamental spacetime structure, but as the coarse-grained clock-flow direction associated with the medium.

10.8 Clock-Sector Effective Action

[ORTHODOX] EFT template: the most general second-order covariant clock-sector action compatible with hypersurface orthogonality is

$$\begin{aligned} S_\chi = \int d^4x \sqrt{-g} \Big[& c_1 (\nabla_\mu u_\nu)(\nabla^\mu u^\nu) + c_2 (\nabla_\mu u^\mu)^2 \\ & + c_3 (\nabla_\mu u_\nu)(\nabla^\nu u^\mu) + c_4 u^\mu u^\nu (\nabla_\mu u_\alpha)(\nabla_\nu u^\alpha) \Big]. \end{aligned} \tag{135}$$

[SPECULATIVE] SST interpretation: Equation (135) is not itself derived from first-principles filament dynamics in the present Canon. It is recorded as the minimal orthodox EFT bridge into a relativistic clock-field description.

10.9 Poisson Limit and Effective Gravity Bridge

[DERIVED] Bridge statement: in the weak-field, slowly varying limit, the scalar clock field is taken to obey a Poisson-type closure

$$\nabla^2 \chi(x) = 4\pi G_{\text{eff}} \, \rho_{\text{matter}}(x), \tag{136}$$

so that outside localized sources

$$\chi(r) \propto \frac{1}{r}, \qquad \mathbf{g}_{\text{eff}} = -\nabla \chi. \tag{137}$$

[ORTHODOX] The $1/r$ scalar potential and $1/r^2$ force law are standard consequences of Eq. (136).

[SPECULATIVE] The identification of χ with the Swirl-Clock and the interpretation of Eq. (136) as an emergent gravity law are SST-specific.

10.10 Observational Constraint from Tensor-Mode Speed

[ORTHODOX] Constraint: If the clock-sector EFT is used, multimessenger gravitational-wave observations require [9]

$$c_{13} \equiv c_1 + c_3 = 0, \tag{138}$$

to enforce luminal tensor-mode propagation at observational precision.

[ORTHODOX] Constraint: any SST realization employing the EFT bridge of Eq. (135) must preserve Eq. (138).

10.11 Interpretive Role in the Canon

This section has a deliberately limited role:

- it upgrades the Swirl-Clock from a purely local kinematic factor to a candidate field variable;
- it records a minimal orthodox EFT bridge for relativistic comparison;
- it supplies the internal reference point needed by Section 11 for the time–gravity link.

Status summary.

- Eq. (134): **[ORTHODOX]** mathematical form; **[DERIVED]** role as an SST bridge variable
- Eq. (135): **[ORTHODOX]** EFT template
- Eq. (136): **[DERIVED]** bridge closure
- gravity interpretation: **[SPECULATIVE]**

10.12 Relation to the Chronos–Kelvin law

Section 2.9 provides the local transport identity of the Swirl-Clock on a material loop. The present section supplies the long-wavelength field-theoretic and analogue-metric reading of the same clock sector. The two layers are therefore complementary:

- Eq. (24) is the loop-level transport law;
- Eq. (134)–Eq. (136) provide the coarse-grained field description.

11 Unified Interpretation

11.1 Overview

Swirl-String Theory (SST) proposes a unified interpretation in which fundamental physical phenomena emerge from the dynamics of a continuous medium supporting stable vortex structures.

This section synthesizes the results of the previous sections into a coherent conceptual framework.

11.2 Ontology of the Theory

[SPECULATIVE] Ontological claim: The fundamental ontology consists of:

- A continuous incompressible medium (Section 3)
- Stable vortex filament configurations (Section 5)
- Circulation as a conserved topological invariant

Interpretation: Particles are not elementary objects, but persistent dynamical structures.

11.3 Mass as Trapped Circulation Energy

From Section 2, and in particular from Eq. (34), the mass sector is organized as the product of a medium scale, a geometric gate, a topological kernel, and a local clock-impedance factor. The canon-level reading is therefore not merely that "energy is stored in rotation," but that persistent inertial scale is assigned only after geometry, topology, and local clocking have all been specified.

$$M_K(x) = \rho_m(x)\, {r_c}^3\, \Pi_K\, \Xi_K\, S^{\circlearrowleft}_{(t)}(x)^{-2}. \tag{139}$$

[SPECULATIVE] Unified interpretation: mass corresponds to localized, self-sustained circulation energy in the medium, modulated by state geometry, topology, and local clock impedance.

11.4 Time as a Relational Quantity

From Section 10 and Section 2, time is not introduced as an independent primitive but as a relational field read from the local state of the medium. The same kinematic quantity that enters the Swirl-Clock also appears in the organizing equation of the mass sector, so the canon treats proper-time modulation and inertial scaling as two projections of one shared substrate state.

$$S^{\circlearrowleft}_{(t)}(x) = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}(x)\|^2}{c^2}}. \tag{140}$$

[SPECULATIVE] Unified interpretation: time is not fundamental, but emerges from the kinematic state of the medium and is read operationally through clock-bearing processes.

11.5 Discreteness without Quantization Postulates

From Section 7:

- Delay introduces temporal nonlocality
- Stability selects discrete modes

[DERIVED] Unified consequence:

> Discrete physical spectra arise dynamically from delay-induced mode selection.

This replaces operator-based quantization with a dynamical mechanism.

11.6 Charge and Circulation

[SPECULATIVE] Provisional bridge only. The canon retains circulation as a structurally important invariant, but it does *not* yet claim a completed derivation of observed gauge charges from circulation data alone. At most, the present document records a working hypothesis that orientation and magnitude of circulation may enter a later charge-assignment program.

11.7 Unification of Interactions

The canon records a *unification program*, not yet a completed unification theorem. In the present document:

- electromagnetic effects are represented by the minimal transverse-field bridge of Section 6,
- gravitational effects are represented by the clock-field and Poisson-limit bridge of Section 10,
- inertial mass is organized by the medium/topology/geometry/clock structure of Eq. (34).

[SPECULATIVE] Programmatic claim: these sectors may ultimately be read as different effective descriptions of one substrate, but their full common derivation remains open.

11.8 Atomic Structure as an Emergent Layer

From Section 8:

- Standard quantum mechanics appears as an effective description
- SST introduces circulation-dependent corrections

[SPECULATIVE] Unified interpretation:

Quantum mechanics is an emergent effective theory of underlying vortex dynamics.

11.9 Consistency with Spectroscopy

From Section 9:

- Internal modes are suppressed by an excitation gap
- Low-energy physics remains unchanged

[ORTHODOX] Constraint: The unified picture does not violate known experimental constraints.

11.10 Hierarchy of Description

The theory admits multiple levels:

1. **Microscopic level:** vortex filament dynamics
2. **Mesoscopic level:** delay-induced mode structure
3. **Macroscopic level:** effective quantum description

11.11 Conceptual Summary

The SST framework can be summarized as:

$$\begin{array}{ll} \textbf{Mass} & \to \text{trapped circulation energy} \\ \textbf{Time} & \to \text{relational swirl-clock field} \\ \textbf{Charge} & \to \text{provisional circulation bridge} \\ \textbf{Quantization} & \to \text{delay-induced stability} \end{array}$$

11.12 Canonical Statement

All observed physical structure emerges from the dynamics of a continuous medium in which stable vortex configurations, governed by quantized circulation and delay dynamics, produce the phenomena conventionally attributed to particles, fields, and quantum mechanics.

11.13 Scope and Limits

[**IMPORTANT**]:

- The framework is not yet a complete fundamental theory
- Several mechanisms remain to be derived rigorously
- Many results are conditional or phenomenological

11.14 Final Consistency Principle

A valid unified interpretation must preserve all experimentally verified results while providing a deeper structural explanation of their origin.

12 Integration of Prior Canon Versions

12.1 Integration policy (v0.5.12 reintegration)

This block records only material from earlier SST canon layers that can be retained inside the v0.8.1 architecture without weakening the explicit classification discipline of Section 4.

Older canon documents are therefore *not* cited as authorities. Transferable content is rewritten here as current-canon statements, each labeled according to its present status.

12.2 Retained v0.5.12 kernel sectors

The present draft now retains the following v0.5.12 kernel sectors in rewritten form:

- the Chronos–Kelvin invariant and clock–radius transport law (Section 2.9);
- the reparametrized zero-parameter corollary and coarse-graining rule (Section 2.10);
- a minimal Swirl–EM bridge and photon / unknot sector (Section 6);
- the swirl pressure law as the Euler-level force bridge (Section 6.3);
- the analogue-metric / clock-field bridge (Section 10).

12.3 Wave–particle phase rule and measurement bridge

A retained qualitative rule from earlier canon layers is that SST distinguishes two limiting phases:

$$\text{R-phase}: \quad \text{extended, unknotted, radiative,} \tag{141}$$
$$\text{T-phase}: \quad \text{localized, knotted, tangible.} \tag{142}$$

A minimal dynamical bookkeeping law for phase conversion is

$$\partial_t P_T = \Gamma_{R\to T}[\mathcal{K}]\, P_R - \Gamma_{T\to R}[\mathcal{K}]\, P_T, \qquad P_R + P_T = 1, \tag{143}$$

where $\mathcal{K}$ denotes an environment- and coherence-dependent kernel functional. The intended interpretation is that "measurement" corresponds not to an axiomatically imposed collapse, but to a medium-induced transfer of support from the extended R-phase to the localized T-phase.

[**ORTHODOX LIMIT**]: in weak coupling and after coarse graining, Eq. (143) should reduce to ordinary environment-induced decoherence models [27].

[**SPECULATIVE**]: the identification of decoherence with an actual R↔T phase transition is SST-specific.

12.4 Gauge and weak-mixing bridge

The retained gauge-level statement is that the effective low-energy algebra should reduce to

$$\mathfrak{g}_{\text{eff}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \tag{144}$$

with gauge potentials interpreted as holonomy data of coarse-grained multi-director transport.

A minimal bridge ansatz for the weak mixing angle is

$$\sin^2 \theta_W^{\text{eff}} = \frac{K_Y}{K_Y + K_W}, \tag{145}$$

where K_Y and K_W are effective Abelian and weak-sector stiffness scales. Eq. (145) is not a completed derivation, but it preserves the v0.5.12 idea that weak mixing should descend from a ratio of medium-response coefficients rather than appear as an unstructured free number.

[**ORTHODOX**]: θ_W is the standard electroweak mixing parameter of the SM [28, 3].

[**SPECULATIVE**]: Eq. (145) is an SST bridge ansatz only.

12.5 What the v0.5.12 reintegration does not import

The present reintegration does *not* import v0.5.12 cosmogony, engineering subsystems, cosmological-term closures, or highly application-specific protocols into the main canon body. Those remain suitable for appendices or standalone papers until they are more tightly tied to the minimal axiom system.

12.6 Purpose of this integration layer

Version 0.8.1 is not a fresh theory written from zero, but an explicit consolidation of earlier canon lines. This section records what has been retained, what has been renamed or reorganized, and what has been demoted from canon status to research-track status.

12.7 Retained from the early canon layers

From the v0.1.x–v0.3.x line, the present canon retains the primitive closure logic:

- a continuum substrate with effective density ρ_f,
- a core scale r_c fixed by Compton anchoring,
- circulation quantization through $\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\|$,
- the interpretation of matter as stable organized structures rather than point primitives.

These elements remain non-negotiable because later sectors depend on them directly.

12.8 Retained from the middle canon layers

From the v0.4.x–v0.6.x line, the canon retains three durable structures:

1. the Rosetta strategy for mapping SST statements into orthodox language,
2. the photon/torsion sector as the minimal radiation bridge,
3. delay-induced mode selection as the main route to discreteness without operator postulates.

These are now distributed across Sections 7, 10, and 11 rather than treated as isolated notes.

12.9 Retained from the late canon layers

From the v0.7.x line, v0.8.1 retains the strongest structurally reusable sectors:

- the Swirl-Clock as the canonical relational-time field,
- the atomic bridge as a benchmark consistency layer rather than a complete atomic derivation,
- the spectroscopic gap logic as a hard consistency filter,
- the topology-first mass program, with topology separated from thermodynamic or radiative dressing,
- the EFT-style foliation and gauge-emergence program as a long-range development track.

12.10 Notation and density cleanup

Earlier canon versions sometimes conflated background density, mass-equivalent density, and saturated core density. The present version adopts the cleaned hierarchy

$$\rho_f \quad \text{(effective fluid density)}, \qquad \rho_E \quad \text{(swirl energy density)}, \qquad \rho_m = \rho_E/c^2, \qquad \rho_{\text{core}} \quad \text{(saturated core densit} \tag{146}$$

Any older passage that identifies a single symbol with more than one of these roles is superseded by the present separation.

12.11 Topology-first reinterpretation of layers

A major consolidation from the recent lepton and knot-mass program is the separation between topology and layering. Knot type is retained as the primary identity label, while Golden-layer or thermal ladder indices are treated as dressing or coherence variables rather than species-defining labels. This prevents the canon from over-reading numerical layer fits as ontological identities.

12.12 Status of optional sectors

The following sectors are retained as important but non-primitive research layers:

- full gauge emergence and charge assignments,
- dual-vacuum phenomenology and Unruh-echo interpretation,
- complete many-electron atomic closure,
- full weak-sector and collider phenomenology,
- explicit topology-to-particle dictionary beyond the best-supported benchmark assignments.

They remain in the canon as organized programs, not as closed theorem-level results.

12.13 Integration principle

The purpose of v0.8.1 is therefore not to maximize novelty per page, but to minimize duplication and to expose the dependency graph of the theory. Whenever two earlier canon fragments make the same structural claim in different language, the present version keeps one canonical form and demotes the others to historical context.

12.14 Research-track companion

The sectors that are useful for continuity but too long or too speculative for a minimal main-canon reading are also collected in the companion file `canon-0.8.1-research-track.tex`. That excerpt includes the thermodynamic-radiation interface, the minimal-coupling QED analogy, the ribbon-invariant chirality refinement, the particle-candidate mapping layer, the hydrodynamic exchange / Pauli-barrier estimate, the numerical-benchmark policy block, the gauge-sector roadmap, and the *dark-sector and galactic canonization execution package* (frozen topology toolkit, benchmark protocol template, Euler-pressure anchor details, galactic branch gate, and probe–equation map). The integration subsections below retain the full in-context treatments with labels and cross-references; the companion file provides a portable standalone copy. Editorial separation does not reject these sectors; it prevents the core narrative from overclaiming closure.

12.15 Dark-sector and galactic canonization status (v0.8.x)

This block records promotion labels for the dark-sector / galactic program consistent with the internal canonization plan. Detailed freezes, the mandatory benchmark-protocol checklist, the galactic profile branch gate (Branch A: legacy-profile refine with benchmark traceability), and the probe falsification table live in `canon-0.8.1-research-track.tex` (Section titled `Dark-sector and galactic canonization execution package`).

[DERIVED] Stage 5 review: cross-item dependencies are respected (pressure anchor and benchmark layer precede full galactic closure; taxonomy strengthens other sectors). No sector is promoted on fit quality alone; each carries explicit falsifiability or reproducibility hooks as in Table 2 and the companion execution package.

12.16 Integration Policy

This section records only material from prior internal SST development that can be retained inside the v0.8.1 architecture *without* weakening the epistemic classification framework of Section 4.

Rule. Older canon documents are *not* cited as external authorities. Instead, transferable content is rewritten here as current-canon statements, each labeled as [**ORTHODOX**], [**DERIVED**], or [**SPECULATIVE**].

Table 2: Master tracker: dark-sector and galactic program (v0.8.x).

Item	Prior state	Target	Status after Stage 0–4 freeze
Dark-sector Euler pressure anchor	BRIDGE	CANON	CANON-CANDIDATE (derivation, assumptions, limits, falsifiers frozen; see Section 6.3)
Dark taxonomy	HOLD	CANON	BRIDGE (symmetry-first classifier + filters; Section 5.3)
Galactic swirl law	BRIDGE	CANON	BRIDGE (Branch A closure; companion execution package)
Experimental probes	HOLD	CANON-BRIDGE	CANON-BRIDGE (equation / null-result map; companion)
Benchmark / reproducibility	BRIDGE	CANON	CANON-CANDIDATE (protocol template frozen; repository-wide reuse pending)
Topological reference identities	CANON-CANDIDATE	CANON	CANON-ready scope/notation (Appendix A)

12.17 Selective Recovery from Pre-v0.7 Canon Layers

The present draft absorbs only those legacy ingredients that can be rewritten in the current epistemic format without appealing to earlier canon versions as authorities.

Analogue line-element bridge. Legacy SST drafts recorded a schematic analogue line element and a metric–clock identification of the form

$$ds^2 = -c^2\, S^{\circlearrowleft}_{(t)}(x)^2\, dt^2 + \delta_{ij}\, dx^i\, dx^j, \tag{147}$$

$$g_{00}(x) = -S^{\circlearrowleft}_{(t)}(x)^2. \tag{148}$$

Recovered legacy sectors include:

- the finite-core atomic bridge, Eqs. (97)–(99);
- the analogue line-element bridge, Eqs. (147)–(148);
- compact research-track interfaces to thermodynamic radiation and minimal-coupling QED analogies;
- topological chirality refinements based on standard ribbon invariants.

Legacy material not imported includes version history, cosmogonic narrative, and phenomenological galaxy-scale laws that are not yet sufficiently integrated with the present minimal axiom system.

12.18 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector.

Let

$$T_{\text{eff}} = T_{\text{eff}}\Big(\rho_E,\, \omega,\, S_{(t)}\Big) \tag{149}$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by T_{eff}.

[ORTHODOX]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[SPECULATIVE]: SST does not yet derive Eq. (149), so this sector remains a research program rather than canonically closed physics.

12.19 Minimal Coupling Interface to Quantum Electrodynamics

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \to \nabla - i\,\frac{m}{\hbar}\,\mathbf{A}_{\rm eff}, \qquad \mathbf{A}_{\rm eff} = \chi\,\mathbf{v}, \tag{150}$$

where χ is a scalar weight and $\mathbf{v}$ is the local carrier velocity field.

[ORTHODOX]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.

[SPECULATIVE]: Eq. (150) is only an analogy template unless a full gauge-covariant derivation is supplied.

12.20 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Călugăreanu–White–Fuller relation

$$Lk = Tw + Wr \tag{151}$$

as a refinement of the knot-based matter/antimatter dictionary.

[ORTHODOX]: Eq. (151) is a standard geometric identity for framed curves and ribbons.

[DERIVED]: it can be used internally to separate writhe-dominated from twist-dominated realizations of the same coarse knot class.

[SPECULATIVE]: the mapping from $\operatorname{sign}(Tw)$ or related chirality measures to particle/antiparticle sectors remains a model hypothesis.

12.21 Framed-Tube Ontology and the Attachment Gate

The particle carrier in SST is not a bare centerline knot alone. The canonical object is a framed, phase-bearing tube,

$$\mathcal{K}_{\rm SST} = (K,\, \mathcal{F},\, \Gamma,\, \theta_{\rm core}), \tag{152}$$

where K is the centerline knot or link, $\mathcal{F}$ is a ribbon/framing, Γ is circulation, and $\theta_{\rm core}$ is a material or order-parameter phase attached to the core.

The current Attachment Lemma is the statement that a local Compton/core cycle can be promoted to a globally closed framed-tube holonomy,

$$\frac{1}{2\pi}\oint_K d\theta_{\rm core} = \chi, \qquad \chi \in \mathbb{Z}. \tag{153}$$

A compensated form allows local swirl-clock redistribution while preserving the exact global integer,

$$\chi_{\rm total} = \chi_{\rm local} + \chi_{\rm attachment} = \chi, \qquad \delta Q = 0. \tag{154}$$

[CANON / OPEN GATE]: Eq. (153) is not yet a theorem of the Euler–SST dynamics. It is a load-bearing gate. Integer energy selection alone is insufficient; exact holonomy is required for canon status.

[CANON / NOTATION]: the framing phase follows the Compton angular cycle ω_c, while the associated local vorticity scale is $2\omega_c$. The factor of two must not be absorbed into χ.

12.22 Particle-Candidate Mapping Layer

[SPECULATIVE] Working dictionary: a minimal knot/composition mapping retained from earlier SST work is summarized in Table 3. This table is organizational rather than definitive; it records current topological assignments and mechanical analogues used by the mass and taxonomy program.

Table 3: Minimal knot-class and analogue summary retained in v0.8.3, with current semantic corrections.

Carrier / composition	Candidate role	Status / interpretation
3_1 trefoil	electron / positron sector	baseline chiral lepton calibra
$T(2,5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2,7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
twist/clasp knots, quasi-unknot loops	quark-like defects	research-track fractional bou
$5_2 + 5_2 + 6_1$ and nearby composites	baryon candidates	legacy mass-functional dicti
triple-gear locked analogue	proton-like charged baryon analogue	mechanical model for nonze
Borromean-type analogue	neutron-like neutral baryon analogue	mechanical/topological mod

[CRITICAL NOTE]: Table 3 is a working dictionary only. Electric charge is not identified with raw Hopf helicity or pairwise linking number. The safer current placeholder is a boundary-winding charge variable q_∂, while knot/link data supply mass, chirality, confinement, and internal topological structure.

[CANON / DECISION]: The trefoil assignment is retained for the electron/positron sector only. Fractional values such as 2/3 belong, if anywhere, to the twist-knot/quark-like boundary-winding research track rather than to the trefoil-electron calibration.

12.23 Hydrodynamic Exchange and the Pauli Barrier

[ORTHODOX] Fluid-mechanical input: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is [6]

$$E_{\text{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\|\mathbf{r}_1 - \mathbf{r}_2\|}. \tag{155}$$

[DERIVED] SST bridge: if two identical electron-candidate loops are forced into the same spatial state, regularization at the physical tube radius a_{core} yields the overlap penalty

$$V_{\text{Pauli}}(\mathbf{d}) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \oint\oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\sqrt{\|\mathbf{r}_1(\theta_1) - \mathbf{r}_2(\theta_2) + \mathbf{d}\|^2 + a_{\text{core}}^2}}, \tag{156}$$

with maximal-overlap scale

$$V_{\text{Pauli}}^{\max} \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{a_{\text{core}}} \right) \mathcal{O}(1). \tag{157}$$

Using the canonical values

$$\begin{aligned} \rho_f &= 7.0 \times 10^{-7}\ \mathrm{kg\,m^{-3}}, \\ r_c &= 1.40897017 \times 10^{-15}\ \mathrm{m}, \\ \Gamma_0 &= 2\pi r_c \|\mathbf{v}_\circlearrowleft\| \approx 9.6836 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}, \end{aligned} \tag{158}$$

and $L \sim 2\pi a_0$ at the hydrogenic scale, the legacy benchmark obtained in the closure limit $a_{\text{core}} = r_c$ is

$$V_{\text{Pauli}}^{\max}(a_{\text{core}} = r_c) \sim 1.23 \times 10^{-18}\ \mathrm{J} \sim 7.6\ \mathrm{eV}. \tag{159}$$

Away from this closure limit the scaling is

$$V^{\max}_{\text{Pauli}}(a_{\text{core}}) \sim V^{\max}_{\text{Pauli}}(r_c)\left(\frac{r_c}{a_{\text{core}}}\right). \tag{160}$$

[CRITICAL NOTE] This is retained as a canonical scale benchmark, not yet as a closed theorem. Its role is to show that the exclusion-energy estimate falls in an atomically relevant window rather than at obviously pathological scales.

As written, Eq. (159) should be interpreted as an order-of-magnitude benchmark attached to the canonical constants and to the closure-limit choice $a_{\text{core}} = r_c$, not yet as a unique parameter-free prediction.

12.24 Numerical Benchmark Layer

[DERIVED] Computational-program requirement: earlier SST work contained explicit benchmark tables for particle and atomic masses. In v0.8.1, the benchmark layer is retained as a reproducibility requirement rather than as a standalone authority claim.

Table 4: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

Particle	m_{exp} (MeV)	m_{SST} (MeV)	rel. error	mode	class
e	0.51099895	0.51099895	0	predictive/closure	3_1
μ	105.6583745	…	…	predictive	candidate torus class
τ	1776.86	…	…	predictive	candidate torus class
p	938.272088	…	…	predictive or closure	composite class
n	939.565420	…	…	predictive or closure	composite class

[CRITICAL NOTE] A future v0.8.x benchmark appendix should include:

- the exact script or package path used for generation,
- archived CSV outputs,
- mode definitions (predictive vs. exact-closure),
- uncertainty propagation on canonical constants and geometric factors,
- theorem-style pass/fail metadata (including tail-quality checks, postchecks, and extrapolation summaries),
- explicit separation between practical best-estimate closure and theorem-level closure,
- normalization of legacy constant names into SST notation (ρ_f, r_c, Γ_0) while preserving numerical values.

This requirement is included to keep the benchmark layer falsifiable and reproducible, rather than merely illustrative.

12.25 Gauge-Sector Roadmap

[SPECULATIVE] Structural roadmap: a retained minimal statement of the gauge program is:

$$\mathfrak{g}_{\text{eff}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1), \tag{161}$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A_\mu^a(x) = \partial_\mu \theta^a(x), \tag{162}$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A_\mu^a \mapsto A_\mu^a + q^a \partial_\mu \chi. \tag{163}$$

[**CRITICAL NOTE**] The gauge layer is retained in v0.8.1 only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.

13 Discussion and Limits

13.1 Status of the recovered v0.5.12 sectors

The material reintroduced here does not have uniform epistemic status.

- The Chronos–Kelvin and pressure-law sectors are orthodox fluid-mechanical structures with SST reinterpretations.
- The analogue-metric and clock-field EFT layers are orthodox comparison tools, but only partially derived within SST.
- The R/T measurement bridge and weak-mixing stiffness ansatz remain explicitly speculative.

13.2 What v0.8.1 now recovers from v0.5.12

After the present patch, v0.8.1 no longer omits the main v0.5.12 kernel sectors that are structurally compatible with the current architecture:

- Chronos–Kelvin / clock transport,
- zero-parameter reparameterization discipline,
- Swirl–EM and photon / unknot bridge,
- swirl pressure law,
- analogue metric and clock-field gravity bridge,
- R/T measurement bridge,
- weak-mixing bridge ansatz.

13.3 Open items that remain outside the present closure

The following remain open and are *not* upgraded to full canonical closure in this draft:

- a first-principles derivation of the full SM gauge sector;
- a completed EWSB derivation tied to the canon constants;
- a rigorous derivation of the R/T measurement kernel from continuum dynamics;
- full CANON promotion of the dark-sector taxonomy and galactic closure beyond the current BRIDGE labels (Section 12.15).

13.4 What is established at canon level

The present document supports the following as canon-level structures:

- the primitive constant chain $(\rho_f, v_{\circlearrowleft}, \omega_c) \to r_c \to \Gamma_0$,
- the Swirl-Clock as the coarse-grained relational time factor,
- delay-induced mode selection as the preferred discreteness mechanism,
- the atomic bridge and spectroscopic sectors as compatibility filters,
- the topology-first program for mass organization.

These claims are not equally proven, but they are the sectors on which the rest of the canon is structurally built.

13.5 What remains closure-level or conjectural

Several central SST ambitions remain open:

1. a rigorous derivation of full particle spectra from topology alone,
2. a first-principles derivation of charge assignments and gauge couplings,
3. a complete many-body derivation of shell filling and exclusion,
4. a full demonstration that delay plus topology reproduces the observed quantum rule set without importing it.

The canon records these as active programs rather than as settled conclusions.

13.6 Compatibility limits

Any viable SST realization must remain compatible with:

- atomic spectroscopy,
- relativistic effective kinematics,
- precision electroweak data where an SST bridge is claimed,
- the absence of large low-energy deviations from standard quantum mechanics.

This is why the canon repeatedly imposes suppression, weak-coupling, and low-energy recovery conditions.

13.7 Minimal falsifiers

The present canon would be materially weakened by any of the following:

- failure of the constant chain to remain dimensionally and numerically self-consistent,
- absence of any workable suppression mechanism for internal modes at atomic energies,
- inability to formulate a non-circular route from delay dynamics to discrete stable spectra,
- systematic contradiction between any SST atomic bridge and precision spectroscopy.

Conversely, strong support would come from a successful theorem ladder showing how discrete spectra, topological stabilization, and low-energy effective quantum structure arise from one shared substrate model.

13.8 Methodological limit

The canon is deliberately hybrid in style. Some sectors are written as exact closures, some as effective models, and some as programmatic roadmaps. This is a feature rather than a flaw, provided the epistemic status of each claim remains explicit. The document should therefore not be judged as though every section were intended to be a stand-alone journal theorem.

13.9 Near-term priorities

The next canon-strengthening tasks are clear:

1. complete the theorem ladder for the mass and clock sectors,
2. expand numerical validation blocks using the canonical constants,
3. refine the atomic bridge into sharper benchmark observables,
4. decide which optional sectors belong in the core canon and which should migrate to separate technical notes or the research-track companion.

13.10 Final limit statement

SST is presently best understood as a structured unification program with several strong internal bridges and several open derivational gaps. The canon is therefore mature enough to organize the theory coherently, but not yet so mature that every phenomenological sector should be presented as theorem-complete.

13.11 Status of Legacy Recoveries

The additional material imported here from older canon layers does not have uniform status.

- The analogue-metric bridge and ribbon identity are mathematically orthodox tools.
- Their SST interpretation is partially derived but not fully closed dynamically.
- The thermal-radiation and QED-interface sectors remain explicitly research-track.

This section therefore serves as a controlled retention layer, not as a blanket validation of all legacy material.

Accordingly, these recoveries should be read as structured extensions of the minimal core, not as fully promoted final doctrine.

13.12 What v0.8.1 Now Retains

After integration, the present Canon retains the following prior-development sectors in current-canon form:

- a particle-candidate topological dictionary (Section 12.22);
- a hydrodynamic exchange-energy estimate for fermionic exclusion (Section 12.23);
- a benchmark / reproducibility layer (Section 12.24);
- a minimal gauge-sector roadmap (Section 12.25);
- a clock-field EFT bridge and Poisson gravity limit (Section 10).

13.13 What is Deliberately Not Claimed

The present document still does *not* claim:

- a completed first-principles derivation of the Standard Model;
- a unique finalized knot-to-particle dictionary;
- a closed proof of the exclusion barrier beyond the current scaling benchmark;
- a fully renormalized relativistic field theory for the gauge and clock sectors.

These omissions are deliberate. v0.8.1 is intended to remain logically strict even when phenomenological sectors are reintroduced.

13.14 Canon Rule Going Forward

Any further reintegrated material must satisfy all of the following:

1. it can be written without citing older internal canon versions as authorities;
2. it can be assigned an explicit epistemic label;
3. it includes either an internal derivation path or a reproducible benchmark route;
4. it preserves compatibility with the orthodox limits recorded earlier in the document.

Appendices

A Topological reference toolkit

The following orthodox items form the mandatory internal mathematical toolkit cited by topology-first SST sectors. Statements are standard; SST-specific mappings appear in the main text with explicit status labels.

1. Gauss linking integral and linking number for disjoint closed curves.
2. Helicity $\mathcal{H} = \int \mathbf{v} \cdot (\nabla \times \mathbf{v})\, dV$ in ideal (inviscid) fluid flow and its conservation in the appropriate limit.
3. Thin-tube reduction of helicity to flux–linkage form in the standard filament limit.
4. Hopf invariant for localized field configurations (degree–type integral) where used as a topological density.
5. Basic torus-knot invariants as needed in the mass program (e.g. minimal crossing number, braid index, genus) with fixed naming consistent with Rolfsen notation in the main text.
6. Călugăreanu–White–Fuller relation $Lk = Tw + Wr$ for framed ribbons (already recalled as Eq. (151) in the integration layer).

[ORTHODOX]: each item is classical mathematics [26, 22, 6]. **[DERIVED]**: scope and notation for v0.8.x are frozen to this list plus house symbols $(\rho_f, \Gamma, r_c, \mathcal{H})$ without parallel aliases.

B Full Derivations

B.1 Canonical constant chain

The present canon uses the closure chain

$$r_e = \frac{\alpha\hbar}{m_e c}, \tag{164}$$
$$r_c = \frac{r_e}{2}, \tag{165}$$
$$\omega_c = \frac{m_e c^2}{\hbar}, \tag{166}$$
$$v_{\circlearrowleft} = \omega_c r_c, \tag{167}$$
$$\Gamma_0 = 2\pi r_c v_{\circlearrowleft}. \tag{168}$$

Substituting $r_c = \alpha\hbar/(2m_e c)$ into $v_{\circlearrowleft} = \omega_c r_c$ gives

$$v_{\circlearrowleft} = \left(\frac{m_e c^2}{\hbar}\right)\left(\frac{\alpha\hbar}{2m_e c}\right) = \frac{\alpha}{2}c, \tag{169}$$

hence the canonical ratio

$$\eta_0 = \frac{v_{\circlearrowleft}}{c} = \frac{\alpha}{2}. \tag{170}$$

B.2 Horn-Envelope Density Normalization

The electron rest-energy normalization may be written as an effective horn-envelope density relation,

$$\rho_{\rm horn}^{\rm eff} \equiv \frac{m_e c^2}{2\pi \|\mathbf{v}_{\circlearrowleft}\|^2 R_{\rm horn}^3}. \tag{171}$$

This relation is dimensionally exact,

$$\left[\frac{m_e c^2}{v^2 r^3}\right] = \mathrm{kg\,m^{-3}}, \tag{172}$$

but its interpretation is now restricted.

[**CANON / SEMANTIC RULE**]: Eq. (171) is an effective horn-envelope or normalization density. It is not a measured local material density of the resolved vortex tube. Any resolved-tube model must use $a_{\rm core}$, a profile function, or ropelength/finite-thickness data rather than silently identifying $a_{\rm core} = r_c$.

[**CANON / MASS-FUNCTIONAL NOTE**]: Existing mass-functionals that use $\rho_{\rm core} {r_c}^3$ should be read as horn-envelope normalizations unless a separate local-core profile is explicitly introduced.

B.3 Geometric gate identity

Using the Compton wavelength $\lambda_c = h/(m_e c)$ and the core closure $r_c = \alpha\hbar/(2m_e c)$, one obtains

$$\frac{\lambda_c}{\pi r_c} = \frac{h/(m_e c)}{\pi\alpha\hbar/(2m_e c)} = \frac{2h}{\pi\alpha\hbar} = \frac{4}{\alpha}. \tag{173}$$

This is the canonical geometric shielding factor used throughout the late mass-functional papers.

B.4 Useful numerical anchors

With the canonical values

$$v_{\circlearrowleft} \approx 1.09384563 \times 10^6\ \mathrm{m\,s^{-1}}, \tag{174}$$

$$r_c \approx 1.40897017 \times 10^{-15}\ \mathrm{m}, \tag{175}$$

$$\Gamma_0 \approx 9.6836 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}, \tag{176}$$

one has the Compton turnover time

$$\tau_c = \frac{2\pi r_c}{\|\mathbf{v}_{\circlearrowleft}\|} = \frac{2\pi}{\omega_c}, \tag{177}$$

and the geometric gate

$$\frac{\lambda_c}{\pi r_c} \approx 5.48 \times 10^2. \tag{178}$$

C Delay Mathematics

C.1 Fixed-point condition for locked modes

Starting from the delayed phase equation

$$\dot{\phi}(t) = \omega_0 + \kappa \sin\big(\phi(t-\tau) - \phi(t)\big), \tag{179}$$

seek a rigidly rotating solution

$$\phi(t) = \Omega t + \phi_\star. \tag{180}$$

Substitution gives the algebraic mode condition

$$\Omega = \omega_0 + \kappa \sin(\Omega\tau), \tag{181}$$

which admits multiple branches labelled by the effective phase winding across the delay interval.

C.2 Linear stability of a locked branch

Let

$$\phi(t) = \Omega t + \epsilon(t), \qquad |\epsilon| \ll 1. \tag{182}$$

Linearizing yields

$$\dot{\epsilon}(t) = \kappa \cos(\Omega\tau)\big(\epsilon(t-\tau) - \epsilon(t)\big). \tag{183}$$

Using the exponential ansatz $\epsilon(t) \propto e^{\lambda t}$ gives the characteristic equation

$$\lambda = \kappa \cos(\Omega\tau)\left(e^{-\lambda\tau} - 1\right). \tag{184}$$

Stability requires $\mathrm{Re}(\lambda) < 0$ for all nontrivial roots. Thus discrete spectra in SST are not arbitrary: only those branches satisfying both the fixed-point equation and the stability inequality survive dynamically.

C.3 Small-delay expansion

For $|\lambda\tau| \ll 1$,

$$e^{-\lambda\tau} = 1 - \lambda\tau + \mathcal{O}((\lambda\tau)^2), \tag{185}$$

so the characteristic equation becomes

$$\lambda \approx -\kappa\tau \cos(\Omega\tau)\, \lambda. \tag{186}$$

Nontrivial damping therefore requires the effective factor $\kappa\tau\cos(\Omega\tau)$ to have the correct sign. This provides a simple branch-selection criterion at weak delay.

C.4 Interpretive consequence

The important canon point is that discreteness is produced in two stages:

$$\text{delay} \;\longrightarrow\; \text{multiple algebraic branches} \;\longrightarrow\; \text{stability filter}. \tag{187}$$

Quantization is therefore not inserted by hand; it is the residue of a dynamical selection problem on a delayed closed carrier.

D Conversation-Derived Insights

D.1 Role of this appendix

This appendix records recurrent insights that emerged during the project-level development of the canon and are useful for guiding future work, even when they do not yet qualify as theorem-level results.

D.2 Topology first, layers second

A central refinement of the recent canon is the separation

$$\text{particle identity} \longleftrightarrow \text{topology}, \qquad \text{excitation or dressing state} \longleftrightarrow \text{layer index}. \tag{188}$$

This prevents Golden-layer fits from being mistaken for a species ontology and keeps the topology program structurally cleaner.

D.3 Spectral origin of $\zeta(3)$

The project discussions repeatedly converged on the conclusion already encoded in Section 5.2: the appearance of $\zeta(3)$ is best interpreted as a spectral invariant of a cubic-weighted closed-carrier mode tower, not as a direct invariant of threefold geometry by itself.

D.4 Atomic bridge status discipline

Another recurring project insight is that the atomic bridge is strongest when treated as a benchmark layer with explicit status tags. Hydrogenic consistency, branch-sensitive couplings, and the spectroscopic gap form a robust bridge; shell filling, exclusion, and full heavy-atom structure remain open programs rather than completed derivations.

D.5 Master-equation viewpoint

A major organizational gain of the recent conversations is the recognition that many SST sectors can be read as different projections of one shared structure:

$$\text{medium scale} \times \text{topology} \times \text{geometry} \times \text{clocking}. \tag{189}$$

Even when this is not written as a single canonical equation inside the present draft, it remains a useful guide for reducing duplication across the mass, time, and thermodynamic sectors.

D.6 State-transition taxonomy from the trefoil-closure discussion

The present conversation also fixes a useful state-transition taxonomy:

$$\text{smooth deformation/binding}: K = 3_1 \text{ preserved}, \tag{190}$$
$$\text{R-to-T closure/nucleation}: K \text{ created at a boundary event}, \tag{191}$$
$$\text{annihilation or weak capture}: K \text{ converted with compensation}. \tag{192}$$

This taxonomy prevents ordinary atomic binding, non-Euler closure, and true particle conversion from being conflated.

D.7 Dual-vacuum and photon sector status

The project-level discussions also clarified a canon policy: the photon/torsion bridge and the dual-vacuum phenomenology are valuable and fertile, but they should not silently override the simpler core canon. They are best treated as phenomenological extensions built on top of the primitive constant chain and Swirl-Clock structure.

D.8 Canon edition notes (v0.8.3)

v0.8.3 integrates framed-tube ontology, attachment gate, trefoil closure discipline, updated particle dictionary, and Pauli a_{core} regularization per conversation, highres, and trefoil diffs.

D.9 Canon edition notes (v0.8.2)

v0.8.2 integrates radius/density semantic correction: $r_c \equiv R_{\text{horn}}$, separate a_{core}, and horn-envelope density normalization per conversation and highres diffs.

D.10 Trefoil closure and persistence discipline

The project-level conclusion is that electron-trefoil creation, if retained, must be formulated as a boundary-supported R-to-T closure event rather than as ordinary reconnection of an already closed ideal-Euler vortex tube. The canon distinction is therefore

$$\text{pre-closure R-phase support} \xrightarrow{\mathcal{K}_{\text{Kairos}}} \text{closed } 3_1 \text{ carrier}, \tag{193}$$

$$\text{closed } 3_1 \text{ carrier} \xrightarrow{\text{smooth Euler}} \text{same knot class } 3_1. \tag{194}$$

This keeps the orthodox frozen-in topology constraint intact while preserving the SST closure program as a clearly labeled bridge.

The radius notation is correspondingly split into the local tube radius a_{core}, the neutral circulation radius r_c, and the classical electron envelope radius $r_e = 2r_c$:

$$a_{\text{core}} \neq r_c, \qquad r_e = 2r_c. \tag{195}$$

Ordinary atomic binding changes the phase envelope of the carrier, not its knot class:

$$e^-_{\text{atom}} = 3_1^{\text{core}} \otimes \Psi^{\text{envelope}}_{n\ell m\sigma}. \tag{196}$$

D.11 Open theorem ladder

The most important unfinished task identified across the project is not adding new sectors, but tightening the derivational ladder between existing ones. In practice this means:

1. turning heuristic closures into explicit lemmas,
2. attaching numerical validation to each nontrivial scaling law,
3. separating benchmark results from full ontological claims.

This appendix therefore serves as a reminder that canon quality increases more by sharpening structure than by increasing conceptual breadth.

E Explicit Topological Mass Closures

E.1 Purpose and status

This appendix supplies an explicit closure layer for the mass program without replacing the canonical master equation of Eq. (34). Its role is to make concrete those invariant choices that were historically used in pre-v0.8 layers whenever actual particle-mass calculations were attempted.

[DERIVED] Canon rule: the abstract kernel Ξ_K of the main text may be specialized to more explicit invariant combinations provided the resulting branch remains dimensionless, benchmarkable, and subordinate to the topology-first hierarchy.

[CRITICAL NOTE] Nothing in this appendix upgrades the explicit kernel to a theorem of first-principles filament dynamics. The present role is closure discipline and reproducible benchmarking.

E.2 Explicit invariant branch of the canonical kernel

A conservative explicit specialization is obtained by resolving the topological kernel into ropelength, braid complexity, and Seifert-genus data:

$$\Xi_K^{(\mathrm{exp})} = \left[\alpha_{\mathcal{L}} \frac{\mathcal{L}_{\mathrm{tot}}(K)}{\mathcal{L}_{\mathrm{tot}}(3_1)} + \alpha_b \big(b(K) - 1\big) + \alpha_g \, g(K)\right] \varphi^{-2k(K)}. \tag{197}$$

Here:

- $\mathcal{L}_{\mathrm{tot}}(K)$ is the total ropelength-like invariant used for the chosen embedding or ideal representative,
- $b(K)$ is the braid index,
- $g(K)$ is the Seifert genus,
- $k(K)$ is the canonical layer / dressing index already used in the main text.

[ORTHODOX] The invariants $\mathcal{L}_{\mathrm{tot}}(K)$, $b(K)$, and $g(K)$ are standard topological or geometric knot descriptors.

[DERIVED] Equation (197) is dimensionless and therefore admissible as a branch of the canonical kernel.

Substituting Eq. (197) into Eq. (34) yields the explicit benchmark mass branch

$$M_K^{(\mathrm{exp})}(x) = \rho_m(x) \, r_c{}^3 \, \Pi_K \, \Xi_K^{(\mathrm{exp})} \, S_{(t)}^{\circlearrowleft}(x)^{-2}. \tag{198}$$

E.3 Baryon benchmark closure

For composite baryons built from constituent topology classes K_u and K_d, a conservative additive benchmark branch is

$$M_B^{(\mathrm{add})} = \Pi_B \, M_0 \left(n_u \, \Xi_{K_u}^{(\mathrm{exp})} + n_d \, \Xi_{K_d}^{(\mathrm{exp})}\right), \tag{199}$$

where n_u and n_d are constituent counts and M_0 is the common medium scale extracted from the main mass functional.

A retained legacy golden-closure benchmark is

$$M_p^{(\varphi)} = \varphi^{-2} 3^{-1/\varphi} \, (2M_u + M_d), \tag{200}$$

$$M_n^{(\varphi)} = \varphi^{-2} 3^{-1/\varphi} \, (M_u + 2M_d). \tag{201}$$

[SPECULATIVE] Equations (200)–(201) are retained as baryon-sector closure benchmarks only. They are not promoted to universal mass laws.

E.4 Canonical use policy

The explicit closures above are intended for:

1. theorem-ladder benchmarking,
2. archived CSV reproduction,
3. side-by-side comparison of topology branches,
4. explicit baryon fits.

They are *not* intended to replace the abstract organizing form of the main text.

F Effective Quantum Bridge

F.1 Purpose and status

The main canon states that quantum mechanics should appear as an effective low-energy description of the medium. The present appendix supplies the minimal bridge formulas needed to make that statement mathematically legible without claiming a completed derivation of the full quantum rule set.

[ORTHODOX] The Madelung decomposition and Schrödinger-form continuum limit are standard.

[DERIVED] SST may use these formulas as a coarse-grained field representation once a local amplitude density and phase have been specified.

[SPECULATIVE] The identification of the resulting fields with branch-resolved swirl carriers remains an SST interpretation.

F.2 Amplitude–phase decomposition

Introduce an effective complex mode field

$$\psi(x,t) = \sqrt{\rho_\psi(x,t)}\, e^{i\theta(x,t)}, \tag{202}$$

with $\rho_\psi \geq 0$ and real phase θ. The associated effective transport velocity is

$$\mathbf{u}_\psi = \frac{\hbar}{m_{\text{eff}}}\nabla\theta. \tag{203}$$

Assuming a Schrödinger-form bridge equation

$$i\hbar\partial_t\psi = \left(-\frac{\hbar^2}{2m_{\text{eff}}}\nabla^2 + V_{\text{eff}}\right)\psi, \tag{204}$$

one obtains the equivalent real system

$$\partial_t\rho_\psi + \nabla\cdot(\rho_\psi\mathbf{u}_\psi) = 0, \tag{205}$$

$$\partial_t\theta + \frac{m_{\text{eff}}}{2}\|\mathbf{u}_\psi\|^2 + V_{\text{eff}} + Q_{\text{eff}} = 0, \tag{206}$$

with effective quantum potential

$$Q_{\text{eff}} = -\frac{\hbar^2}{2m_{\text{eff}}}\frac{\nabla^2\sqrt{\rho_\psi}}{\sqrt{\rho_\psi}}. \tag{207}$$

[ORTHODOX] Equations (205)–(207) are the standard Madelung system associated with Eq. (204) [14, 15].

F.3 SST reading of the bridge

The conservative SST interpretation is:

$$\rho_\psi \leftrightarrow \text{coarse-grained mode-support density,} \tag{208}$$
$$\theta \leftrightarrow \text{effective circulation phase,} \tag{209}$$
$$\mathbf{u}_\psi \leftrightarrow \text{phase-transport velocity of the low-energy envelope.} \tag{210}$$

At this level, the branch-resolved atomic sector of Section 8 is read not as a replacement of the orthodox wavefunction, but as a candidate microscopic underpinning for the effective fields ρ_ψ and θ.

F.4 Two-component branch bookkeeping

A minimal two-branch extension consistent with the $\sigma = \pm 1$ structure of the atomic sector is

$$\Psi = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}, \qquad \psi_\pm = \sqrt{\rho_\pm}\, e^{i\theta_\pm}, \tag{211}$$

with total coarse-grained density

$$\rho_{\text{tot}} = \rho_+ + \rho_-. \tag{212}$$

[**SPECULATIVE**] This two-component structure is the minimal SST bridge to spinor-like bookkeeping. It is not yet a first-principles derivation of fermionic spin statistics.

F.5 Topological reading of the bridge

The most conservative topology statement retained at appendix level is that nontrivial phase maps may encode nontrivial carrier organization. In particular, the effective phase field may carry nontrivial winding or linking data even when the coarse-grained description is written in wave language. This is the appendix-level reason why the canon treats topology and wave structure as compatible rather than as competing descriptions.

G Dark-Sector and Galactic Continuum Law

G.1 Purpose and status

The main canon already retains the dark-knot taxonomy and the radial swirl pressure law. This appendix collects the next phenomenological layer: the galaxy-scale force law and the simplest rotation-curve ansatz inherited from earlier SST development.

[**ORTHODOX**] The underlying force law is ordinary Euler balance in steady swirl.

[**DERIVED**] SST reinterprets that balance as a candidate large-scale acceleration source.

[**SPECULATIVE**] The explicit galaxy-fit profile below is a phenomenological closure ansatz, not a theorem.

G.2 Effective dark acceleration law

Starting from the steady radial pressure law of Eq. (85),

$$\frac{1}{\rho_f}\frac{dp_{\text{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}, \tag{213}$$

define the effective SST acceleration by

$$a_{\mathrm{SST}}(r) := \frac{1}{\rho_f}\frac{dp_{\mathrm{swirl}}}{dr}. \tag{214}$$

Then

$$a_{\mathrm{SST}}(r) = \frac{v_\theta(r)^2}{r}. \tag{215}$$

[DERIVED] Equations (214)–(215) are immediate consequences of the retained pressure law.

G.3 Two-component galactic rotation ansatz

A retained phenomenological velocity profile is

$$v_{\mathrm{swirl}}(r) = \frac{C_{\mathrm{core}}}{\sqrt{1+\left(r_c^{\mathrm{gal}}/r\right)^2}} + C_{\mathrm{tail}}\left(1-e^{-r/r_c^{\mathrm{gal}}}\right), \tag{216}$$

with corresponding total circular speed

$$v_{\mathrm{tot}}^2(r) = v_{\mathrm{bar}}^2(r) + v_{\mathrm{swirl}}^2(r). \tag{217}$$

[SPECULATIVE] Equation (216) is a phenomenological fit ansatz and should be used only at the appendix / research-track level.

G.4 Relation to dark-knot taxonomy

The taxonomy of Eq. (77) classifies low-helicity, low-instability knot sectors as dark or quasi-dark candidates. The present galactic law does *not* assume that all galactic anomalies are produced by localized dark knots alone. Rather, it provides the complementary continuum statement: long-range swirl pressure gradients may produce effective extra acceleration even when the underlying microscopic carriers are topologically sparse or weakly luminous.

G.5 Use policy

The galactic law belongs in the canon only as:

1. an appendix-level phenomenological sector,
2. a separate astrophysics / dark-sector track,
3. a fit target for future rotation-curve studies.

It is deliberately not promoted in v0.8.1 to the same status as the primitive constant chain, the Swirl-Clock, or the delay-selection sector.

H Research-track sectors migrated from the main canon

This companion document collects those sectors that remain useful for continuity and future development but are too long, too provisional, or too phenomenology-specific for the core body of *Swirl-String-Theory_Canon-v0.8.3*. Their migration out of the main canon is editorial rather than negative: each subsection remains part of the broader SST research program, but none is required for the minimal logical closure of the core canon.

Editorial note (v0.8.3): the maximal swirl tension sector ($F_{\rm swirl}^{\rm max}$, Rydberg bridge identities, Gibbons-class maximum-tension comparison, and the threefold interpretation table) is developed in the core document under `been_processed/v0.8.3/SST_CANON-v0.8.3.tex`, subsection `Maximal Swirl Tension` (label `sec:Fmax`); it is not duplicated here.

H.1 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector. Let

$$T_{\rm eff} = T_{\rm eff}\big(\rho_E,\, \omega,\, S_{(t)}^{\circlearrowleft}\big) \tag{218}$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by $T_{\rm eff}$.

[**ORTHODOX**]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[**SPECULATIVE**]: SST does not yet derive the constitutive relation above, so this sector remains a research program rather than canonically closed physics.

H.2 Minimal Coupling Interface to QED

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \to \nabla - i\frac{m}{\hbar}A_{\rm eff}, \qquad A_{\rm eff} = \chi\, v, \tag{219}$$

where χ is a scalar weight and v is the local carrier velocity field.

[**ORTHODOX**]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.

[**SPECULATIVE**]: the above substitution is only an analogy template unless a full gauge-covariant derivation is supplied.

H.3 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Calugareanu–White–Fuller relation

$$Lk = Tw + Wr \tag{220}$$

as a refinement of the knot-based matter/antimatter dictionary.

[**ORTHODOX**]: this is a standard geometric identity for framed curves and ribbons.

[**DERIVED**]: it can be used internally to separate writhe-dominated from twist-dominated realizations of the same coarse knot class.

[**SPECULATIVE**]: the mapping from $\mathrm{sign}(Tw)$ or related chirality measures to particle/antiparticle sectors remains a model hypothesis.

H.4 Particle-Candidate Mapping Layer

[SPECULATIVE] Working dictionary: a minimal knot/composition mapping retained from earlier SST work is summarized in Table 5. This table is organizational rather than definitive; it records the current topological assignments used by the mass and taxonomy program.

Knot class / composition	Candidate	Status / role
3_1 trefoil	electron / positron sector	baseline chiral lepton calibra
$T(2,5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2,7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
twist/clasp knots, quasi-unknot loops	quark-like defects	research-track fractional bou
$5_2 + 5_2 + 6_1$ and nearby composites	baryon candidates	legacy mass-functional dicti
triple-gear locked analogue	proton-like charged baryon analogue	mechanical model for nonze
Borromean-type analogue	neutron-like neutral baryon analogue	mechanical/topological mod

Table 5: Minimal knot-class and analogue summary (v0.8.3 research-track), aligned with main canon semantic corrections.

[**CRITICAL NOTE**]: Table 5 is a working dictionary only. A full canonical assignment requires an explicit invariant set and a mass-functional comparison program.

H.5 Hydrodynamic Exchange and the Pauli Barrier

[**ORTHODOX**]: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is

$$E_{\text{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\ell_1 \cdot d\ell_2}{\|r_1 - r_2\|}. \tag{221}$$

[**DERIVED**]: if two identical electron-candidate loops are forced into the same spatial state, regularization at the physical tube radius a_{core} yields the overlap penalty

$$V_{\text{Pauli}}(d) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \iint \frac{d\ell_1 \cdot d\ell_2}{\sqrt{\|r_1(\theta_1) - r_2(\theta_2) + d\|^2 + a_{\text{core}}^2}}, \tag{222}$$

with maximal-overlap scale

$$V_{\text{Pauli}}^{\max} \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{a_{\text{core}}} \right) \mathcal{O}(1). \tag{223}$$

Using the canonical values

$$\rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}, \tag{224}$$

$$r_c = 1.40897017 \times 10^{-15} \text{ m}, \tag{225}$$

$$\Gamma_0 = 2\pi r_c \,\|\mathbf{v}_{\circlearrowleft}\| \approx 9.6836 \times 10^{-9} \text{ m}^2\,\text{s}^{-1}, \tag{226}$$

and $L \sim 2\pi a_0$ at the hydrogenic scale, one obtains the benchmark estimate

$$V_{\text{Pauli}}^{\max} \sim 1.23 \times 10^{-18} \text{ J} \sim 7.6 \text{ eV}. \tag{227}$$

[**CRITICAL NOTE**]: this is retained as a canonical scale benchmark, not yet as a closed theorem. Its role is to show that the exclusion-energy estimate falls in an atomically relevant window rather than at obviously pathological scales.

Particle	m_{exp} (MeV)	m_{SST} (MeV)	rel. error	mode class
e	0.51099895	0.51099895	0	predictive/closure
μ	105.6583745	...	...	predictive candidate
τ	1776.86	...	...	predictive candidate
p	938.272088	...	...	predictive or closure
n	939.565420	...	...	predictive or closure

Table 6: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

H.6 Numerical Benchmark Layer

[**DERIVED**] **Computational-program requirement:** earlier SST work contained explicit benchmark tables for particle and atomic masses. In v0.8.1, the benchmark layer is retained as a reproducibility requirement rather than as a stand-alone authority claim.

[**CRITICAL NOTE**]: a future v0.8.x benchmark appendix should include the exact script or package path used for generation, archived CSV outputs, mode definitions (predictive vs. exact-closure), and uncertainty propagation on canonical constants and geometric factors.

H.7 Helicity-by-base archive (diagnostic layer)

[**DERIVED**] **Archive role:** the SST helicity-by-base outputs are retained as a measurable balance diagnostic for taxonomy support, not as an independent law. Define the midpoint-distance diagnostic

$$\Delta_H(K) = \left|\widehat{\mathcal{H}}_b(K) + \frac{1}{2}\right|, \tag{228}$$

where $\widehat{\mathcal{H}}_b(K)$ is the archived normalized helicity-like base indicator for class K.

Near-balanced values with $\Delta_H(K) \ll 1$ indicate candidate dark or quasi-dark sectors only when symmetry and instability filters are also satisfied.

[**CRITICAL NOTE**]: no single helicity-like scalar is sufficient for sector assignment; classifier decisions remain conjunction rules over symmetry, helicity balance, and low-order stability content.

[**CANON WORKFLOW NOTE**]: if SST-labelled and parallel VAM-labelled helicity outputs are numerically identical, the SST-labelled archive is the only required source for canon workflows.

H.8 Trefoil-closure archive (benchmark discipline)

[**DERIVED**] **Benchmark role:** ideal-trefoil robustness sweeps, final-best-estimate tables, and theorem-style reports are retained as benchmark-discipline infrastructure.

[**CRITICAL NOTE**]: practical best-estimate closure and theorem-level closure are distinct. A result may remain useful for calibration while still failing strict theorem-level tail or postcheck requirements.

[**REQUIRED METADATA**]: benchmark reports should include explicit pass/fail logic, postchecks, extrapolation summaries, and local-curve diagnostics so that closure claims are auditable.

H.9 ϕ_{3_1} transform archive (deferred)

[**SPECULATIVE**] **Status:** retained as research-track spectral evidence for later zeta/spectral programs.

[SCOPE RULE]: this archive is not part of the current dark-sector or galactic canonization closure path in v0.8.x.

H.10 Gauge-Sector Roadmap

[SPECULATIVE] Structural roadmap: a retained minimal statement of the gauge program is

$$g_{\text{eff}} = su(3) \oplus su(2) \oplus u(1), \tag{229}$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A_\mu^a(x) = \partial_\mu \theta^a(x), \tag{230}$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A_\mu^a \mapsto A_\mu^a + q^a \partial_\mu \chi. \tag{231}$$

[CRITICAL NOTE]: the gauge layer is retained only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.

I Dark-sector and galactic canonization execution package (v0.8.x)

I.1 Stage-0 freeze: topology toolkit and benchmark protocol

[DERIVED] Scope freeze: the mandatory topology toolkit for v0.8.x canonization is fixed to:

1. Gauss linking integral,
2. helicity in ideal fluids,
3. thin-tube helicity reduction,
4. Hopf invariant,
5. basic torus-knot invariants (c, b, g),
6. Calugareanu–White–Fuller relation $Lk = Tw + Wr$.

[DERIVED] Notation freeze: one notation family is used in this package:

$$\rho_f, \quad v_\theta(r), \quad p_{\text{swirl}}(r), \quad \mathcal{H}(K), \quad N_u^{(1)}(K), \quad \Gamma, \quad \kappa, \quad S_{(t)}^{\circlearrowleft}.$$

No parallel symbol aliases are introduced for the same quantity.

[DERIVED] Mandatory benchmark protocol template:

1. **Inputs:** canonical constants, topology identifiers, profile parameters, and dataset/version tags.
2. **Procedure:** exact script or package path, run command, and branch/commit hash.
3. **Outputs:** archived CSV files with fixed column schema and an attached metadata note.
4. **Uncertainty:** first-order propagation for calibrated constants and sensitivity sweep on profile parameters.
5. **Rerun:** deterministic rerun steps and acceptance tolerances.

[CANON WORKFLOW NOTE]: every promoted quantitative claim in this package is invalid without an explicit input-to-CSV trace.

I.2 Stage-1 freeze: dark-sector Euler pressure anchor

[ORTHODOX] Euler radial balance (steady, axisymmetric, azimuthal-dominant):

$$\frac{1}{\rho_f}\frac{dp_{\text{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}. \tag{232}$$

[DERIVED] Flat-tail limit (constant tail speed):

$$v_\theta(r) \to v_0 \quad \Longrightarrow \quad p_{\text{swirl}}(r) = p_0 + \rho_f v_0^2 \ln\left(\frac{r}{r_0}\right). \tag{233}$$

[DERIVED] Assumption block: incompressible, inviscid, stationary, axisymmetric, azimuthal-dominant flow, with validity restricted to regions where radial drift and stratification corrections are subleading.

[DERIVED] Dimensional check:

$$\left[\frac{1}{\rho_f}\frac{dp}{dr}\right] = \frac{\mathrm{kg\,m^{-2}\,s^{-2}}}{\mathrm{kg\,m^{-3}}} = \mathrm{m\,s^{-2}} = \left[\frac{v^2}{r}\right].$$

[SPECULATIVE] SST interpretation: Eq. (232) is interpreted as a classifier/driver layer for dark-sector support. This interpretation is model-level and separate from the orthodox derivation.

[DERIVED] Attached observables/falsifiers:

1. Rotation-curve consistency after pressure reconstruction from archived $v_\theta(r)$ data.
2. Null falsifier: systematic mismatch in the reconstructed pressure gradient sign or scale across validated radial windows.

I.3 Stage-2 freeze: measurable dark taxonomy

[DERIVED] Minimal classifier variables: symmetry class (including amphichirality status), chirality sign/state, helicity-balance indicator $\Delta_H(K)$, and low-order instability count $N_u^{(1)}(K)$.

[DERIVED] Decision rules (symmetry-first, then stability-filtered):

$$\text{Dark}: \ \text{amphichiral candidate} \ \wedge \ \Delta_H(K) \ll 1 \ \wedge \ N_u^{(1)}(K) = 0, \tag{234}$$

$$\text{Quasi-dark}: \ \text{near-balanced symmetry/helicity} \ \wedge \ 0 < N_u^{(1)}(K) \leq 2, \tag{235}$$

$$\text{Visible}: \ \text{chiral-dominant and/or helicity-unbalanced sectors.} \tag{236}$$

[DERIVED] Ambiguity tie-break: if classes overlap, choose the class with the strongest conjunction score in the order

$$\text{symmetry gate} \to \text{helicity balance gate} \to \text{stability gate.}$$

[DERIVED] Baseline templates and falsifiers:

- **Visible baseline:** trefoil-like chiral template; falsifier = persistent amphichiral-plus-balanced signature under validated data updates.
- **Quasi-dark baseline:** Borromean-type weak-coupling template; falsifier = stable zero-instability assignment across independent scans.
- **Dark baseline:** figure-eight/amphichiral template; falsifier = robust nonzero chirality or high-instability assignment.

[CANON WORKFLOW NOTE]: amphichiral $\Rightarrow$ dark remains a classifier proposal tested against observables, not a closed theorem.

I.4 Stage-3 freeze: galactic law branch gate and closure route

[DERIVED] Branch-gate decision: Branch A (legacy profile refine) is selected for current v0.8.x closure because it preserves tested limits while allowing benchmark traceability.

Working profile:

$$v_\theta(r) = \frac{C_{\rm core}}{\sqrt{1 + (r_c/r)^2}} + C_{\rm tail}\left(1 - e^{-r/r_c}\right). \tag{237}$$

[DERIVED] Required checks: small-r behavior, flat-tail behavior, parameter semantics, and pressure reconstruction via Eq. (232).

[DERIVED] Failure modes:

1. non-physical parameter combinations (wrong sign or unstable growth),
2. failure to satisfy tail-limit behavior in benchmark windows,
3. inability to reconstruct a consistent pressure profile from the same archived inputs.

I.5 Stage-4 freeze: probe falsification map

[DERIVED] probe classes are fixed as astrophysical, laboratory pressure/swirl analog, HV-coil/thrust-inspired, optical/torsional coupling, and null dark-sector tests.

Probe class	Equation under test	Null-result meaning	Primary confounder	Expected sign/trend
Astrophysical rotation curves	Eq. (232) + profile closure	weakens pressure-law applicability window	baryonic-model bias	reconstructed dp/dr consistent with v_θ^2/r
Lab swirl analog	Euler radial balance scaling	weakens scale-transfer claim	viscosity/edge forcing	monotone pressure rise with imposed swirl
HV-coil/thrust-inspired	circulation-pressure bridge claims	kills quantized-bridge claim if no scaling	EM/thermal artifacts	response tracks circulation proxy
Optical/torsional coupling	swirl-clock bridge signatures	weakens coupling interpretation	detector drift/noise	phase/timing shift with controlled swirl proxy
Null dark tests	taxonomy conjunction rules	kills classifier branch if conjunction fails	mislabelled topology class	class score decreases under contradictory diagnostics

Table 7: Frozen equation-to-probe map for falsification-first workflow in the v0.8.x dark-sector program.

[CANON WORKFLOW NOTE]: probe outcomes can support or reject already-stated equations, but they do not replace derivation-level closure.

I.6 Promotion snapshot after execution

[**DERIVED**] post-freeze status in this package:

- Item 6 (topology toolkit): CANON-ready in scope and notation.
- Item 5 (benchmark protocol): CANON-CANDIDATE pending repository-wide protocol reuse.
- Item 1 (Euler pressure anchor): CANON-CANDIDATE with explicit assumptions, limits, and falsifier path.
- Item 2 (dark taxonomy): BRIDGE with measurable classifier gates and falsifiers.
- Item 3 (galactic swirl law): BRIDGE with branch gate closed and benchmark route fixed.
- Item 4 (probe map): CANON-BRIDGE with equation/null-result mapping frozen.

J Candidate CANON-v0.8.x Modules from Swirl-Flux, Envelope, and Reconnection Analysis

J.1 Status Classification

The following modules are proposed as CANON-v0.8.x candidates.

1. The Schrödinger kinetic coefficient is retained as an effective envelope stiffness,
$$\mathcal{S}_e = \frac{\hbar^2}{2M_e}.$$
The replacement $-\mathbf{v}_{\circlearrowleft}^2\nabla^2/2$ is rejected on dimensional grounds.
2. The Meissner effect is interpreted as macroscopic swirl-flux phase locking in a coherent paired condensate.
3. R/T wave–string interaction is represented by conservation of radiative pseudomomentum plus vorticity impulse.
4. Reconnection is excluded in the ideal sector but allowed in measurement, decay, and dissipative sectors, where helicity is redistributed between topology, writhe, twist, radiation, and dissipation.

J.2 Envelope Stiffness and the Rejection of the $\mathbf{v}_{\circlearrowleft}^2\nabla^2$ Kinetic Term

For an effective R/T envelope amplitude $\Psi(\mathbf{r}, t)$, the kinetic coefficient must have dimension $\mathrm{J\,m^2}$. The correct envelope stiffness is therefore

$$\mathcal{S}_e \equiv \frac{\hbar^2}{2M_e}, \qquad [\mathcal{S}_e] = \mathrm{J\,m^2}.$$

The effective low-energy envelope equation is

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[-\mathcal{S}_e\nabla^2 + V_{\mathrm{SST}}(\mathbf{r}, t)\right]\Psi.$$

The operator $-\mathbf{v}_{\circlearrowleft}^2\nabla^2/2$ is not a Schrödinger kinetic energy operator because

$$\left[\mathbf{v}_{\circlearrowleft}^2\nabla^2\right] = \frac{\mathrm{m}^2}{\mathrm{s}^2}\frac{1}{\mathrm{m}^2} = \mathrm{s}^{-2},$$

which is not an energy.

For tunneling through a one-dimensional effective SST barrier,

$$T \sim \exp\left(-2\int_{x_1}^{x_2} \kappa_{\mathrm{SST}}(x)\,dx\right),$$

where

$$\kappa_{\mathrm{SST}}(x) = \sqrt{\frac{V_{\mathrm{SST}}(x) - E}{\mathcal{S}_e}} = \sqrt{\frac{2M_e\,[V_{\mathrm{SST}}(x) - E]}{\hbar^2}}.$$

This is dimensionally consistent because

$$\left[\frac{V - E}{\mathcal{S}_e}\right] = \mathrm{m}^{-2}, \qquad [\kappa_{\mathrm{SST}}] = \mathrm{m}^{-1}.$$

J.3 Numerical Bridge Anomaly

A numerically close but dimensionally non-identical bridge is

$$\mathcal{B}_e = \frac{F_{\mathrm{swirl}}^{\max} {r_c}^3}{5\lambda_c\|\mathbf{v}_{\circlearrowleft}\|}.$$

Using

$$\lambda_c = \frac{h}{M_e c},$$

one obtains

$$\frac{\hbar^2}{2M_e} = 6.10426431 \times 10^{-39}\,\mathrm{J\,m^2},$$

whereas

$$\mathcal{B}_e = 6.12394931 \times 10^{-39}\,\mathrm{J\,s}.$$

The numerical ratio is

$$\left.\frac{\mathcal{B}_e}{\mathcal{S}_e}\right|_{\mathrm{SI\ numbers}} = 1.00322479.$$

This is classified as a Research Track numerical bridge, not a canonical identity, because

$$[\mathcal{B}_e] = \mathrm{J\,s} \neq \mathrm{J\,m^2} = [\mathcal{S}_e].$$

The dimensionally meaningful relation is instead

$$5\lambda_c\|\mathbf{v}_{\circlearrowleft}\| = 5\alpha\frac{h}{2M_e},$$

with both sides having dimension $\mathrm{m^2\,s^{-1}}$.

J.4 Swirl-Flux Locking and the Meissner Effect

For a coherent paired electronic condensate of mass $2M_e$ and charge $q_{\mathrm{pair}} = -2e$, phase single-valuedness gives

$$\oint_{\partial\Sigma} (2M_e\mathbf{v}_{\mathrm{pair}} + q_{\mathrm{pair}}\mathbf{A}) \cdot d\boldsymbol{\ell} = nh.$$

Since $q_{\mathrm{pair}} = -2e$,

$$2M_e\Gamma_{\mathrm{pair}} - 2e\Phi_B = nh.$$

Thus

$$\Phi_B = \frac{M_e}{e}\Gamma_{\mathrm{pair}} - n\frac{h}{2e}.$$

In magnitude,

$$|\Phi_B| = \left| n\Phi_0 - \frac{M_e}{e}\Gamma_{\text{pair}} \right|, \qquad \Phi_0 = \frac{h}{2e}.$$

For a non-rotating coherent bulk,

$$\Gamma_{\text{pair}} \to 0,$$

and therefore

$$|\Phi_B| = n\Phi_0.$$

The Meissner effect is then interpreted in SST as exclusion of non-quantized magnetic flux from a phase-locked coherent R/T condensate.

J.5 R/T Pseudomomentum–Impulse Exchange

For a closed R/T interaction domain,

$$\frac{d}{dt}\left(\mathbf{P}_R + \mathbf{I}_T\right) = 0.$$

Here $\mathbf{P}_R$ denotes radiative R-phase pseudomomentum and $\mathbf{I}_T$ denotes T-phase vorticity impulse,

$$\mathbf{I}_T = \frac{\rho_f}{2} \int_\Omega \mathbf{x} \times \boldsymbol{\omega}\, d^3x.$$

The dimension is

$$[\mathbf{I}_T] = \mathrm{kg\,m\,s^{-1}},$$

so $\mathbf{I}_T$ is a true momentum-like quantity.

For a thin circular swirl string of circulation Γ and radius R,

$$\mathbf{I}_T = \rho_f \Gamma \pi R^2 \hat{\mathbf{n}}.$$

The core circulation scale is

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_\circlearrowleft\| = 9.68361920 \times 10^{-9}\,\mathrm{m^2\,s^{-1}}.$$

J.6 Reconnection and Helicity Redistribution

In the ideal sector, topology is conserved:

$$\frac{D\mathcal{K}}{Dt} = 0,$$

under the assumptions

$$\nabla \cdot \mathbf{v} = 0, \qquad \nu = 0, \qquad \text{barotropic}, \qquad \text{no reconnection}.$$

In the dissipative, measurement, or decay sector, reconnection is allowed:

$$\frac{D\mathcal{K}}{Dt} \neq 0.$$

The helicity budget is written

$$\Delta H_{\text{topology}} + \Delta H_{\text{writhe}} + \Delta H_{\text{twist}} + \Delta H_{\text{radiative}} = -\Delta H_{\text{diss}}.$$

In the weak-dissipation limit,

$$\Delta H_{\text{topology}} + \Delta H_{\text{writhe}} + \Delta H_{\text{twist}} + \Delta H_{\text{radiative}} \approx 0.$$

For filamentary structures,

$$H_{\text{centerline}} = \sum_{i \neq j} \Gamma_i \Gamma_j \mathrm{Lk}_{ij} + \sum_i \Gamma_i^2 \left(\mathrm{Wr}_i + \mathrm{Tw}_i\right).$$

J.7 Spherical Pressure Envelopes

For inviscid incompressible SST flow,

$$\nabla p_{\mathrm{SST}} = \rho_f (\mathbf{v} \cdot \nabla)\mathbf{v}.$$

For an azimuthal swirl profile $v_\theta(r)$,

$$\frac{dp}{dr} = \rho_f \frac{v_\theta^2(r)}{r}.$$

If

$$v_\theta(r) = \frac{\Gamma}{2\pi r},$$

then

$$\frac{dp}{dr} = \rho_f \frac{\Gamma^2}{4\pi^2 r^3},$$

and hence

$$p(r) = p_\infty - \frac{\rho_f \Gamma^2}{8\pi^2 r^2}.$$

Therefore

$$\Delta p(r) = p_\infty - p(r) = \frac{\rho_f \Gamma^2}{8\pi^2 r^2} = \frac{1}{2}\rho_f v_\theta^2(r).$$

At the canonical swirl speed,

$$\frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2 = 4.18774392 \times 10^5\,\mathrm{Pa}.$$

At the core-density stress scale,

$$\rho_{\mathrm{core}} \|\mathbf{v}_{\circlearrowleft}\|^2 = 4.65848920 \times 10^{30}\,\mathrm{Pa}.$$

The canonical maximum swirl force is recovered as

$$F_{\mathrm{swirl}}^{\mathrm{max}} = \rho_{\mathrm{core}} \|\mathbf{v}_{\circlearrowleft}\|^2 \pi {r_c}^2 = 29.053507\,\mathrm{N}.$$

J.8 Golden-Layer Scaling

The universal rule

$$E_n = \varphi^n E_0$$

is rejected as too strong. The weaker Research Track ansatz is

$$E_{K,k} = E_K^{(0)} \varphi^{-2k}, \qquad k \in \mathbb{Z}_{\geq 0}.$$

Here K labels a knot family and k labels a relaxation layer inside that family.

J.9 Non-Holonomic Gravity Claims

Podkletnov-type anti-gravity claims are not adopted as Canon. At most, the formal analogy

$$\text{non-holonomic twist} \longleftrightarrow \text{foliation twist/vorticity}$$

may be retained in a historical or speculative appendix. No experimental anti-gravity claim is inferred.

References

[1] F. London and H. London, "The Electromagnetic Equations of the Supraconductor," *Proceedings of the Royal Society A*, vol. 149, no. 866, pp. 71–88, 1935. DOI: 10.1098/rspa.1935.0048.

[2] M. Tinkham, *Introduction to Superconductivity*, 2nd ed., McGraw–Hill, New York, 1996. ISBN: 978-0486435039.

[3] L. Onsager, "Statistical hydrodynamics," *Il Nuovo Cimento*, vol. 6, Suppl. 2, pp. 279–287, 1949. DOI: 10.1007/BF02780991.

[4] R. P. Feynman, "Application of Quantum Mechanics to Liquid Helium," in *Progress in Low Temperature Physics*, vol. 1, pp. 17–53, Elsevier, 1955. DOI: 10.1016/S0079-6417(08)60077-3.

[5] W. F. Vinen, "The Physics of Superfluid Helium," CERN Yellow Reports / Lecture Notes, 2004. Permalink: https://cds.cern.ch/record/808382/files/p363.pdf.

[6] O. Bühler and M. E. McIntyre, "Wave capture and wave–vortex duality," *Journal of Fluid Mechanics*, vol. 534, pp. 67–95, 2005. DOI: 10.1017/S0022112005004374.

[7] D. Kleckner and W. T. M. Irvine, "Creation and dynamics of knotted vortices," *Nature Physics*, vol. 9, pp. 253–258, 2013. DOI: 10.1038/nphys2560.

[8] D. Kleckner, L. H. Kauffman, and W. T. M. Irvine, "How superfluid vortex knots untie," *Nature Physics*, vol. 12, pp. 650–655, 2016. DOI: 10.1038/nphys3679.

[9] J. M. Keller and A. F. Cheviakov, "Exact spherical vortex-type equilibrium flows in fluids and plasmas," *Fundamental Plasma Physics*, vol. 11, article 100063, 2024. DOI: 10.1016/j.fpp.2024.100063.

[10] N. F. Brickman, "Rotating Superconductors," *Physical Review*, vol. 184, pp. 460–470, 1969. DOI: 10.1103/PhysRev.184.460.

K Mode-Locked Swirl-Coil Excitation and $\|\mathbf{v}_{\circlearrowleft}\| = f\Delta x$ Metrology

Status. [RESEARCH TRACK / CANON-CANDIDATE]. This appendix formulates a falsifiable experimental protocol for testing whether externally driven multi-phase coil geometries can inject coherent swirl modes into a bounded measurement region. The construction is not assumed as a new force law. It is a mode-selection and metrology rule for identifying when the canonical characteristic swirl speed

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}} \tag{238}$$

is sampled by a laboratory-scale frequency–length product.

K.1 Motivation

The basic dimensional identity

$$\|\mathbf{v}_{\circlearrowleft}\| = f\,\Delta x \tag{239}$$

has the units of speed and is therefore not, by itself, a dynamical claim. Its SST use is metrological: a driven system is said to probe a swirl-resonant channel only when a temporal frequency f and a geometrically defined spatial period Δx satisfy Eq. (239) within experimental bandwidth.

For a circular or quasi-circular coil, the relevant spatial period is generally not an arbitrary external distance. It is the azimuthal wavelength selected by the coil radius and spatial mode number. Hence the physical coil radius enters the resonance condition directly.

K.2 Ring-mode closure condition

Let R_{coil} be the effective radius of the active winding, f_0 the electrical base frequency, $n \in \mathbb{N}$ the temporal harmonic, and $m \in \mathbb{Z} \setminus \{0\}$ the azimuthal spatial mode number around the coil. The spatial wavelength of mode m along the coil circumference is

$$\Delta x_m = \frac{2\pi R_{\text{coil}}}{|m|}. \tag{240}$$

The (n)-th temporal harmonic has frequency

$$f_n = n f_0. \tag{241}$$

The dimensionless swirl-lock parameter is therefore

$$\boxed{\eta_{n,m} = \frac{f_n \Delta x_m}{\|\mathbf{v}_\circlearrowleft\|} = \frac{n f_0}{\|\mathbf{v}_\circlearrowleft\|} \frac{2\pi R_{\text{coil}}}{|m|}}. \tag{242}$$

A ring mode is swirl-locked when

$$\boxed{\eta_{n,m} = 1.} \tag{243}$$

Equivalently, the resonant coil radius for a given (n, m, f_0) is

$$\boxed{R_{\text{coil}}^{(n,m)} = \frac{|m| \, \|\mathbf{v}_\circlearrowleft\|}{2\pi n f_0}}. \tag{244}$$

Dimensional check.

$$[R_{\text{coil}}] = \frac{\text{m}\,\text{s}^{-1}}{\text{s}^{-1}} = \text{m}.$$

Thus Eq. (244) is dimensionally well-defined.

K.3 Numerical calibration examples

For the canonical value

$$\|\mathbf{v}_\circlearrowleft\| = 1.09384563 \times 10^6\,\text{m}\,\text{s}^{-1},$$

and a base drive frequency

$$f_0 = 150\,\text{kHz},$$

the first three $m = 1$ radius locks are

$$R_{\text{coil}}^{(1,1)} = \frac{1.09384563 \times 10^6}{2\pi(1)(150000)} = 1.1606\,\text{m}, \tag{245}$$

$$R_{\text{coil}}^{(2,1)} = 0.5803\,\text{m}, \tag{246}$$

$$R_{\text{coil}}^{(3,1)} = 0.3869\,\text{m}. \tag{247}$$

For a smaller experimental coil with

$$R_{\text{coil}} = 0.150\,\text{m},$$

the $(n = 3,\ m = 1)$ lock frequency is

$$f_0 = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{2\pi n R_{\text{coil}}} = \frac{1.09384563 \times 10^6}{2\pi(3)(0.150)} = 3.8687 \times 10^5\,\text{Hz}. \tag{248}$$

Hence a 0.150 m coil driven at 150 kHz does not satisfy the $(n = 3, m = 1)$ ring-lock condition. Its lock parameter is

$$\eta_{3,1} = \frac{3(150000)\,2\pi(0.150)}{1.09384563 \times 10^6} = 0.3877. \tag{249}$$

This is sub-resonant for the $(3, 1)$ ring mode, although it may still couple to different (n, m) channels.

K.4 Three-phase selection rule

Consider a three-phase coil drive with phase index $q = 0, 1, 2$. If the q-th phase is shifted by $2\pi q/3$, the discrete phase sum for temporal harmonic n and spatial azimuthal mode m is

$$T^{(3\phi)}_{n,m} = \frac{1}{3}\left|\sum_{q=0}^{2} \exp!\left[iq\frac{2\pi}{3}(m-n)\right]\right|. \tag{250}$$

The exact selection rule is

$$\boxed{T^{(3\phi)}_{n,m} = \begin{cases} 1, & m \equiv n \pmod 3, \\ 0, & m \not\equiv n \pmod 3. \end{cases}} \tag{251}$$

Thus a three-phase winding does not merely "add power"; it filters the allowed temporal–spatial mode pairs.

K.5 Double-stack phase-shift filter

Let the coil be built as two vertically separated stacks, with the second stack rotated azimuthally by

$$\delta = 30^\circ = \frac{\pi}{6}. \tag{252}$$

Let ψ denote the electrical inter-stack phase offset. The two-stack spatial filter is

$$L_m(\delta, \psi) = |1 + \exp[i(m\delta + \psi)]|\,. \tag{253}$$

For equal stack polarity and zero additional electrical offset ($\psi = 0$), the destructive condition is

$$m\delta = (2q + 1)\pi, \qquad q \in \mathbb{Z}. \tag{254}$$

With $\delta = \pi/6$, this gives

$$\boxed{m = 6, 18, 30, \ldots} \tag{255}$$

Therefore a 30° double-stack geometry exactly suppresses the sixth azimuthal spatial harmonic and its odd multiples in this idealized equal-amplitude model.

K.6 Golden-duty PWM spectrum

Let the duty cycle be the inverse golden-square fraction

$$d = \varphi^{-2} = \left(\frac{1+\sqrt{5}}{2}\right)^{-2} = 0.381966011\ldots. \tag{256}$$

For an ideal unipolar rectangular pulse train, the magnitude of the n-th Fourier coefficient, ignoring the DC normalization convention, is proportional to

$$P_n(d) = \left| \frac{\sin(\pi n d)}{\pi n} \right|. \tag{257}$$

For $d = \varphi^{-2}$, the first seven relative amplitudes normalized to P_1 are approximately

n	$P_n(d)$	$P_n(d)/P_1(d)$
1	0.296675	1.000
2	0.107508	0.362
3	0.046948	0.158
4	0.079273	0.267
5	0.017794	0.060
6	0.042102	0.142
7	0.038864	0.131

Thus golden-duty PWM strongly suppresses the fifth harmonic, but does not by itself null the sixth and seventh harmonics. In the proposed three-phase double-stack geometry, the suppression of the fifth, sixth, and seventh bands must be understood as a composite filter effect:

$$\boxed{\mathcal{C}_{n,m} = P_n(d)\, T^{(3\phi)}_{n,m}\, S_m(\mathrm{P40}, +11, -9)\, L_m(\delta, \psi)\, G_m(k_n R_{\mathrm{coil}})}. \tag{258}$$

Here:

- $P_n(d)$ is the PWM harmonic amplitude;
- $T^{(3\phi)}_{n,m}$ is the three-phase selection factor;
- $S_m(\mathrm{P40}, +11, -9)$ is the spatial winding factor of the SawShape path;
- $L_m(\delta, \psi)$ is the double-stack filter;
- $G_m(k_n R_{\mathrm{coil}})$ is the geometric near-field coupling factor.

The wave number of the n-th temporal harmonic is

$$k_n = \frac{2\pi n f_0}{\|\mathbf{v}_{\circlearrowleft}\|}. \tag{259}$$

K.7 SawShape winding factor for $\mathrm{P40}, +11, -9$

For a discrete winding on $N = 40$ angular sites, define the node angle

$$\theta_j = \frac{2\pi p_j}{40}, \qquad p_j \in \mathbb{Z}/40\mathbb{Z}. \tag{260}$$

A SawShape path with alternating skip sequence $(+11, -9)$ can be represented by

$$p_{j+1} = p_j + s_j \pmod{40}, \qquad s_j \in \{+11, -9\}. \tag{261}$$

The normalized spatial Fourier factor is then

$$\boxed{S_m(\mathrm{P40}, +11, -9) = \left| \frac{1}{N_s} \sum_{j=0}^{N_s - 1} w_j \exp(-i m \theta_j) \right|}. \tag{262}$$

Here w_j encodes segment weight, orientation, current direction, layer index, and local segment length. This factor is experimentally accessible: it can be inferred by measuring the spatial FFT of the near-field magnetic map over a circular sampling path.

K.8 Canonical interpretation

The proposed coil does not directly generate a canonical force. Instead, it prepares a structured, rotating, mode-filtered boundary condition for the local field. In SST language, the coil acts as a laboratory-scale swirl-mode injector.

The relevant energy-density comparison is

$$\rho_E = \frac{1}{2}\rho_f \|\mathbf{v}_{\circlearrowleft}\|^2. \tag{263}$$

Using

$$\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}, \qquad \|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6\,\mathrm{m\,s^{-1}},$$

one obtains

$$\rho_E = \frac{1}{2}(7.0 \times 10^{-7})(1.09384563 \times 10^6)^2 = 4.187 \times 10^5\,\mathrm{J\,m^{-3}}. \tag{264}$$

The corresponding mass-equivalent density is

$$\rho_m = \frac{\rho_E}{c^2} = \frac{4.187 \times 10^5}{(2.99792458 \times 10^8)^2} = 4.659 \times 10^{-12}\,\mathrm{kg\,m^{-3}}. \tag{265}$$

These values should be understood as canonical scale references, not as a claim that a laboratory coil reaches the full canonical swirl speed in ordinary electromagnetic matter.

K.9 Experimental falsification protocol

A coil experiment is SST-relevant only if it passes the following null hierarchy.

1. **Electrical null:** repeat with identical RMS current and power but randomized phase ordering.
2. **Geometric null:** replace SawShape $(\mathrm{P}40, +11, -9)$ by a symmetry-preserving circular winding with the same resistance and inductance.
3. **Duty-cycle null:** compare $d = 0.381966$ against $d = 0.500$, $d = 0.333$, and $d = 0.250$ at fixed RMS current.
4. **Stack null:** compare 30° double-stack shift against 0°, 15°, 45°, and 60°.
5. **Mode-lock scan:** sweep f_0 and test whether anomalous coherence peaks occur at

$$f_0^{(n,m)} = \frac{|m|\,\|\mathbf{v}_{\circlearrowleft}\|}{2\pi n R_{\mathrm{coil}}}.$$

6. **Environmental null:** repeat in still air, shielded enclosure, and, where safely possible, reduced pressure to remove ion wind and acoustic coupling.
7. **Force-decomposition null:** decompose any measured force into electromagnetic, thermal, acoustic, EHD, and mechanical vibration contributions before assigning a residual.

Only a residual signal satisfying both

$$\eta_{n,m} \approx 1 \tag{266}$$

and

$$\mathcal{C}_{n,m} \neq 0 \tag{267}$$

is admissible as an SST research-track candidate.

K.10 Vorticity-force decomposition layer

Any measured force must first be written schematically as

$$\mathbf{F}_{\text{meas}} = \mathbf{F}_{\text{EM}} + \mathbf{F}_{\text{EHD}} + \mathbf{F}_{\text{thermal}} + \mathbf{F}_{\text{acoustic}} + \mathbf{F}_{\text{mechanical}} + \mathbf{F}_{\omega} + \mathbf{F}_{\text{res}}. \tag{268}$$

Here $\mathbf{F}_{\omega}$ denotes force reconstructed from ordinary vorticity transport and impulse balance. Only the remaining term $\mathbf{F}_{\text{res}}$, after conventional reconstruction and uncertainty propagation, can be tested against the SST mode-locking rule. The canonical research criterion is therefore

$$\boxed{\mathbf{F}_{\text{res}} = \mathbf{F}_{\text{res}}(\eta_{n,m}, \mathcal{C}_{n,m}) \quad \text{with a peak near} \quad \eta_{n,m} = 1.} \tag{269}$$

K.11 Recommended measured observables

The minimum observable set is:

1. drive current $I_A(t), I_B(t), I_C(t)$;
2. voltage $V_A(t), V_B(t), V_C(t)$;
3. real input power $P_{\text{in}}(t)$;
4. local magnetic field map $\mathbf{B}(\mathbf{x}, t)$;
5. pickup-coil FFT around the active circumference;
6. accelerometer spectrum on the coil frame;
7. air temperature and pressure field if operated in air;
8. force sensor output with synchronized phase reference.

A positive SST candidate requires a phase-stable spectral feature at a predicted (n, m) lock that disappears under at least one of the symmetry-breaking null tests.

K.12 Conclusion

The 3-phase SawShape $(\text{P}40, +11, -9)$ double-stack coil with 30° inter-stack rotation and $d = \varphi^{-2}$ PWM is canonically useful as a mode-filtered swirl-excitation apparatus. Its SST value is not that it assumes a new force, but that it makes the frequency–length criterion

$$\|\mathbf{v}_{\circlearrowleft}\| = f\Delta x$$

experimentally precise.

The essential design rule is

$$\eta_{n,m} = \frac{nf_0}{\|\mathbf{v}_{\circlearrowleft}\|} \frac{2\pi R_{\text{coil}}}{|m|} \approx 1,$$

with observable coupling only when

$$\mathcal{C}_{n,m} = P_n(d)\, T^{(3\phi)}_{n,m}\, S_m(\text{P}40, +11, -9)\, L_m(\delta, \psi)\, G_m(k_n R_{\text{coil}})$$

is nonzero. This converts the coil concept into a falsifiable SST metrology protocol.

L Research Appendix: Non-Abelian Topological Charge for Protected Swirl Strings

Canon status. [SPECULATIVE] Research Track. This appendix does not promote a new particle assignment to canon. It records a canon-compatible extension of the SST topology sector: knot type and helicity may be insufficient to determine dynamical stability when reconnection and strand-crossing moves are allowed. A non-Abelian coloring invariant may supply an additional obstruction to decay.

L.1 Motivation

The current SST knot sector uses geometric and topological descriptors such as crossing number, ropelength, helicity, chirality, and hyperbolic-volume proxies. A minimal effective form is

$$E_{\text{eff}}[K] = \alpha C(K) + \beta L(K) + \gamma \mathcal{H}(K), \tag{270}$$

where $C(K)$ is a crossing-complexity proxy, $L(K)$ is a ropelength or length proxy, and $\mathcal{H}(K)$ denotes helicity or helicity-derived topological data.

Recent work on non-Abelian topological defects shows that some linked structures cannot be removed by local reconnections and strand crossings because the strands carry non-commuting elements of an order-parameter fundamental group. This suggests that SST should distinguish two layers:

1. the geometric knot type K of the swirl string centerline;
2. an internal topological coloring q valued in a discrete or continuous group G.

L.2 Group-colored swirl-string sector

Let a colored swirl-string configuration be denoted

$$\mathcal{K}_G = (K, q), \tag{271}$$

where

$$q : \pi_1\Big(\mathbb{R}^3 \setminus K\Big) \longrightarrow G \tag{272}$$

assigns a group element to each Wirtinger generator of the complement. For a diagram with arc generators a_i, the coloring must satisfy the usual crossing consistency relations. In SST language, $q(a_i)$ labels an internal director or phase-sector attached to a local swirl-string strand.

[ORTHODOX] Constraint. For a non-Abelian group G, strand exchange is obstructed when the corresponding labels do not commute:

$$[q(a_i), q(a_j)] = q(a_i)q(a_j)q(a_i)^{-1}q(a_j)^{-1} eqe_G. \tag{273}$$

[SPECULATIVE] SST interpretation. Equation (273) is interpreted as a local reconnection barrier in the swirl medium. If two strands belong to non-commuting internal sectors, a direct crossing or reconnection requires an additional defect cord or phase bridge. In SST terms this is a candidate mechanism for topological protection beyond ordinary helicity conservation.

L.3 Protected-sector invariant

Define a protection marker

$$\mathcal{P}_G(\mathcal{K}_G) = \begin{cases} 1, & \text{if no allowed local reconnection/crossing sequence reduces } \mathcal{K}_G \text{ to the trivial sector,} \\ 0, & \text{otherwise.} \end{cases} \tag{274}$$

The marker $\mathcal{P}_G$ is not an energy term by itself. It is a sector label. Energetically, it enters as a barrier rule:

$$\Delta E_{\rm rec}(\mathcal{K}_G \to \mathcal{K}'_G) = \begin{cases} < \infty, & Q_G(\mathcal{K}_G) = Q_G(\mathcal{K}'_G), \\ \infty \text{ or macroscopically large}, & Q_G(\mathcal{K}_G) \neq Q_G(\mathcal{K}'_G), \end{cases} \tag{275}$$

where Q_G is a conserved coloring invariant under the permitted move set.

A useful research-track extension of Eq. (270) is therefore

$$E_{\rm eff}^{(G)}[\mathcal{K}_G] = \alpha C(K) + \beta L(K) + \gamma \mathcal{H}(K) + \Delta_G \, \mathcal{P}_G(\mathcal{K}_G), \tag{276}$$

where Δ_G is not a fitted mass term but a symbolic protection scale measuring the minimum energy required to violate or bypass the coloring constraint.

L.4 Dimensional check

The quantities $C(K)$, $\mathcal{H}(K)$ after normalization, Q_G, and $\mathcal{P}_G$ are dimensionless. Therefore α, γ, and Δ_G must carry units of energy if Eq. (276) is used as an energy functional. If instead $E_{\rm eff}$ is a dimensionless scoring functional, all coefficients are dimensionless and must later be multiplied by an SST energy scale such as $\rho_E V_K$.

L.5 Minimal falsifier

If two SST candidate particles have the same $(C, L, \mathcal{H})$ but different observed stability, the model predicts that a missing internal sector label exists:

$$(C_1, L_1, \mathcal{H}_1) = (C_2, L_2, \mathcal{H}_2), \qquad \tau_1 \neq \tau_2 \quad \Rightarrow \quad Q_G(\mathcal{K}_1) \neq Q_G(\mathcal{K}_2). \tag{277}$$

Failure to find such a sector in simulation or experiment would demote this appendix from Research Track to rejected model branch.

M Research Appendix: Reconnection Decay as a Stability Falsifier

Canon status. [ORTHODOX] Constraint + [SPECULATIVE] SST interpretation. This appendix records a negative constraint on SST. A single-component Gross–Pitaevskii-like medium with permitted reconnections does not automatically protect knotted filaments. Therefore, SST particle stability requires additional structure beyond generic phase-defect topology.

M.1 Baseline statement

In a dissipative or reconnection-permitting medium, knotted filament topology can decay. The relevant orthodox lesson is not that SST is false, but that topology alone is insufficient unless the move set is restricted by the dynamics.

Let $K(t)$ denote the topology of a swirl-string centerline. A reconnection path is a sequence

$$K_0 \longrightarrow K_1 \longrightarrow \cdots \longrightarrow K_N, \tag{278}$$

where each arrow is a local reconnection or strand-crossing event. If K_N is an unlink or collection of unknotted loops, then K_0 is dynamically unprotected under that move set.

M.2 SST stability criterion

SST should not claim that every closed filament is stable. A canon-compatible stability criterion is

$$\tau_{\text{life}}(K) \sim \tau_0 \exp\left(\frac{\Delta E_{\text{rec}}(K)}{E_{\text{drive}}}\right), \tag{279}$$

where $\Delta E_{\text{rec}}(K)$ is the lowest reconnection barrier out of the topological sector, and E_{drive} is the effective environmental or internal driving energy available to trigger reconnection.

The stable-particle condition is therefore not merely

$$K \neq \bigcirc, \tag{280}$$

but rather

$$\Delta E_{\text{rec}}(K) \gg E_{\text{drive}} \quad \text{and/or} \quad Q_G(K) \neq Q_G(\bigcirc). \tag{281}$$

M.3 Relation to Chronos–Kelvin transport

For ideal SST evolution without reconnection or external source, the Chronos–Kelvin invariant remains

$$\frac{D}{Dt}\left(R^2\omega\right) = 0. \tag{282}$$

Reconnection is precisely a non-ideal event where this transport law is locally invalid or discontinuous. Thus a κ-type transition in the canon may be interpreted as the event layer at which topology can change.

[SPECULATIVE] Identification. A reconnection event is an SST Kairos event when it changes the topological sector:

$$\kappa: \quad K_i \to K_{i+1}, \qquad Q_G(K_i) \neq Q_G(K_{i+1}) \quad \text{or} \quad \mathcal{H}(K_i) \neq \mathcal{H}(K_{i+1}). \tag{283}$$

M.4 Dimensionless diagnostic

Define the stability ratio

$$\Xi_{\text{rec}}(K) = \frac{\Delta E_{\text{rec}}(K)}{E_{\text{tube}}(K)}. \tag{284}$$

For a slender swirl tube, the standard leading envelope energy scale is

$$E_{\text{tube}}(K) \simeq \frac{\rho_f \Gamma^2}{4\pi}\, \ell_K \ln\left(\frac{R}{r_c}\right), \tag{285}$$

where Γ is circulation, ℓ_K is centerline length, and R is the outer cutoff. The units check:

$$[\rho_f \Gamma^2 \ell_K] = \frac{\text{kg}}{\text{m}^3}\frac{\text{m}^4}{\text{s}^2}\text{m} = \text{kg}\,\text{m}^2\text{s}^{-2} = \text{J}. \tag{286}$$

A practical SST classification rule is

$$\Xi_{\text{rec}}(K) \lesssim 1 \Rightarrow \text{hydrodynamically fragile}, \qquad \Xi_{\text{rec}}(K) \gg 1 \Rightarrow \text{candidate stable T-phase state}. \tag{287}$$

M.5 Minimal simulation protocol

A candidate SST knot assignment should be tested under three levels of dynamics:

1. single-component phase-defect evolution;
2. constrained incompressible evolution without reconnection;
3. multi-director or non-Abelian colored evolution with reconnection rules.

Only states stable under the third protocol should be promoted as particle candidates.

N Research Appendix: Fractional Torus-Phase Winding and Swirl-Clock Phase

Canon status. [SPECULATIVE] Research Track. This appendix records a bridge between coordinated rotational phase mechanics and SST R-phase torsional packets. It does not assert that optical polarization knots are literal SST particles. It supplies a useful mathematical template for phase winding, internal rotation, and torus-knot closure.

N.1 Coordinated phase generator

Let θ be an external azimuthal coordinate and let ϕ be an internal director angle. A coordinated rotation acts as

$$\theta \mapsto \theta + \alpha, \qquad \phi \mapsto \phi + \gamma\alpha. \tag{288}$$

The generator has the schematic form

$$J_\gamma = L_{\mathrm{orb}} + \gamma S_{\mathrm{int}}, \tag{289}$$

where L_{orb} shifts external azimuthal phase and S_{int} shifts internal swirl-director phase.

For two commensurate internal frequencies $p\Omega$ and $q\Omega$ carrying azimuthal mode numbers m_1 and m_2, the coordinated-winding parameter is

$$\gamma = \frac{m_2 p - m_1 q}{p + q}. \tag{290}$$

The corresponding torus-winding pair is

$$(m, n) = \left(m_2 p - m_1 q,\ p + q\right). \tag{291}$$

If $\gcd(m, n) = 1$, the internal lobe trajectory is a single (m, n) torus knot. If $d = \gcd(m, n) > 1$, the trajectory splits into d linked components.

N.2 SST phase-action translation

To avoid notation conflict with the Swirl-Clock factor $S^{\circlearrowleft}_{(t)}$, define a phase-action field $\mathscr{S}(\vec{x}, t)$. In an R-phase torsional packet, write the reduced phase on a toroidal carrier as

$$\Theta(\theta, \phi, t) = n\phi - m\theta - \Omega t. \tag{292}$$

The closure condition is

$$\oint d\Theta = 2\pi N, \qquad N \in \mathbb{Z}. \tag{293}$$

The phase-action field may be represented locally as

$$\mathscr{S}(\vec{x}, t) = \hbar_{\mathrm{SST}}\, \Theta(\vec{x}, t), \tag{294}$$

so that the circulation quantization condition becomes

$$\oint \nabla \mathscr{S} \cdot d\vec{\ell} = 2\pi N \hbar_{\mathrm{SST}}. \tag{295}$$

N.3 Connection to Swirl-Clock modulation

The local Swirl-Clock factor remains the kinematic clock factor

$$S_{(t)}^{\circlearrowleft} = \sqrt{1 - \frac{\|\mathbf{v}_{\circlearrowleft}\|^2}{c^2}}. \tag{296}$$

The phase-action field controls wave-like periodicity, while $S_{(t)}^{\circlearrowleft}$ controls local clock rate. The research-track bridge is that a stable R-phase packet may require simultaneous closure of both:

$$\oint \nabla \mathscr{S} \cdot d\vec{\ell} = 2\pi N \hbar_{\mathrm{SST}}, \tag{297}$$

$$\frac{D}{Dt}\left(R^2\omega\right) = 0. \tag{298}$$

[SPECULATIVE] Interpretation. Equation (290) suggests a route to fractional internal winding without fractional circulation. The circulation quantum remains integer-valued, but the internal phase orientation can return only after multiple external turns. This may be useful for modeling R-phase torsional photons, spin-orbit-like packet structure, or generation-family phase layering.

N.4 Dimensional check

The variables m, n, p, q, N and γ are dimensionless. The phase Θ is dimensionless. Therefore $\mathscr{S} = \hbar_{\mathrm{SST}}\Theta$ has units of action, and Eq. (295) has units of action because $\nabla\mathscr{S} \cdot d\vec{\ell}$ has units $(\mathrm{J\,s\,m^{-1}})\mathrm{m} = \mathrm{J\,s}$.

O Research Appendix: Topological Fluid Mechanics and the SST Energy Functional

Canon status. [ORTHODOX] Mathematical background + [SPECULATIVE] SST synthesis. This appendix records the topological-fluid mechanics background behind the SST knot-energy program. The orthodox part is that ideal incompressible flow preserves topology and helicity under the usual assumptions. The speculative part is the identification of selected relaxed knot classes with particle sectors.

O.1 Frozen-field topology

In an ideal incompressible medium, topology is transported by the flow map. For SST this supports the use of closed swirl strings as dynamically persistent structures under non-reconnecting evolution.

The primary invariants are

$$\Gamma = \oint_C \mathbf{v}_{\circlearrowleft} \cdot d\vec{\ell}, \tag{299}$$

$$\mathcal{H} = \int_V \mathbf{v}_{\circlearrowleft} \cdot \vec{\omega}\, d^3x, \tag{300}$$

$$\vec{\omega} = \nabla \times \mathbf{v}_{\circlearrowleft}. \tag{301}$$

For linked tubes with circulations Γ_i, the leading topological helicity decomposition is

$$\mathcal{H} \simeq \sum_i \Gamma_i^2\, SL_i + 2\sum_{i<j} Lk_{ij}\Gamma_i\Gamma_j, \tag{302}$$

where SL_i is self-linking and Lk_{ij} is Gauss linking number.

O.2 Energy relaxation under topological constraints

For a slender swirl tube, the leading hydrodynamic energy scale is

$$E_{\rm env} \simeq \frac{\rho_f \Gamma^2}{4\pi}\,\ell_K \ln\!\left(\frac{R}{r_c}\right), \tag{303}$$

where ℓ_K is the tube centerline length. The core contribution can be represented by a line-tension term

$$E_{\rm core} \simeq T_{\rm core}\ell_K. \tag{304}$$

Near contacts and curvature corrections may be encoded by

$$E_{\rm geom}[K] = T_{\rm core}\ell_K + \frac{\rho_f \Gamma^2}{4\pi}\ell_K \ln\!\left(\frac{R}{r_c}\right) + \beta_\kappa \int_K \kappa_s^2\, ds + \beta_{\rm nc}\, C_{\rm near}(K). \tag{305}$$

Here κ_s is centerline curvature and $C_{\rm near}$ is a near-contact or crossing-complexity proxy.

O.3 Reduction to the SST normal form

After nondimensionalization by a reference energy E_0, Eq. (305) reduces to the canon-style scoring form

$$\frac{E_{\rm geom}[K]}{E_0} \rightsquigarrow \alpha C(K) + \beta L(K) + \gamma \mathcal{H}(K) + \cdots. \tag{306}$$

This justifies the SST normal form as a coarse-grained energy-ordering tool, not as a finished mass theorem.

O.4 Canon discipline

The functional in Eq. (306) should be used as follows:

1. **[ORTHODOX]** Use Γ, $\mathcal{H}$, Lk, writhe, twist, and ropelength as established geometric/topological descriptors.
2. **[DERIVED]** Within a chosen tube model, derive the units and leading energy scaling from ρ_f, Γ, ℓ_K, and r_c.
3. **[SPECULATIVE]** Only after this, map a relaxed topological class to an SST particle sector.

O.5 Minimal falsifier

If two distinct knots have the same SST particle assignment but relax to incompatible energy minima or different protected sectors under the same admissible dynamics, the assignment must be split or rejected:

$$K_1 \sim_{\rm SST} K_2 \quad \text{but} \quad \min E[K_1] \neq \min E[K_2] \quad \Rightarrow \quad \text{taxonomy refinement required.} \tag{307}$$

References

[1] T. Annala, R. Zamora-Zamora, and M. Möttönen, "Topologically protected vortex knots and links," *Communications Physics* **5**, 309 (2022). DOI: 10.1038/s42005-022-01071-2.

[2] D. Kleckner, L. H. Kauffman, and W. T. M. Irvine, "How superfluid vortex knots untie," *Nature Physics* **12**, 650–655 (2016). DOI: 10.1038/nphys3679.

[3] E. Pisanty, G. Jiménez, V. Vicuña-Hernández, A. Picón, A. Celi, J. P. Torres, and M. Lewenstein, "Knotting fractional-order knots with the polarization state of light," *arXiv:1808.05193* (2018). Permalink: arXiv:1808.05193.

[4] R. L. Ricca, "Applications of knot theory in fluid mechanics," *Banach Center Publications* **42**, 321–346 (1998). Permalink: EuDML:252224.

[5] H. K. Moffatt, "The degree of knottedness of tangled vortex lines," *Journal of Fluid Mechanics* **35**, 117–129 (1969). DOI: 10.1017/S0022112069000991.

[6] V. I. Arnold and B. A. Khesin, *Topological Methods in Hydrodynamics*, Springer (1998). DOI: 10.1007/978-1-4612-0645-3.

P Resonance Selection Principle

Status: [RESEARCH-TRACK; candidate Canon principle].

A swirl-string state K is assigned a causal response function $\chi_K(\omega)$. An external perturbation $F(t)$, with Fourier spectrum $\widetilde{F}(\omega)$, excites the state K with amplitude

$$\mathcal{A}_K[F] = \int_0^\infty \widetilde{F}(\omega)\,\chi_K(\omega)\,d\omega. \tag{308}$$

The excitation probability is

$$P_K[F] = |\mathcal{A}_K[F]|^2. \tag{309}$$

For isolated narrow resonances, the response function may be written as

$$\chi_K(\omega) = \sum_n \frac{B_{Kn}\gamma_{Kn}}{\omega - \omega_{Kn} + i\gamma_{Kn}}, \tag{310}$$

where ω_{Kn} are internal swirl-string eigenfrequencies, γ_{Kn} are linewidths, and B_{Kn} are coupling weights.

The natural SST frequency scale is

$$\Omega_{\mathrm{SST}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}, \tag{311}$$

so that internal modes may be parametrized by

$$\omega_{Kn} = \eta_{Kn}\Omega_{\mathrm{SST}}, \tag{312}$$

with dimensionless mode factors η_{Kn}.

This principle unifies delay-loop selection, Swirl-Clock phase-locking, mode-locking, and knot-state excitation as instances of one spectral overlap law.

Q Resonance Selection Principle for Swirl–String Excitation

Status and purpose

Status: [RESEARCH-TRACK]. This appendix promotes the frequency-overlap idea from a specific beam–swirl model to a general SST excitation principle. It is not a proof of nuclear activation or fusion enhancement. It is a controlled response ansatz compatible with the Canon-v0.8.1 language of swirl strings, mode selection, circulation, helicity, and Kelvin-compatible dynamics.

Claim

A swirl-string state K is excited most efficiently when the spectrum of an external perturbation overlaps the internal response spectrum of K. Define a normalized external spectral density

$$p_{\text{ext}}(\omega) \geq 0, \qquad \int_0^\infty p_{\text{ext}}(\omega)\, d\omega = 1, \tag{313}$$

where p_{ext} has units of seconds, because $d\omega$ has units s^{-1}. Let the dimensionless response function of the swirl-string state be

$$R_K(\omega) = \sum_n B_{Kn} \frac{\gamma_{Kn}^2}{(\omega - \omega_{Kn})^2 + \gamma_{Kn}^2}, \tag{314}$$

where:

- ω_{Kn} is the n-th internal mode frequency,
- γ_{Kn} is the corresponding linewidth,
- B_{Kn} is a dimensionless coupling weight.

The dimensionless excitation yield is then

$$\boxed{\mathcal{Y}_K = \int_0^\infty p_{\text{ext}}(\omega)\, R_K(\omega)\, d\omega} \qquad \text{[RESEARCH-TRACK]}. \tag{315}$$

Dimensional check

Since

$$[p_{\text{ext}}] = \mathrm{s}, \qquad [d\omega] = \mathrm{s}^{-1}, \qquad [R_K] = 1,$$

we obtain

$$[\mathcal{Y}_K] = \mathrm{s} \cdot 1 \cdot \mathrm{s}^{-1} = 1.$$

Thus $\mathcal{Y}_K$ is a dimensionless excitation efficiency.

Canonical scale

The natural SST angular-frequency scale is

$$\Omega_{\text{SST}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}. \tag{316}$$

Using the Canon values

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6\ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\ \mathrm{m},$$

gives

$$\Omega_{\text{SST}} = 7.76344066 \times 10^{20}\ \mathrm{s}^{-1}. \tag{317}$$

The corresponding cycle period is

$$T_{\text{SST}} = \frac{2\pi}{\Omega_{\text{SST}}} = 8.09329985 \times 10^{-21}\ \mathrm{s}. \tag{318}$$

Interpretation

Equation (315) does not assert that all nuclear, atomic, or particle transitions are already derived in SST. It states a weaker and safer rule:

> External excitation is strongest when the perturbing spectrum overlaps an allowed internal swirl-string response spectrum.

This principle can later be specialized to:

- radiative transitions,
- qubit-like R/T phase transitions,
- knot-mode excitation,
- delayed mode selection,
- controlled response experiments in optical, plasma, or condensed-matter analogues.

Falsifiability

For a fixed target state K, SST predicts that changing the spectral center ω_0 of a narrow-band perturbation should change the activation yield according to $R_K(\omega_0)$, up to normalization and detector response. If no resonance-like dependence appears after controlling for beam power, fluence, and thermal effects, then Eq. (315) is falsified as a useful excitation law.

R Time-Domain Kernel Form of Swirl–String Response

Status and purpose

Status: [RESEARCH-TRACK]. This appendix converts the frequency-domain selection rule into a time-domain response law. It is intended to connect SST mode selection to pulsed experiments, delay-loop systems, and Swirl-Clock phase locking.

Pulse-response formulation

Let $f(t)$ be a real external driving pulse. A Gaussian-windowed monochromatic pulse may be written as

$$f(t) = f_0 \exp\left(-\frac{t^2}{2\tau_p^2}\right) \cos(\omega_0 t + \phi_0), \tag{319}$$

where τ_p is the pulse duration and ω_0 is the carrier angular frequency.

Let $k_K(t)$ be the causal impulse-response kernel of the swirl-string state K, with

$$k_K(t < 0) = 0. \tag{320}$$

The induced response is

$$\boxed{s_K(t) = \int_{-\infty}^{t} f(t')\, k_K(t - t')\, dt'} \qquad \text{[ORTHODOX form; RESEARCH-TRACK SST interpretation].} \tag{321}$$

Frequency-domain equivalence

With the Fourier convention

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{i\omega t}\,dt,$$

the convolution theorem gives

$$\hat{s}_K(\omega) = \hat{f}(\omega)\,\chi_K(\omega), \tag{322}$$

where $\chi_K(\omega)$ is the complex response function of the swirl-string state. The measurable spectral response is proportional to

$$R_K(\omega) \propto |\chi_K(\omega)|^2. \tag{323}$$

Bandwidth–duration relation

A Gaussian pulse obeys the scaling

$$\Delta\omega\,\tau_p \sim 1. \tag{324}$$

Therefore:

- short pulses have broad spectra and can excite many nearby modes;
- long pulses have narrow spectra and selectively excite one mode;
- coherent excitation requires phase stability over at least one internal response time.

For the canonical SST scale,

$$T_{\mathrm{SST}} = 8.09329985 \times 10^{-21}\ \mathrm{s}.$$

Thus, a pulse with duration $\tau_p \gg T_{\mathrm{SST}}$ is narrow relative to the core-scale carrier, whereas $\tau_p \lesssim T_{\mathrm{SST}}$ is broadband at the canonical core scale.

Connection to Swirl Clock

If the internal phase of a swirl string is denoted by $\theta_K(t)$, then a resonant perturbation can be represented as a phase-locking term

$$\frac{d\theta_K}{dt} = \omega_K + \epsilon\, f(t)\sin(\omega_0 t - \theta_K), \tag{325}$$

where ϵ is a coupling parameter. Phase locking occurs when

$$|\omega_0 - \omega_K| \lesssim \epsilon |f_0|. \tag{326}$$

This gives a direct bridge between frequency-selective excitation and Swirl-Clock synchronization.

Falsifiability

For a given K, changing only the pulse duration τ_p while keeping total pulse energy fixed should alter the response through the bandwidth $\Delta\omega$. A null result after controlling for heating and total fluence would falsify this time-domain response model.

S Swirl Impedance and Mode Linewidth

Status and critical correction

Status: [RESEARCH-TRACK]. The earlier expression

$$Z_{\text{eff}} \sim \frac{P_{\text{core}} r_c}{\|\mathbf{v}_{\circlearrowleft}\| \hbar}$$

should *not* be canonised as written. Its dimensions are

$$\left[\frac{P_{\text{core}} r_c}{\|\mathbf{v}_{\circlearrowleft}\| \hbar}\right] = \mathrm{m}^{-3},$$

so it is not an impedance and not dimensionless.

A Canon-compatible replacement is to define either:

1. a dimensionless rigidity ratio,
2. or an acoustic-like pressure-to-speed impedance.

Dimensionless swirl-rigidity ratio

Let V_K be the effective active volume of a swirl-string configuration and P_K the pressure scale associated with its deformation. Define

$$\boxed{\Pi_K(\omega) = \frac{P_K V_K}{\hbar\omega}} \qquad \text{[RESEARCH-TRACK]}. \tag{327}$$

This compares stored mechanical energy $P_K V_K$ to the quantum energy scale $\hbar\omega$. It is dimensionless:

$$[P_K V_K] = \mathrm{J}, \qquad [\hbar\omega] = \mathrm{J}.$$

Acoustic-like swirl impedance

An alternative intensive impedance is

$$\boxed{Z_{\text{swirl}} = \frac{P_K}{v_K}} \tag{328}$$

with units

$$[Z_{\text{swirl}}] = \frac{\mathrm{kg\,m^{-1}\,s^{-2}}}{\mathrm{m\,s^{-1}}} = \mathrm{kg\,m^{-2}\,s^{-1}},$$

the same units as acoustic impedance. This object measures resistance to velocity-driven pressure excitation.

Connection to linewidth and quality factor

A resonant mode with angular frequency ω_K and quality factor Q_K has linewidth

$$\gamma_K = \frac{\omega_K}{2Q_K}. \tag{329}$$

A minimal phenomenological relation is

$$Q_K = Q_0\, \Pi_K^{\eta}, \qquad \eta > 0, \tag{330}$$

so that larger stored-energy rigidity produces narrower response lines.

Numerical check using canonical scales

Using

$$F_{\rm swirl}^{\max} = 29.053507 \text{ N}, \qquad r_c = 1.40897017 \times 10^{-15} \text{ m},$$

define the core pressure scale

$$P_{\max} = \frac{F_{\rm swirl}^{\max}}{\pi {r_c}^2} = 4.65848920 \times 10^{30} \text{ Pa}. \tag{331}$$

For a horn-torus active volume

$$V_{\rm horn} = 2\pi^2 {r_c}^3 = 5.52122107 \times 10^{-44} \text{ m}^3, \tag{332}$$

the stored pressure energy is

$$P_{\max} V_{\rm horn} = 2.57205487 \times 10^{-13} \text{ J}. \tag{333}$$

At the canonical angular frequency

$$\Omega_{\rm SST} = 7.76344066 \times 10^{20} \text{ s}^{-1},$$

we have

$$\hbar\Omega_{\rm SST} = 8.18710572 \times 10^{-14} \text{ J} = 510998.946 \text{ eV}. \tag{334}$$

Therefore,

$$\Pi_{\rm horn} = \frac{P_{\max} V_{\rm horn}}{\hbar\Omega_{\rm SST}} = 3.14159235 \approx \pi. \tag{335}$$

Interpretation

Equation (335) is not yet a derivation of a particle mass or coupling constant. It is a useful dimensional coincidence produced by canonical SST scales and the horn-torus active-volume choice. It should be labelled:

$$\boxed{\text{[CALIBRATED / RESEARCH-TRACK NUMERICAL STRUCTURE]}}$$

until derived from the SST action or from a topological minimization principle.

Falsifiability

If Π_K controls linewidth, then families of analog swirl-string excitations should satisfy

$$\gamma_K \propto \omega_K \Pi_K^{-\eta}.$$

A measured absence of correlation between active-volume rigidity and linewidth would falsify Eq. (330).

T Layered Relaxation and Multi-Mode Decay

Status and warning

Status: [RESEARCH-TRACK]. This appendix preserves the useful part of the delayed-decay idea: a multi-layer swirl-string system may relax through multiple modes. It explicitly rejects the naive identification

$$\tau_i = \frac{r_i}{\|\mathbf{v}_{\circlearrowleft}\|}$$

with observed delayed-neutron group lifetimes. For nuclear-scale radii, this gives $\tau_i \sim 10^{-21}$ s, whereas delayed-neutron groups are macroscopic nuclear decay times. Therefore direct radial-transit mapping is dimensionally possible but physically wrong in scale.

General multi-mode relaxation

If a perturbed swirl-string state has several metastable channels, its observable relaxation can be expanded as

$$\boxed{A(t) = \sum_{i=1}^{N} A_i e^{-t/\tau_i} \cos(\omega_i t + \phi_i)} \qquad \text{[ORTHODOX form; RESEARCH-TRACK SST interpretation].} \tag{336}$$

The non-oscillatory limit is

$$A(t) = \sum_{i=1}^{N} A_i e^{-t/\tau_i}. \tag{337}$$

Pole interpretation

Equation (336) corresponds to a response function with poles

$$\omega = \omega_i - \frac{i}{\tau_i}. \tag{338}$$

Thus, each decay time τ_i is a lifetime of a response pole, not necessarily a physical radial crossing time.

Allowed SST interpretation

In SST, a layered or nested swirl-string configuration may contain:

- a core-scale response,
- a sheath-scale response,
- a linking/helicity response,
- a global Swirl-Clock phase response.

The decay constants τ_i should therefore be treated as *effective metastable relaxation times*. A safe parametrization is

$$\tau_i = \frac{Q_i}{\omega_i}, \tag{339}$$

where Q_i is the quality factor of the i-th mode.

Critical scale check

For the canonical circulation scale,

$$\tau_c = \frac{r_c}{\|\mathbf{v}_{\circlearrowleft}\|} = 1.28808868 \times 10^{-21}\ \text{s}. \tag{340}$$

Therefore, an observed lifetime of 1 s would require

$$Q \sim \omega_i \tau_i \sim 7.76 \times 10^{20}. \tag{341}$$

Such a huge Q-factor is not impossible as a formal metastability parameter, but it cannot be interpreted as a simple geometric shell crossing.

Experimental analogy

Multi-exponential decay models are common in nuclear delayed-neutron kinetics and in linear response theory. SST may use this only as an analogy unless a topological barrier calculation derives the Q_i or τ_i values independently.

Falsifiability

The layered relaxation ansatz predicts that perturbing a structured target should produce reproducible decay constants τ_i independent of total pulse amplitude, provided the response remains linear. If the apparent τ_i vary strongly with drive amplitude, heating, or detector threshold, the model is likely observing instrumental or thermal relaxation rather than swirl-string metastability.

U Summary of Canon Placement

The four additions should be placed as follows:

Appendix	**Canon status**	**Recommended placement**
Resonance Selection Principle	[RESEARCH-TRACK] strong candidate	CANON-v0.8.1 companion appendix; possible future main-canon principle after validation.
Time-Domain Kernel Response	[RESEARCH-TRACK] strong candidate	Near Swirl Clock, delay-mode selection, or quantum-response sections.
Swirl Impedance and Linewidth	[RESEARCH-TRACK] with corrected dimensions	Use as a response/linewidth ansatz; do not canonise the older dimensionally incorrect formula.
Layered Relaxation	[RESEARCH-TRACK] analogy only	Keep as metastable multi-mode response; reject direct delayed-neutron shell-crossing interpretation.

References

[1] R. Kubo. (1957). *Statistical-Mechanical Theory of Irreversible Processes. I. General Theory and Simple Applications to Magnetic and Conduction Problems*. Journal of the Physical Society of Japan **12**, 570–586. doi: 10.1143/JPSJ.12.570.

[2] G. Breit and E. Wigner. (1936). *Capture of Slow Neutrons*. Physical Review **49**, 519–531. doi: 10.1103/PhysRev.49.519.

[3] G. K. Batchelor. (1967). *An Introduction to Fluid Dynamics*. Cambridge University Press.

[4] P. G. Saffman. (1992). *Vortex Dynamics*. Cambridge University Press. doi: 10.1017/CBO9780511624063.

[5] H. von Helmholtz. (1858). *Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen*. Journal für die reine und angewandte Mathematik **55**, 25–55. doi: 10.1515/crll.1858.55.25.

[6] W. Thomson, Lord Kelvin. (1867). *On Vortex Atoms*. Proceedings of the Royal Society of Edinburgh **6**, 94–105.

[7] L. Onsager. (1949). *Statistical Hydrodynamics.* Il Nuovo Cimento **6**, 279–287. doi: 10.1007/BF02780991.

[8] R. P. Feynman. (1955). *Application of Quantum Mechanics to Liquid Helium.* In C. J. Gorter (ed.), *Progress in Low Temperature Physics*, Vol. 1, pp. 17–53. North-Holland. doi: 10.1016/S0079-6417(08)60077-3.

[9] R. N. Bracewell. (2000). *The Fourier Transform and Its Applications.* 3rd ed., McGraw–Hill.

[10] G. R. Keepin. (1965). *Physics of Nuclear Kinetics.* Addison-Wesley.

[11] M. Dhibar, I. Mazumdar, G. Anil Kumar, S. P. Weppner, A. K. Rhine Kumar, S. M. Patel, and P. B. Chavan. (2018). *Study of* $^{12}\mathrm{C}(p,p'\gamma)^{12}\mathrm{C}$ *reaction.* arXiv:1806.02126. https://arxiv.org/abs/1806.02126.

[12] P. B. Rose Jr. et al. (2018). *High-energy gamma rays resulting from low-energy nuclear reactions.* Physical Review C **97**, 064305. doi: 10.1103/PhysRevC.97.064305.

V Research Track: Nested Vortex–Antivortex States as Entropy-Minimizing Swirl Structures

V.1 Motivation

Consider a coherent swirl structure consisting of an outer circulation region with vorticity $\omega_{\text{out}} > 0$ and an embedded inner core with opposite vorticity

$$\omega_{\text{in}} < 0.$$

Such configurations appear naturally in fluid dynamics, magnetic flux ropes, and vortex-ring interactions.

Within SST, opposite swirl chirality corresponds to opposite temporal orientation of the Swirl Clock:

$$S_t^{\circlearrowleft} \quad \leftrightarrow \quad \text{matter},$$

$$S_t^{\circlearrowright} \quad \leftrightarrow \quad \text{antimatter}.$$

V.2 Helicity Cancellation

The total helicity may be decomposed as

$$H_{\text{tot}} = H_{\text{out}} + H_{\text{in}}.$$

Because the embedded structure carries opposite chirality,

$$H_{\text{in}} < 0.$$

Hence

$$|H_{\text{tot}}| < |H_{\text{out}}|.$$

The configuration therefore possesses reduced net topological complexity.

V.3 Entropy Hypothesis

We introduce the conjecture

$$S_{\text{top}} \propto |H|,$$

where S_{top} denotes topological entropy.

Consequently,

$$S_{\text{top}}^{\text{nested}} < S_{\text{top}}^{\text{single}},$$

suggesting that nested vortex–antivortex states may represent preferred low-energy attractors of the swirl medium.

V.4 Status

Research Track.

Not yet derived from the canonical SST Lagrangian. Proposed as a candidate mechanism for stability enhancement and matter–antimatter coexistence.

W Research Track: Helicity Conversion and Torsional Radiation

W.1 Motivation

Experiments and simulations of knotted vortex structures indicate that reconnection events frequently preserve a substantial fraction of helicity.

Rather than disappearing, helicity is redistributed between knotting, linking, writhe, and twist.

W.2 Helicity Partition

Write total helicity as

$$H = H_{\text{knot}} + H_{\text{twist}} + H_{\text{link}}.$$

During reconnection:

$$H_{\text{knot}} \downarrow$$

while

$$H_{\text{twist}} \uparrow.$$

Thus

$$H_{\text{tot}} \approx \text{const.}$$

W.3 SST Interpretation

Within SST, mass-bearing T-phase structures correspond to localized knot helicity.

Radiative R-phase excitations correspond to propagating torsional modes.

We therefore conjecture

$$H_{\text{knot}} \to H_{\text{twist}} \to \gamma_R,$$

where γ_R denotes an R-phase torsional pulse.

W.4 Mass–Radiation Conversion Chain

The proposed conversion pathway is

$$M \leftrightarrow H_{\mathrm{knot}} \leftrightarrow H_{\mathrm{twist}} \leftrightarrow \gamma_R.$$

In this picture:

- stable particles store helicity as topology,
- reconnections release helicity as torsional twist,
- torsional twist propagates as photon-like R-phase radiation.

W.5 Connection to Photon Ontology

This hypothesis is consistent with the SST identification

$$\text{photon} = \text{torsional R-phase pulse},$$

and provides a possible mechanism for

- particle decay,
- annihilation,
- radiation emission,
- energy transport.

W.6 Falsifiable Prediction

If the hypothesis is correct, vortex reconnection experiments should show

$$\Delta H_{\mathrm{knot}} \approx -\Delta H_{\mathrm{twist}}$$

with emitted torsional wave packets carrying the missing helicity.

W.7 Status

Research Track.

Motivated by helicity-conservation studies in knotted vortex systems. Requires derivation from the SST radiation sector.

X Research Track: Kairos Surfaces as Chirality Transition Layers

X.1 Definition

The Temporal Ontology introduces the Kairos event

$$\kappa,$$

representing an irreversible topological transition.

We extend this concept spatially.

Let

$$\omega(\mathbf{x}) = 0$$

define a surface separating opposite chiral domains:

$$\omega > 0 \quad \leftrightarrow \quad S_t^{\circlearrowleft},$$

$$\omega < 0 \quad \leftrightarrow \quad S_t^{\circlearrowright}.$$

We define the locus

$$\Sigma_\kappa = \{\mathbf{x} \mid \omega(\mathbf{x}) = 0\}$$

as a *Kairos Surface.*

X.2 Physical Interpretation

Across Σ_κ the swirl orientation reverses sign.

The surface therefore acts as a boundary between distinct temporal sectors.

$$S_t^{\circlearrowleft} \longleftrightarrow S_t^{\circlearrowright}.$$

Such regions may act as:

- annihilation interfaces,
- vortex reconnection sites,
- phase-transition boundaries,
- chirality-conversion layers.

X.3 Kairos Criterion

A local Kairos transition is conjectured whenever

$$\omega \to 0$$

and

$$\nabla\omega \neq 0.$$

The gradient supplies a preferred transition direction.

X.4 Status

Research Track.

Consistent with the Temporal Ontology but not yet derived from the SST field equations.

Y Euler-Decomposed Swirl Gravity and Probe Transport

Y.1 Purpose and status

This appendix separates the canonical scalar pressure acceleration from orientation-dependent transverse transport effects. The pressure law is retained as the canon-level bridge, while Magnus-type terms are treated as probe-level corrections.

Y.2 Canonical pressure acceleration

$$\mathbf{a}_P = -\frac{1}{\rho_f}\nabla p_{\text{swirl}}. \tag{342}$$

Y.3 Euler decomposition

For an incompressible inviscid background flow,

$$\frac{\partial \mathbf{u}}{\partial t} + \boldsymbol{\omega} \times \mathbf{u} + \nabla\left(\frac{u^2}{2}\right) = -\frac{1}{\rho_f}\nabla p_{\text{swirl}}. \tag{343}$$

Y.4 Filament Magnus transport

For a circulation-carrying swirl-string probe K,

$$\mathbf{F}_{\text{M}}^{(K)} = \int_K \rho_f \Gamma_K \mathbf{T}(s) \times [\mathbf{U}_K(s) - \mathbf{u}(\mathbf{X}(s))]\, ds. \tag{344}$$

Y.5 Curvature self-transport

In the local-induction approximation,

$$\mathbf{f}_\kappa \simeq -\frac{\rho_f \Gamma_K^2}{4\pi}\Lambda_K \kappa \mathbf{N}. \tag{345}$$

Y.6 Status

The scalar pressure term is canon-level. The Magnus and curvature terms are research-track transport corrections until a universal coupling law for macroscopic matter is derived.

Z Research Track: Swirl–Electromagnetic Normalization by Vorticity

Z.1 Purpose and Status

This appendix records a research-track extension of Swirl–String Theory (SST) in which electromagnetic field variables are interpreted as SI-normalized coarse-grained measures of swirl kinematics. The result is not proposed as a replacement of Maxwell's equations at the level of observed electrodynamics. Instead, it is a constitutive bridge between the SST swirl-medium variables and the ordinary electromagnetic variables **E**, **B**, **A**, **D**, and **H**.

The central observation is that a dimensionally correct identification of the vector potential is obtained by scaling the local swirl velocity by the electron mass-to-charge ratio:

$$\boxed{\mathbf{A}_{\text{SI}} = \frac{m_e}{e}\mathbf{v}_\circlearrowleft}. \tag{346}$$

Since $[\mathbf{A}] = \text{kg}\,\text{m}\,\text{C}^{-1}\,\text{s}^{-1}$ and $[(m_e/e)\mathbf{v}_\circlearrowleft] = \text{kg}\,\text{C}^{-1} \cdot \text{m}\,\text{s}^{-1}$, this equation has the correct SI units.

The associated magnetic field is then

$$\boxed{\mathbf{B}_{\text{SST}} = \nabla \times \mathbf{A}_{\text{SI}} = \frac{m_e}{e}\boldsymbol{\omega}}, \qquad \boldsymbol{\omega} = \nabla \times \mathbf{v}_\circlearrowleft. \tag{347}$$

This is the cleanest form of the statement that magnetic flux arises from structured vorticity, not merely from charge flow. The charge-current description remains the effective macroscopic description; the SST interpretation places the microscopic origin of the magnetic field in quantized circulation and vorticity.

Z.2 Why Direct Vorticity Source Terms Are Rejected

Earlier trial equations attempted to write vorticity as an additional source term directly inside Maxwell's equations, for example terms of the schematic form

$$\nabla \cdot \mathbf{E} \sim \frac{\rho}{\varepsilon_0} + \frac{F_{\max}\omega}{\|\mathbf{v}_{\circlearrowleft}\| {r_c}^2}, \qquad \nabla \times \mathbf{B} \sim \mu_0 \mathbf{J} + \frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{q{r_c}^2}\mathbf{\Omega}. \tag{348}$$

These forms are not retained as canonical equations because the added terms do not generally have the SI dimensions required by Maxwell's source equations. In particular, $\nabla \cdot \mathbf{E}$ has units $\mathrm{V\,m^{-2}}$, while the proposed vorticity-force expression does not reduce to this unit without an additional calibrated constitutive coefficient. Likewise, the Ampere–Maxwell equation requires units of $\mathrm{T\,m^{-1}}$, not a force-per-charge expression multiplied by an unfixed vorticity vector.

Therefore, the research-track correction is:

> Vorticity should not be inserted as an uncalibrated extra Maxwell source. It should enter through a constitutive identification of the potential and field variables.

The retained bridge is thus Eq. (346), not Eq. (348).

Z.3 Electric Field from Time-Dependent Swirl

With the vector potential bridge established, the electric field follows from the usual potential relation:

$$\boxed{\mathbf{E}_{\mathrm{SST}} = -\frac{\partial \mathbf{A}_{\mathrm{SI}}}{\partial t} - \nabla\Phi_{\mathrm{SST}} = -\frac{m_e}{e}\frac{\partial \mathbf{v}_{\circlearrowleft}}{\partial t} - \nabla\Phi_{\mathrm{SST}}}. \tag{349}$$

Here Φ_{SST} is the SI-normalized scalar potential associated with compressive, pressure-like, or boundary-induced swirl response. In strictly incompressible regions, Φ_{SST} should be interpreted as an effective potential imposed by boundary constraints, topology, and coarse-grained pressure balance rather than by local volume compression.

Equation (349) gives the SST analogue of Faraday induction: time-varying swirl produces an electric field. Taking the curl gives

$$\nabla \times \mathbf{E}_{\mathrm{SST}} = -\frac{\partial}{\partial t}\left(\nabla \times \mathbf{A}_{\mathrm{SI}}\right) = -\frac{\partial \mathbf{B}_{\mathrm{SST}}}{\partial t}, \tag{350}$$

since $\nabla \times \nabla\Phi_{\mathrm{SST}} = 0$. Thus the ordinary Faraday law is preserved exactly.

Z.4 Maxwell Equations as Effective Medium Equations

The correct research-track formulation is to retain Maxwell's equations in their macroscopic form while adding swirl-induced polarization and magnetization terms:

$$\nabla \cdot \mathbf{D}_{\mathrm{SST}} = \rho_{\mathrm{free}}, \tag{351}$$

$$\nabla \cdot \mathbf{B}_{\mathrm{SST}} = 0, \tag{352}$$

$$\nabla \times \mathbf{E}_{\mathrm{SST}} = -\frac{\partial \mathbf{B}_{\mathrm{SST}}}{\partial t}, \tag{353}$$

$$\nabla \times \mathbf{H}_{\mathrm{SST}} = \mathbf{J}_{\mathrm{free}} + \frac{\partial \mathbf{D}_{\mathrm{SST}}}{\partial t}. \tag{354}$$

The constitutive relations are

$$\mathbf{D}_{\mathrm{SST}} = \varepsilon_0 \mathbf{E}_{\mathrm{SST}} + \mathbf{P}_{\mathrm{swirl}}, \tag{355}$$

$$\mathbf{H}_{\mathrm{SST}} = \frac{1}{\mu_0}\mathbf{B}_{\mathrm{SST}} - \mathbf{M}_{\mathrm{swirl}}. \tag{356}$$

Here $\mathbf{P}_{\text{swirl}}$ and $\mathbf{M}_{\text{swirl}}$ encode the coarse-grained response of bound swirl structures, knot cores, circulating R-phase modes, and boundary-induced polarization.

This formulation has an important advantage: it does not violate the observed Maxwell structure. It instead interprets charge and current as effective descriptions of deeper circulation and vorticity organization.

Z.5 Canonical Circulation-Scale Calibration and the QED Critical Field

At the SST canonical circulation scale the characteristic angular frequency scale is

$$\Omega_{\text{core}} = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}. \tag{357}$$

Using the canonical SST values

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\ \text{m}\,\text{s}^{-1}, \qquad r_c = 1.40897017 \times 10^{-15}\ \text{m}, \tag{358}$$

one obtains

$$\Omega_{\text{core}} = 7.76344066 \times 10^{20}\ \text{s}^{-1}. \tag{359}$$

The corresponding SST magnetic field scale is

$$B_{\text{core}}^{\text{SST}} = \frac{m_e}{e}\Omega_{\text{core}}. \tag{360}$$

Numerically,

$$\frac{m_e}{e} = 5.68563010 \times 10^{-12}\ \text{kg}\,\text{C}^{-1}, \tag{361}$$

and therefore

$$\boxed{B_{\text{core}}^{\text{SST}} = 4.41400519 \times 10^{9}\ \text{T}}. \tag{362}$$

This is not an arbitrary scale. The standard QED critical magnetic field is

$$B_{\text{QED}} = \frac{m_e^2 c^2}{e\hbar} = 4.41400522 \times 10^{9}\ \text{T}. \tag{363}$$

Thus

$$\frac{B_{\text{core}}^{\text{SST}}}{B_{\text{QED}}} = 0.999999993. \tag{364}$$

The equality follows analytically from the SST Compton-core relation

$$\lambda_c = \frac{2\pi c r_c}{\|\mathbf{v}_{\circlearrowleft}\|}, \tag{365}$$

together with the standard identity

$$\lambda_c = \frac{2\pi\hbar}{m_e c}. \tag{366}$$

Equating these two expressions gives

$$\frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = \frac{m_e c^2}{\hbar}. \tag{367}$$

Substitution into Eq. (360) yields

$$B_{\text{core}}^{\text{SST}} = \frac{m_e}{e}\frac{m_e c^2}{\hbar} = \frac{m_e^2 c^2}{e\hbar} = B_{\text{QED}}. \tag{368}$$

This is the strongest numerical result of this appendix: the SI-normalized vorticity bridge naturally reproduces the Schwinger/QED magnetic field scale at the SST core.

The associated electric field scale is

$$E_{\text{core}}^{\text{SST}} = cB_{\text{core}}^{\text{SST}} = 1.32328547 \times 10^{18}\ \text{V}\,\text{m}^{-1}, \tag{369}$$

which matches the usual Schwinger electric field

$$E_{\text{Schwinger}} = \frac{m_e^2 c^3}{e\hbar}. \tag{370}$$

Z.6 Corrected Maximum Force Relation

The same calibration fixes the correct factor in the SST maximum-force relation. The expression

$$\frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{{r_c}^2} \tag{371}$$

has units of force, but it is twice the canonical force scale. The consistent relation is

$$\boxed{F^{\text{swirl}}_{\max} = \frac{\|\mathbf{v}_{\circlearrowleft}\|\hbar}{2{r_c}^2}}. \tag{372}$$

Using the canonical constants gives

$$F^{\text{swirl}}_{\max} = 29.0535098\ \text{N}, \tag{373}$$

in agreement with the canonical value

$$F^{\text{swirl}}_{\max} \simeq 29.053507\ \text{N}. \tag{374}$$

The electron rest energy relation is then

$$\boxed{m_e c^2 = 2F^{\text{swirl}}_{\max} r_c}. \tag{375}$$

Equivalently,

$$m_e = \frac{2F^{\text{swirl}}_{\max} r_c}{c^2}. \tag{376}$$

Numerically,

$$\frac{2F^{\text{swirl}}_{\max} r_c}{c^2} = 9.10938 \times 10^{-31}\ \text{kg}, \tag{377}$$

which matches the electron mass within the stated calibration precision.

Z.7 Photon and R-Phase Normal Modes

A second research-track result is that photon frequency should not be introduced through an ad hoc energy-density law of the form

$$u(r,\omega,T) \propto \frac{F_{\max}\omega^3}{\|\mathbf{v}_{\circlearrowleft}\|r^2}\frac{1}{\exp(\hbar\omega/k_B T) - 1}. \tag{378}$$

This expression is not retained as a fundamental energy density because its bare prefactor does not have the units of energy per volume without additional geometric or constitutive factors.

Instead, the research-track photon sector is formulated as a normal-mode condition on a closed or quasi-closed R-phase path:

$$\boxed{k_n = \frac{2\pi n}{L_{\text{eff}}(K)}, \qquad \omega_n = c_\gamma k_n = \frac{2\pi n c_\gamma}{L_{\text{eff}}(K)}}. \tag{379}$$

Here $L_{\text{eff}}(K)$ is the effective optical/swirl path length of the supporting configuration K, and c_γ is the propagation speed of the R-phase excitation. In the vacuum limit one expects $c_\gamma = c$. For a circular ring of radius R,

$$L_{\text{eff}} = 2\pi R, \tag{380}$$

so that

$$\boxed{\omega_n = \frac{n c_\gamma}{R}}. \tag{381}$$

This gives a clean formulation of the statement:

Photon frequency depends on the supporting swirl-ring or path structure.

For knotted, linked, or boundary-deformed paths, $L_{\text{eff}}(K)$ may include writhe, torsion, self-linking, and medium-response corrections:

$$L_{\text{eff}}(K) = L(K) + \Delta L_{\text{writhe}} + \Delta L_{\text{torsion}} + \Delta L_{\text{boundary}} + \Delta L_{\text{medium}}. \tag{382}$$

This places the photon sector in the same mathematical family as the SST delay-loop and mode-selection principles.

Z.8 Kelvin Impulse Bridge

For a thin circular swirl ring with circulation Γ, major radius R, and core radius a, the classical ring scalings are

$$E_{\text{ring}} = \frac{1}{2}\rho_f \Gamma^2 R \left[\ln\left(\frac{8R}{a}\right) - \alpha_{\text{ring}}\right], \tag{383}$$

$$P_{\text{ring}} = \rho_f \Gamma \pi R^2, \tag{384}$$

$$V_{\text{ring}} = \frac{\Gamma}{4\pi R}\left[\ln\left(\frac{8R}{a}\right) - \beta_{\text{ring}}\right]. \tag{385}$$

These relations provide a bridge between Kelvin's impulse theory and an SST momentum assignment. A research-track identification may be written as

$$\boxed{\hbar k \leftrightarrow \eta_I P_{\text{swirl}}}, \qquad P_{\text{swirl}} = \rho_f \Gamma \pi R^2. \tag{386}$$

The coefficient η_I is not a free fit parameter in the final theory. It must be determined from the same SI-normalization that fixes Eqs. (346)–(368), or from an independently specified R-phase normalization.

This suggests a useful research direction:

> Planck momentum may correspond to normalized Kelvin impulse of a circulating R-phase excitation.

Z.9 Topology, Helicity, and Reconnection

The ideal SST sector assumes conservation of circulation, linkage, and helicity in the absence of reconnection. This corresponds to the Helmholtz–Kelvin regime:

$$\frac{D\Gamma}{Dt} = 0, \qquad \frac{D\mathcal{H}}{Dt} = 0, \qquad \mathcal{H} = \int \mathbf{v}_{\circlearrowleft} \cdot \boldsymbol{\omega}\, d^3x. \tag{387}$$

In this ideal sector, knot class is preserved.

However, laboratory and numerical studies of knotted and linked fluid structures show that reconnection can transfer helicity between linking, twist, and writhe. Therefore, SST should distinguish two regimes:

$$\text{Ideal sector:} \quad K(t) = K(0), \qquad \Gamma = \text{constant}, \qquad \mathcal{H} = \text{constant}, \tag{388}$$

$$\text{Transition sector:} \quad K_i \to K_j, \qquad \Delta\mathcal{H} = \Delta\mathcal{L} + \Delta\mathcal{T} + \Delta\mathcal{W}, \tag{389}$$

where $\mathcal{L}$, $\mathcal{T}$, and $\mathcal{W}$ denote linking, twist, and writhe contributions respectively.

This is relevant for the SST interpretation of measurement, decay, and mode conversion. A stable particle-like excitation belongs to the ideal sector. A decay or transition corresponds to a controlled failure of ideal topological conservation, usually through reconnection or boundary interaction.

Z.10 Pressure Envelopes and Spherical Equilibrium

The electromagnetic bridge above should be kept distinct from the pressure-envelope sector of SST. The pressure sector is governed by the Bernoulli-type swirl balance

$$p + \frac{1}{2}\rho_f \|\mathbf{v}_\circlearrowleft\|^2 = p_0, \tag{390}$$

or, for a perturbation,

$$\Delta p = -\frac{1}{2}\rho_f \Delta\left(\|\mathbf{v}_\circlearrowleft\|^2\right). \tag{391}$$

This pressure defect can generate attraction-like behavior in the gravitational/swirl-clock sector. It should not be confused with the magnetic vorticity bridge

$$\mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}. \tag{392}$$

Thus the research-track separation is:

$$\text{EM-like sector:} \quad \mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\mathbf{v}_\circlearrowleft, \qquad \mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}, \tag{393}$$

$$\text{pressure/gravity sector:} \quad \Delta p = -\frac{1}{2}\rho_f \Delta(\|\mathbf{v}_\circlearrowleft\|^2), \qquad d\tau/dt = \sqrt{1 - \|\mathbf{v}_\circlearrowleft\|^2/c^2}. \tag{394}$$

Z.11 Negative-Temperature and Entropy Analogy

A further research-track idea is that bounded swirl ensembles may possess entropy-ordering behavior analogous to negative-temperature vortex states. This is not required for the electromagnetic normalization, but it may be useful in the classification of high-energy knot ensembles.

A possible entropy functional is

$$S_{\mathrm{swirl}} = k_B \ln \Omega_{\mathrm{topo}}(E, \Gamma, \mathcal{H}), \tag{395}$$

where Ω_{topo} counts accessible configurations at fixed energy, circulation, and helicity. A negative-temperature-like regime would satisfy

$$\frac{\partial S_{\mathrm{swirl}}}{\partial E} < 0. \tag{396}$$

Such a regime would describe increasing order with increasing energy, for example through clustering, alignment, or coherent large-scale swirl organization. This idea should remain in the research track until it is connected to a concrete SST Hamiltonian and a well-defined finite phase space.

Z.12 Research Claims Retained

The following claims are retained for research-track development:

1. Magnetic field strength can be SI-normalized as vorticity:

$$\mathbf{B}_{\mathrm{SST}} = \frac{m_e}{e}\boldsymbol{\omega}.$$

2. The vector potential is the SI-normalized swirl velocity:

$$\mathbf{A}_{\mathrm{SI}} = \frac{m_e}{e}\mathbf{v}_\circlearrowleft.$$

3. The core-scale SST magnetic field equals the QED critical field:

$$B_{\rm core}^{\rm SST} = B_{\rm QED} = \frac{m_e^2 c^2}{e\hbar}.$$

4. The associated electric field equals the Schwinger scale:

$$E_{\rm core}^{\rm SST} = cB_{\rm core}^{\rm SST} = \frac{m_e^2 c^3}{e\hbar}.$$

5. Photon frequency should be modeled by normal-mode closure:

$$\omega_n = \frac{2\pi n c_\gamma}{L_{\rm eff}(K)}.$$

6. Kelvin impulse may provide the bridge to photon momentum:

$$\hbar k \leftrightarrow \eta_I \rho_f \Gamma \pi R^2.$$

7. Reconnection belongs to a separate transition sector and should not be mixed with ideal knot conservation.

Z.13 Claims Not Retained as Canonical

The following statements are not retained as canonical equations:

1. Direct insertion of uncalibrated vorticity terms into Maxwell source equations.
2. The statement that charge is unnecessary in all electromagnetic phenomena. In this appendix charge remains the effective macroscopic source variable.
3. The claim that Podkletnov-type gravitational shielding is established evidence for SST. Such material may be mentioned only as speculative historical motivation, not as support for the canon.
4. Photon energy-density laws containing $F_{\rm max}\omega^3/(\|\mathbf{v}_{\circlearrowleft}\| r^2)$ without a complete dimensional and geometric derivation.

Z.14 Minimal Experimental Tests

The vorticity-normalization bridge suggests several falsifiable directions.

1. Structured plasmonic or polaritonic media. If electromagnetic fields are coupled to structured circulation-like degrees of freedom, then anisotropic or hyperbolic media should show geometry-dependent emission and propagation consistent with an effective path length $L_{\rm eff}(K)$.

2. Superfluid or quantum-fluid circulation. Quantized circulation structures should generate measurable effective magnetization only when the system contains charged, spinful, or electromagnetically coupled degrees of freedom. A neutral inviscid fluid alone should not emit ordinary electromagnetic fields unless an additional coupling channel exists.

3. Ring-resonator delay tests. The photon/R-phase normal-mode law predicts that frequency selection should scale as

$$\omega_n \propto \frac{1}{L_{\rm eff}}.$$

Boundary deformation, writhe-like path changes, or torsional loading should shift the selected resonances.

4. Core-scale extrapolation. The equality

$$B_{\text{core}}^{\text{SST}} = B_{\text{QED}}$$

is not directly reachable in ordinary laboratory magnetic fields, but it can be tested indirectly by checking whether the same normalization reproduces Compton, Schwinger, and electron-rest-energy scales without additional tuning.

Z.15 Numerical Verification Table

Using the canonical SST constants

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15}\ \mathrm{m},$$

one obtains:

Quantity	Expression	Numerical value
Core angular scale	$\Omega_{\text{core}} = \|\mathbf{v}_{\circlearrowleft}\|/r_c$	$7.76344066 \times 10^{20}\ \mathrm{s^{-1}}$
Mass-charge ratio	m_e/e	$5.68563010 \times 10^{-12}\ \mathrm{kg\,C^{-1}}$
SST core magnetic field	$(m_e/e)\Omega_{\text{core}}$	$4.41400519 \times 10^{9}\ \mathrm{T}$
QED critical magnetic field	$m_e^2c^2/(e\hbar)$	$4.41400522 \times 10^{9}\ \mathrm{T}$
Ratio	$B_{\text{core}}^{\text{SST}}/B_{\text{QED}}$	0.999999993
SST electric field scale	$cB_{\text{core}}^{\text{SST}}$	$1.32328547 \times 10^{18}\ \mathrm{V\,m^{-1}}$
Circulation quantum	$2\pi r_c\|\mathbf{v}_{\circlearrowleft}\|$	$9.68361920 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}$
Corrected force ceiling	$\|\mathbf{v}_{\circlearrowleft}\|\hbar/(2{r_c}^2)$	29.0535098 N
Electron mass from force	$2F_{\text{max}}^{\text{swirl}}r_c/c^2$	$9.10938 \times 10^{-31}\ \mathrm{kg}$

Z.16 Conclusion

The strongest result of this appendix is the SI-normalized vorticity bridge

$$\mathbf{A}_{\text{SI}} = \frac{m_e}{e}\mathbf{v}_{\circlearrowleft}, \qquad \mathbf{B}_{\text{SST}} = \frac{m_e}{e}\boldsymbol{\omega}.$$

It preserves Maxwell's equations as effective field equations, avoids dimensionally invalid source terms, and reproduces the QED critical field at the SST core scale. For this reason it is suitable for inclusion in the research-track sections of CANON-v0.8.x.

The proposed canonical status is:

Result	Status
$\mathbf{A}_{\text{SI}} = (m_e/e)\mathbf{v}_{\circlearrowleft}$	Research Track, strong candidate
$\mathbf{B}_{\text{SST}} = (m_e/e)\boldsymbol{\omega}$	Research Track, strong candidate
$B_{\text{core}}^{\text{SST}} = B_{\text{QED}}$	Numerically verified calibration result
Direct vorticity source terms in Maxwell equations	Rejected / not canonical
Photon frequency from $L_{\text{eff}}(K)$	Research Track, compatible with delay-loop sector
Kelvin impulse to $\hbar k$ bridge	Research Track, unresolved normalization
Negative-temperature swirl entropy	Speculative Research Track

References

[1] J. K. Hale and S. M. Verduyn Lunel, *Introduction to Functional Differential Equations*, Springer (1993).

[2] CMS Collaboration, High-precision measurement of the W boson mass with the CMS experiment, Nature **652** (2026), 321–337, DOI: 10.1038/s41586-026-10168-5.

[3] S. Navas et al. (Particle Data Group), Review of Particle Physics, Phys. Rev. D **110** (2024), 030001, DOI: 10.1103/PhysRevD.110.030001.

[4] J. de Blas et al., Global analysis of electroweak data in the Standard Model, Phys. Rev. D **106** (2022), 033003, DOI: 10.1103/PhysRevD.106.033003.

[5] CDF Collaboration, High-precision measurement of the W boson mass with the CDF II detector, Science **376** (2022), 170–176, DOI: 10.1126/science.abk1781.

[6] G. K. Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University Press (1967).

[7] T. Jacobson, Einstein-Æther gravity: a status report, PoS QG-PH (2007) 020, DOI: 10.22323/1.043.0020.

[8] D. Blas, O. Pujolàs, and S. Sibiryakov, Models of non-relativistic quantum gravity: the good, the bad and the healthy, JHEP **04** (2011) 018, DOI: 10.1007/JHEP04(2011)018.

[9] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), GW170817: Observation of gravitational waves from a binary neutron star inspiral, Phys. Rev. Lett. **119** (2017) 161101, DOI: 10.1103/PhysRevLett.119.161101.

[10] W. G. Unruh, Experimental black-hole evaporation?, Phys. Rev. Lett. **46** (1981), 1351–1353, DOI: 10.1103/PhysRevLett.46.1351.

[11] M. Visser, Acoustic black holes: horizons, ergospheres and Hawking radiation, Class. Quantum Grav. **15** (1998), 1767–1791, DOI: 10.1088/0264-9381/15/6/024.

[12] P. Painlevé, La mécanique classique et la théorie de la relativité, C. R. Acad. Sci. (Paris) **173** (1921), 677–680.

[13] A. Gullstrand, Allgemeine Lösung des statischen Einkörperproblems in der Einsteinschen Gravitationstheorie, Arkiv. Mat. Astron. Fys. **16**(8) (1922), 1–15.

[14] E. Schrödinger, Quantisierung als Eigenwertproblem, Annalen der Physik **384** (1926), 361–376, DOI: 10.1002/andp.19263840404.

[15] E. Madelung, Quantentheorie in hydrodynamischer Form, Z. Phys. **40** (1927), 322–326, DOI: 10.1007/BF01400372.

[16] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley (1999).

[17] M. Planck, Ueber das Gesetz der Energieverteilung im Normalspectrum, Annalen der Physik **309** (1901), 553–563, DOI: 10.1002/andp.19013090310.

[18] D. Rolfsen, *Knots and Links*, Publish or Perish (1976); AMS Chelsea reprint (2003).

[19] J. H. Conway, An enumeration of knots and links, and some of their algebraic properties, in *Computational Problems in Abstract Algebra*, Pergamon Press (1970), pp. 329–358, DOI: 10.1016/B978-0-08-012975-4.50034-5.

[20] W. Wien, Ueber die Energieverteilung im Emissionsspectrum eines schwarzen Körpers, Annalen der Physik **294** (1896), 662–669, DOI: 10.1002/andp.18962940310.

[21] G. Călugăreanu, Sur les classes d'isotopie des noeuds tridimensionnels et leurs invariants, Czech. Math. J. **11** (1961), 588–625.

[22] J. H. White, Self-linking and the Gauss integral in higher dimensions, Amer. J. Math. **91** (1969), 693–728, DOI: 10.2307/2373348.

[23] P. G. Saffman, *Vortex Dynamics*, Cambridge University Press (1992).

[24] S. Kida and M. Takaoka, Vortex reconnection, Annual Review of Fluid Mechanics **26** (1994), 169–189, DOI: 10.1146/annurev.fl.26.010194.001125.

[25] J. Koplik and H. Levine, Vortex reconnection in superfluid helium, Physical Review Letters **71** (1993), 1375–1378, DOI: 10.1103/PhysRevLett.71.1375.

[26] H. K. Moffatt, The degree of knottedness of tangled vortex lines, J. Fluid Mech. **35** (1969), 117–129, DOI: 10.1017/S0022112069000991.

[27] W. H. Zurek, Decoherence, einselection, and the quantum origins of the classical, Rev. Mod. Phys. **75** (2003), 715–775, DOI: 10.1103/RevModPhys.75.715.

[28] S. Weinberg, A model of leptons, Phys. Rev. Lett. **19** (1967), 1264–1266, DOI: 10.1103/PhysRevLett.19.1264.

[29] G. W. Gibbons, The maximum tension principle in general relativity, Found. Phys. **32** (2002), 1891–1901, DOI: 10.1023/A:1022370717626.